# The Effects of Service Integration on Platform Competition and Welfare: Evidence from Fulfillment by Amazon

Kyungho Lee*

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**Abstract**

I build an equilibrium model of competition between Amazon and Walmart to measure the welfare effects of integrated logistics, Fulfillment by Amazon (FBA), in the U.S. e-commerce market. While integration eliminates double marginalization arising from marketplace commissions and logistics fees, it allows a platform to make its service exclusive to its own marketplace or more costly to use on a rival platform. Such foreclosure weakens cross-platform competition, with the magnitude of this effect depending on how much sellers' adoption of integrated logistics raises their costs of multihoming (i.e., selling on both platforms). The model captures this trade-off: platforms set both commission rates and logistics fees, accounting for consumer choice and sellers' multi-platform entry, logistics adoption, and pricing. I construct a unique dataset linking products and sellers across the two platforms, estimate the model, and document evidence of both pro- and anti-competitive effects of integration: the platforms hold logistics fees down substantially to increase commission profits, yet sellers who use FBA are much less likely to sell on Walmart, a finding that is further supported by a regression discontinuity design. On net, counterfactual separation of FBA from Amazon would reduce consumer welfare by 19%: although separation increases sellers' multihoming by 92%, it raises logistics fees, commission rates, and consumer prices (+25%). A policy that removes exclusivity while preserving integration, such as making integrated logistics interoperable, would improve welfare.

**Keywords**: digital platforms, vertical integration, multihoming, cross-platform competition, e-commerce, logistics, antitrust

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# 1 Introduction

Amazon delivered 6.7 billion packages to U.S. households in 2025, becoming the largest logistics provider in the United States in addition to being the dominant e-commerce platform (Gürbes, 2026).[1] A major part of its logistics operations is Fulfillment by Amazon (FBA),[2] which stores and delivers sellers' products and has played a pivotal role in the growth of Amazon's e-commerce business (Bezos, 2019).[3] More broadly, integrating complementary services has become a key strategy among digital platforms, extending their role beyond that of marketplace intermediaries: Walmart and TikTok operate their own e-commerce logistics; Google and Apple provide payment systems for their app stores; and Google offers ad tech tools alongside its ad exchange.

To what extent does a platform's vertical integration of complementary services benefit or harm consumers and sellers? How would regulatory policies affect welfare in an integrated platform economy? Because large digital platforms serve as gatekeepers for a wide range of digital transactions, whether such integration restricts competition or undermines efficiency has become a global regulatory issue (*Epic v. Apple*, 2020; *Epic v. Google*, 2020; AGCM, 2021; EC, 2022; *United States v. Google*, 2023; CMA, 2023).[4] However, research on a platform's integration is scarce, partly because examining platform competition involves a complex equilibrium among consumers, sellers, and platforms, and because data revealing the multi-platform behavior of consumers and sellers, necessary for studying cross-platform competition, are rarely available. In this paper, I focus on the U.S. e-commerce market and build an equilibrium model of competition between Amazon and Walmart. I construct a dataset linking products and sellers across the two platforms, estimate the model, and quantify the welfare effects of Amazon's FBA integration. This analysis is timely, as the role of FBA in competition and welfare is at the heart of an active antitrust case—*FTC v. Amazon* (2023).

A platform's integration creates a welfare trade-off between eliminating double marginalization (Spengler, 1950) and foreclosing rivals, or raising their costs, to divert sales toward the integrated platform (Salop and Scheffman, 1983; Salinger, 1988).[5] On the one hand, integration lets the platform internalize the profit from service provision. Whereas an independent marketplace owner and an independent logistics provider would each add its own markup to the commission rate and to the logistics fee respectively, the integrated platform sets both jointly. Integration therefore creates in-

[1] This volume exceeds that of USPS (6.6 billion), UPS (4.4 billion), and FedEx (3.6 billion), and it is equivalent to 18.3 million packages per day, or 212 every second.

[2] I use the terms Amazon logistics and FBA interchangeably.

[3] As Amazon's CEO wrote in his letter to shareholders, "... of great importance are Fulfillment by Amazon and the Prime membership program ..."

[4] FBA is a key focus of investigations by competition authorities in Europe (AGCM, 2021; EC, 2022; CMA, 2023). In app markets, which have a similar platform structure, the role of integrated payment systems is central to antitrust cases (*Epic v. Apple*, 2020; *Epic v. Google*, 2020). There is also an antitrust case concerning the integration of complementary services in the ad exchange market, *United States v. Google* (2023).

[5] Although I focus on logistics integration, the welfare trade-off is also applicable to other platform markets.

centives for the platform to lower its logistics fees in order to increase sales on its own marketplace.[6] The lower fees make e-commerce more profitable for sellers, encourage their entry, and ultimately benefit consumers via pass-through to prices and greater within-platform competition.

On the other hand, integration allows a platform to make its service exclusive to its own marketplace, or more costly to use on a rival platform. Such foreclosure weakens cross-platform competition. The size of the foreclosure effect depends on how much sellers' adoption of integrated service raises their costs of multihoming (i.e., selling on both platforms) and hence restricts cross-platform substitution. This channel relates to the policy concern about anti-competitive lock-in.[7] To be specific, when sellers use the integrated logistics, the platform stores their products. If the integrated logistics is exclusive, sellers need another logistics provider to operate on a rival platform, which raises their cost of entering the rival platform. Even if sellers can use the integrated logistics to fulfill rival platform orders, they pay higher logistics fees on such orders than on integrated platform orders. These higher fees make selling on rival platforms less profitable, discourage sellers' multihoming, and reduce cross-platform competition. Therefore, the welfare effects of integration are theoretically ambiguous, and the key force in determining the magnitude of the effects is how much integration affects cross-platform substitution and platforms' pricing.

I construct a novel dataset that sheds light on competition that Amazon faces from a rival platform—a margin that is important but understudied.[8] I match product and seller data across the two platforms and observe key variables: sellers' logistics usage, multihoming, and logistics fees.[9] There are three main parts to the dataset. The first is the *full-category panel*, which tracks about 100,000 bestsellers across nine product categories on Amazon from January 2021 through August 2025 and records whether these products and their sellers are also listed on Walmart. The second is the *kitchen appliance panel*, a more detailed web-scraped panel covering bestsellers within the kitchen appliance category on *both* Amazon and Walmart from April to June 2025 with rich characteristics. This category is well suited to the empirical analysis because the diverse sizes and weights of kitchen appliances generate large variation in logistics fees. The third is the *Numerator consumer panel*. This panel contains information on consumers' income, platform memberships (Prime and Walmart+), and shopping behavior across retailers, including the two platforms. I use the first part to examine empirical patterns across broad categories and the second and third parts

[6] This is consistent with the pricing of complementary goods (Cournot, 1838). Linnemer (2022) discusses the relation between Cournot's complements pricing and the elimination of double marginalization (Spengler, 1950).

[7] Competition concerns about lock-in extend beyond e-commerce to ad exchange and cloud computing markets. In ad markets, competition authorities have raised concerns that vertically integrated tools can limit users' ability to use competing ad exchanges (CMA, 2020; EC, 2025). In cloud computing, regulators have argued that data-transfer fees and technical barriers to switching and interoperability as potential sources of customer lock-in that can weaken competition (AdlC, 2023; CMA, 2025).

[8] For example, Lee and Musolff (2025) address cross-platform competition and multihoming of sellers as a research direction that should be examined.

[9] I match products across platforms using universal product codes (UPCs) and sellers using business names, addresses, and phone numbers. Section 3 introduces the data and summary statistics, and Appendix B provides further details. The dataset also contains the logistics fees and commission rates that each seller pays to the platforms as both are publicly posted as menu. These charges account for a large share of sellers' costs.

to estimate the equilibrium model.

Using these data, I conduct descriptive and regression discontinuity (RD) analyses. I document three stylized facts about sellers' logistics usage and multihoming. First, the majority of Amazon sellers use Amazon logistics, and most sellers store either all or none of their products with Amazon. Second, multihoming is rare among Amazon sellers. Third, there is a positive correlation between a seller's scale and its likelihood of multihoming. The correlation is stronger among sellers who use independent logistics services than among those who use Amazon logistics. To examine the impact of logistics usage on market outcomes, including multihoming, I exploit discontinuities in Amazon's logistics fee schedule: Amazon classifies products into tiers by size and weight, and logistics fees jump by 30–50% at tier thresholds. Using the RD design, I find that sellers facing the higher fees just above a threshold are more likely to drop Amazon logistics and more likely to sell on Walmart. These results are consistent with the foreclosure effect: adopting Amazon logistics hinders cross-platform competition by discouraging sellers' multihoming. In addition, I find discontinuous increases in consumer prices at the thresholds, indicating that logistics fees are passed through to prices and, in turn, affect consumer welfare.

I then build a model. Accounting for the responses of consumers and sellers, Amazon and Walmart compete by setting logistics fees and commission rates while also pricing their own retail products. In particular, the platforms' joint pricing of fees and rates captures the elimination of double marginalization (EDM). On the demand side, I focus on three aspects that are important for consumer welfare and the strategic decisions of sellers and platforms: price sensitivity, logistics valuation, and cross-platform substitution. To capture them, I let preferences for prices, logistics, and platforms flexibly vary with consumers' platform memberships, income, and random taste shocks (Berry *et al.*, 1995; Petrin, 2002; Berry and Haile, 2014). On the seller side, the key is to capture how logistics adoption affects sellers' costs and, thereby, their entry. I specify a seller's marginal cost function to depend on logistics fees and scale economies.[10] If a seller uses integrated logistics, it pays the posted fees but cannot pool inventory across platforms, leading to a loss of scale. In contrast, if a seller uses independent logistics, it pays the fees charged by independent logistics providers but can pool inventory across platforms to achieve greater scale.[11] Sellers also incur fixed entry costs that depend on the combination of platforms entered and logistics adopted. This specification captures whether adopting integrated logistics increases the fixed costs of multihoming. Sellers set prices after entry and logistics adoption, incorporating pass-through of variable costs to consumers.

I estimate the model and decompose the first-order conditions of the platforms' pricing. The main finding is that both platforms hold logistics fees down substantially to increase commission

[10]The modeling of economies of scale is motivated by the FTC's case against Amazon. The complaint *FTC v. Amazon* (2023) states: "... many sellers would prefer to use a single fulfillment provider for all their online orders, on and off Amazon...," reflecting the scale benefits of using a single logistics provider across platforms.

[11]I use the UPS ground rate as a proxy for the logistics fees of independent logistics providers.

profits, consistent with the pro-competitive EDM (Spengler, 1950). The estimates also reveal the trade-offs sellers face in choosing which logistics to use and where to enter, reflecting both consumer behavior and the cost-side impacts of adopting logistics. The demand estimates reveal three key features:[12] consumers are highly price sensitive;[13] Prime subscribers substantially value FBA, creating strong incentives for sellers to adopt Amazon logistics;[14] and cross-platform substitution is limited, indicating that multihoming is a way to expand into an additional retail channel.[15] On the cost side, I find that, in terms of logistics fees, Walmart logistics is the cheapest option, followed by Amazon logistics, and independent logistics is the most expensive.[16] Amazon's relatively low fees further incentivize sellers to adopt FBA, and the simulated expected variable profit of using FBA on Amazon is 70% larger than that of using independent logistics. However, sellers who adopt Amazon logistics face significant scale losses and large fixed costs of entering Walmart.[17] These cost-side penalties, combined with relatively low expected variable profits on Walmart, discourage FBA sellers from multihoming, consistent with the pattern of low multihoming and high FBA adoption and the anti-competitive effects of integration.

On net, counterfactual separation of FBA from Amazon would reduce consumer welfare and increase profit inequality among sellers.[18] Consumer welfare falls by 19%, despite separation's pro-competitive effects—non-exclusive access to high-quality logistics without discriminatory fees, inventory pooling, and the increase in multihoming (+92%). This is because separation introduces double marginalization: the separated Amazon logistics raises its fee by 41%, and Amazon and Walmart simultaneously raise their commission rates by 16% and 18%, respectively. Sellers pass the higher fees and rates through to consumers, and prices rise by 25% on average. While aggregate seller profits change little (+3%), separation substantially reduces the profits of small and medium-sized sellers and increases those of top earners, who tend to multihome, pool inventory, and benefit from scale. Walmart is the clear winner from separation. Its profit increases by 50%, primarily because its commission profit rises with higher prices, higher rates, and more multihoming.

Although I find that Amazon's logistics integration benefits consumers, there remains room for policy intervention to improve welfare. I propose a policy that I call *full interoperability*, which re-

[12] I use aggregate and micro moments to estimate consumer demand (Petrin, 2002; Berry *et al.*, 2004; Berry and Haile, 2024). To address the endogeneity of prices, I use logistics fees as a cost-shifter instrument.

[13] The average own-price elasticity is $-4.2$. This estimate is consistent with the range reported in the literature (Yu, 2024; Farronato *et al.*, 2025).

[14] Sellers are strongly incentivized to use Amazon logistics, as the majority of U.S. households (80%) subscribe to Amazon Prime, which allows subscribers to get fast shipping from FBA.

[15] The estimates indicate that platform memberships primarily determine which platform consumers choose.

[16] Marginal costs are recovered from the sellers' pricing model, and I use excluded demand shifters of a seller's other products and of rival products as instruments for the scale variables.

[17] I estimate fixed costs using revealed-preference inequalities, extending the binary-choice framework (enter or not) of Eizenberg (2014) to multiple discrete alternatives. Identification comes from sellers' observed choices: the fixed costs rationalize each choice given the variable profits of all alternatives.

[18] In the simulation, the separated FBA sets its logistics fee, and the Amazon marketplace sets its commission rate and its own retail prices. Walmart remains integrated and sets all three. Sellers can use the separated FBA to fulfill Walmart orders at the same fee as for Amazon orders and at the same fixed cost as with independent logistics.

moves exclusivity and discriminatory fees while preserving integration.[19] Under full interoperability, sellers can use integrated logistics to fulfill every order at the same quality and cost, regardless of the platform on which the order is placed. In the counterfactual simulation, full interoperability could raise consumer and seller welfare by 38% and 30%, respectively, relative to the status quo. This result supports applying the interoperability obligations that European regulators enforce under the Digital Markets Act (European Union, 2022) to integrated logistics in e-commerce markets.

I conduct two further counterfactual analyses to understand platform competition and integrated logistics. To quantify the extent to which a competing platform disciplines the dominant platform's pricing, I counterfactually remove Walmart's e-commerce marketplace. Without Walmart, Amazon, as a monopolist, would raise its commission rate by 23% but lower its logistics fee by 6%. Overall welfare falls by 20%, with consumer welfare down 28% and seller profits down 29%, while Amazon's profit rises by 10%. These results suggest that a policy promoting platform entry or preserving rival platforms, such as the recent court order in *Epic v. Google* (2025), can generate consumer and seller gains by providing an additional sales channel and by restoring competitive pressure. The second counterfactual measures how much of the cost savings from Amazon's logistics investment (Houde *et al.*, 2023) is passed through to consumers and sellers. I find that a 1% decrease in Amazon's logistics cost raises consumer welfare by about 0.15%, so pass-through is modest; seller profits change little.

**Related Literature** The main contribution of this paper is to develop and estimate a framework for evaluating the *cross-platform* competitive and welfare effects of a platform's integration of complementary services, which, to my knowledge, is the first in the literature. This paper is related to three strands of literature: the economics of Amazon, vertical integration, and multi-sided markets.

First, I contribute to the literature on the economics of Amazon.[20] Spurred by "Amazon's Antitrust Paradox" (Khan, 2017), a growing literature examines competition and welfare related to Amazon's business conduct. Existing work focuses on competition *within* the Amazon marketplace, especially Amazon's dual role as retailer and intermediary: self-preferencing and steering in search and recommendation algorithms (Chen and Tsai, 2024; Lam, 2021; Raval, 2023; Farronato *et al.*, 2023; Reimers and Waldfogel, 2023; Waldfogel, 2024; Lee and Musolff, 2025; Bennati, 2025), Amazon's entry into its sellers' product spaces (Zhu and Liu, 2018), sponsored advertising (Yu, 2024), regulatory effects (Gutierrez, 2022), and private brands (Farronato *et al.*, 2025).[21] My paper

[19] This differs from Amazon's multi-channel fulfillment service, which charges higher fees for orders placed outside Amazon than for those placed on Amazon.

[20] Theoretical work on Amazon's dual role asks whether a marketplace should be allowed to sell its own products (Hagiu *et al.*, 2022); Etro (2024) surveys this literature. The closest theoretical paper to mine is Li *et al.* (2025), who model competition between a platform with integrated fulfillment and one without. Beyond providing empirical evidence, my paper incorporates platforms' strategic pricing of commission rates and logistics fees and the endogenous responses of consumers and sellers to them.

[21] One paper that sheds light on cross-platform competition through consumer-side behavior is Farronato *et al.* (2025). They document that hiding Amazon's private brands from search results has little effect on consumers' cross-

expands the scope of the market to incorporate strategic competition *between* platforms and the competitive impacts of logistics integration, an essential part of Amazon's e-commerce business. Relatedly, Houde *et al.* (2023) study the distortionary effects of tax laws on the geographic expansion of Amazon's fulfillment centers. Building on their finding of substantial gains to Amazon from its investment, this paper asks whether such gains reach consumers and sellers despite the presence of market power.

Second, this paper is related to empirical work on the welfare effects of vertical integration. Vertical integration is a classic antitrust issue across many industries, including platform markets,[22] and theory predicts both pro- and anti-competitive effects.[23] A large empirical literature examines the welfare effects of vertical integration; examples include Hortaçsu and Syverson (2007); Crawford *et al.* (2018); Luco and Marshall (2020); Hodgson and Sun (2025). The evidence is mixed; see Lafontaine and Slade (2007) and Beck and Scott Morton (2021) for surveys. Although a platform's vertical integration has prompted active scrutiny, relatively little is known about its welfare effects, which can be substantial. For example, Athey and Scott Morton (2021) discuss the potential for a platform to use integrated complementary services to discourage platform competition. The closest paper to mine is Lee (2013), which studies the welfare effects of exclusive integration between video game platforms and software developers. I contribute to this literature by examining a different notion of vertical integration (here, integration of complementary services that affect multihoming costs) and related platforms' strategic price competition.

Third, I contribute to the literature on competition in multi-sided markets. Theory highlights multihoming as central to competition among platforms (Rochet and Tirole, 2003; Armstrong, 2006; Bakos and Halaburda, 2020; Teh *et al.*, 2023); Jullien *et al.* (2021) survey this literature. The theoretical predictions of how multihoming affects equilibrium platform prices are mixed, so the question necessitates an empirical examination.[24] Yet, even setting aside integrated services, few empirical studies incorporate both platform pricing and two-sided multihoming; notable exceptions are Wang (2025) on the U.S. payment industry and Sullivan (forthcoming) on the U.S. food delivery industry. This paper studies platforms' pricing, incorporating the cross-platform effects of integrating complementary services that make multihoming more costly for sellers.

platform search, consistent with my demand estimates indicating limited cross-platform substitution in Section 6.1.

[22] For instance, vertical mergers have received regulatory attention, reflected in the 2020 Vertical Merger Guidelines and the 2023 Merger Guidelines.

[23] Classic theoretical work (Spengler, 1950; Williamson, 1971; Salop and Scheffman, 1983; Hart and Tirole, 1990) highlights the tension between efficiency gains and foreclosure (i.e., raising rivals' costs) incentives. The elimination of double marginalization (Spengler, 1950) and the alignment of investment incentives (Grossman and Hart, 1986; Ciliberto, 2006; Yang, 2020) are highlighted as key pro-competitive benefits of vertical integration. Conversely, the literature on vertical foreclosure underscores the potential for integrated firms to harm rivals. Early theoretical work focuses on input foreclosure, where an integrated upstream firm restricts supply or raises costs for downstream competitors (Salop and Scheffman, 1983; Salinger, 1988; Ordover *et al.*, 1990).

[24] Armstrong (2006); Armstrong and Wright (2007) propose a competitive bottleneck: when sellers multihome and consumers single-home, a seller can reach a platform's consumers only by joining that platform, so each platform holds monopoly power over access to its own consumers. In contrast, Belleflamme and Peitz (2019) show that a change from single-homing to multihoming does not necessarily harm the multihoming side and can even benefit both sides.

# 2 Institutional Background

I study e-commerce platform markets in the U.S., where the platforms, Amazon and Walmart, operate an integrated logistics service for sellers who sell products through a platform's marketplace. This section describes the market structure, the role of logistics in e-commerce, and the key features of integrated logistics.

## 2.1 Market Structure

The U.S. e-commerce industry is highly concentrated. In 2024, Amazon recorded $447.5 billion in sales revenue, capturing 39.7% of the market, while Walmart recorded $120.9 billion, accounting for 10.6% (Digital Commerce 360, 2025). The gap between these two and the other platforms is large; no other U.S. e-commerce marketplace exceeds a 4% market share according to an industry report (eShipz, 2026). These statistics imply that Amazon and Walmart are effectively each other's main competitors.[25] Although both are major players in e-commerce markets, the platforms differ in seller participation: Amazon hosts approximately 2 million active independent sellers (Seller Labs, 2025; Red Stag Fulfillment, 2026), compared to approximately 200,000 on Walmart (Red Stag Fulfillment, 2025) in 2025.

The two platforms' core business is marketplace intermediation: each connects consumers with sellers and charges sellers a commission proportional to the price of each sale (i.e., ad valorem). These commission rates have been stable for a long period. A seller lists products on one or multiple marketplaces and sets its own product prices. Consumers visit the online shops, compare listed products, and choose among them.

In addition to intermediation, both platforms are integrated into further stages of the e-commerce supply chain. First, each functions as a logistics provider that offers logistics services to sellers. Second, each acts as an own-product retailer—also known as a first-party retailer—that sells products directly to consumers. Together with the intermediation, these roles define a platform's payoff: each platform earns commission revenue from independent sellers' total sales, logistics profits from sales processed by the integrated logistics, and own-product retail profits by selling products. Therefore, each platform charges three broad categories of prices: commission rates, logistics fees, and own-product prices, and each price reflects its impact on the other profit margins.

## 2.2 E-commerce Logistics

E-commerce sellers use a logistics (or fulfillment) service to store inventory and deliver products to consumers after an order is placed. A logistics service therefore combines two main functions: warehousing, in which sellers hold inventories of their products, and shipping, in which individual orders are picked, packed, and delivered to consumers. Logistics providers differ in service quality,

[25] I use this institutional detail and model marketplace competition as a duopoly between Amazon and Walmart.

including shipping speed and reliability, while logistics fees typically depend on product weight and size. These features make sellers' logistics choices relevant for both the cost of serving consumers and the quality of the purchasing experience.

An important feature of e-commerce logistics is the presence of economies of scale from pooling inventory. A seller operating on multiple marketplaces can, in principle, use a single logistics provider to serve orders originating from each platform. Doing so allows the seller to maintain a common inventory pool rather than splitting inventory across multiple logistics services. Pooling can reduce operational costs and allow sellers to benefit from volume discounts. Consistent with these incentives, the FTC notes that many sellers would prefer to use a single fulfillment provider for their orders both on and off Amazon (*FTC v. Amazon*, 2023). Thus, the ability of a logistics provider to serve multiple marketplaces can directly affect the cost of multihoming and therefore the extent of cross-platform selling.

### 2.3 Platform Integrated Logistics

Amazon operates its own logistics service, FBA, alongside its online marketplace.[26] Sellers have an incentive to adopt Amazon logistics to attain the demand-side gains from using it and to increase their sales on Amazon. Using Amazon logistics gives sellers access to Prime shipping, which is known to be a high-quality shipping service that Amazon provides to its Prime subscribers.[27] Because Prime eligibility can make an offer more attractive to consumers and a majority of U.S. households subscribe to Prime, sellers have a strong incentive to adopt FBA.[28]

Using Amazon's integrated logistics raises sellers' cost of selling across platforms. This is because sellers either cannot use Amazon's logistics to fulfill orders from competing platforms or must pay higher fees for those orders than for Amazon orders. Specifically, an FBA seller needed a separate logistics provider for its Walmart sales until mid-2025, which prevented inventory pooling and made multihoming costly.[29] Even when Amazon's logistics is available for orders from competing platforms, FBA sellers face higher fees for those orders than for Amazon orders.[30] Because my

[26] Amazon does not separately disclose FBA revenue. In 2024, its "Third-party seller services" segment, which bundles referral commissions with FBA fulfillment and shipping fees, generated $156.2 billion, roughly 63% of the $247.0 billion generated by its first-party online stores (Amazon.com, Inc., 2025). Given that 82% of active Amazon sellers use FBA (Red Stag Fulfillment, 2026), FBA accounts for the majority of this segment.

[27] The literature also documents the fast shipping of Amazon logistics (Edgel *et al.*, 2023; Etumnu and Foster, 2026).

[28] I treat FBA as effectively the only route to Prime eligibility on Amazon. In principle, Amazon's Seller Fulfilled Prime (SFP) program lets sellers using independent logistics become Prime eligible after passing a one-month performance evaluation without Prime eligibility. Amazon resumed SFP in October 2023 after a multi-year suspension. However, SFP participation is effectively negligible in my data: fewer than 1% of sellers use independent logistics and hold Prime status. The evaluation period likely discourages adoption, because sellers must forgo Prime-induced revenue for at least a month, which acts as a cost of joining SFP.

[29] Walmart's shipping policy prohibited sellers from using Amazon's logistics services for Walmart orders until May 15, 2025, and Amazon officially announced in September 2025 that sellers could use its logistics services to fulfill Walmart orders.

[30] According to the Amazon's fee policy, fulfilling an order outside of Amazon costs 66% to 183% more than fulfilling an Amazon order, depending on shipping speed and product weight. Figure A12 compares the posted fees of Amazon's

dataset, introduced in Section 3, almost entirely predates policy changes with respect to Walmart orders, I treat Amazon's logistics as unavailable for Walmart orders throughout the analysis.

Walmart operates an analogous service, Walmart Fulfillment Services (WFS), which functions similarly to FBA in its basic operations, yet WFS is not tightly tied to Walmart's consumer-side membership program, Walmart+. WFS orders ship within two business days regardless of whether the consumer subscribes to Walmart+, and non-members receive free shipping on orders above $35. Therefore, adopting Walmart logistics confers a smaller incremental demand advantage than adopting Amazon logistics does.

Logistics fees on both platforms are publicly posted. Amazon calculates the per-unit shipping fee (called the "logistics fee") in two steps. First, it assigns each product to a *size tier* based on size and shipping weight. Second, within each tier, it applies a fee schedule under which fees increase monotonically with shipping weight. Notably, fees jump discretely at size-tier boundaries: a product that exceeds a tier threshold is assigned to the larger tier and faces substantially higher fees. I exploit these discontinuities in an RD design in Section 3.3.1.

# 3 Data and Evidence

## 3.1 Data

For descriptive and RD analyses, I first study bestsellers on the Amazon marketplace across broad categories, while verifying the presence of products and sellers on Walmart. I call this dataset, which covers broad product categories, the "full-category panel." For model estimation and counterfactual analysis, I focus on the kitchen appliance category and construct a dataset that covers rich characteristics on both platforms. I refer to this as the "kitchen appliance panel." I augment the model estimation with the Numerator consumer panel, which records consumers' multi-retailer shopping behavior along with their income and membership subscription information.

**Full-category Panel** I build the full-category panel as follows. I obtain Amazon's bestseller lists for nine broad categories at two points in time, April 2024 and August 2025; together, the lists cover approximately 100,000 products.[31] For each product, I collect historical product and seller characteristics from Keepa. I then use two data services, DataSpark and BlueCart, to determine by UPC whether each product is also listed on Walmart and, if so, to collect the corresponding Walmart-side product and seller characteristics.[32] I match sellers across the two platforms using display name, business name, address, and phone number. The panel spans January 2021 to August

multi-channel fulfillment (MCF), which apply to orders placed outside Amazon, with the FBA fees that apply to Amazon orders.

[31] The nine categories are Appliances; Electronics; Office Products; Tools & Home Improvement; Home & Kitchen; Sports & Outdoors; Pet Supplies; Beauty & Personal Care; and Clothing, Shoes & Jewelry.

[32] I find a Walmart listing for about half of the products, a share similar to that reported by Hunold *et al.* (2022).

Table 1: Summary Statistics of the Full-Category Panel

| Variable | Mean | SD | Observations |
|---|---|---|---|
| *Panel A: Product–month panel (106,583 products)* | | | |
| Price ($) | 55.97 | 106.23 | 1,599,657 |
| FBA fee ($) | 7.83 | 13.76 | 1,589,115 |
| Shipping weight (lb) | 7.03 | 27.32 | 1,479,075 |
| Has FBA seller | 0.67 | 0.47 | 1,599,657 |
| Has non-FBA seller | 0.36 | 0.48 | 1,599,657 |
| Has multihoming seller | 0.08 | 0.27 | 849,250 |
| *Panel B: Seller–month panel (185,683 sellers)* | | | |
| Products offered | 4.17 | 47.89 | 1,557,204 |
| FBA seller | 0.61 | 0.49 | 1,557,204 |
| Multihoming seller | 0.02 | 0.15 | 1,066,053 |

*Notes.* This table shows means and standard deviations (SD) over product–month observations (Panel A) and seller–month observations (Panel B) in the full-category panel, January 2021–August 2025. The statistics here are based on independent (third-party) sellers and their products. Prices include shipping charges to consumers. "Has FBA seller," "Has non-FBA seller," and "Has multihoming seller" equal one if at least one seller of the product uses Amazon logistics (FBA), uses independent logistics, or also lists the product on Walmart, respectively. "FBA seller" equals one if at least half of a seller's products use FBA; "Multihoming seller" equals one if the seller lists any product on Walmart. "Has multihoming seller" and "Multihoming seller" are defined on the subperiod matched to Walmart, spanning from January 2021–April 2024.

2025, but Walmart-side information is available only through April 2024, when DataSpark discontinued its service. Appendix Section B describes the dataset construction in detail, and Table A2 summarizes the data sources.

Table 1 reports summary statistics at two levels of aggregation: a monthly product panel and a monthly seller panel. Amazon logistics (FBA) is the dominant logistics choice among independent sellers: 67% of products have at least one Amazon logistics seller, and 61% of sellers primarily use Amazon logistics. The average Amazon logistics fee is $7.8 per unit, and the average product weighs about 7 pounds. Within the subperiod matched to Walmart, 7.6% of products have a multihoming seller, and 2.3% of sellers list any product on Walmart in a given month.[33] This low multihoming rate suggests that multihoming is costly for sellers.

**Kitchen Appliance Panel** I additionally construct a second panel, which complements the full-category panel with richer product characteristics, especially logistics service usage and sales estimates on the Walmart side. I focus on the kitchen appliance category because products in this category, such as air fryers and microwave ovens, exhibit large variation in product weights and sizes, and thus in logistics fees. This variation is important for the subsequent empirical analyses.

[33]These low rates partly reflect the size difference between the two seller bases: Walmart hosted roughly 150,000 sellers in 2024, about 7.5% of the active Amazon seller base (Red Stag Fulfillment, 2026).

I take weekly snapshots of bestsellers of each subcategory in the kitchen appliance category to assemble the panel and aggregate them to the monthly-level.[34] I scrape approximately 140 bestsellers on Amazon and 40 on Walmart each week. I also collect similar numbers of featured products on the platforms to obtain information about organic search ranks. In addition to bestseller ranks in the scraped data, I use two data services, JungleScout and Bizmetrica, to get the sales estimates as a proxy for sales.[35] I match products and sellers based on UPC and seller identifiers, as in the full-category panel. I explain details of the data construction in the Appendix Section B.

Table 2: Summary Statistics of the Kitchen Appliance Panel

| | Amazon | | | Walmart | | |
|---|---|---|---|---|---|---|
| Logistics service | All | FBA | Independent | All | WFS | Independent |
| Sales | 2,751.2 | 2,841.1 | 2,022.2 | 3,114.5 | 3,035.2 | 3,411.8 |
| | (5,265.1) | (5,407.3) | (3,852.2) | (8,051.7) | (7,547.5) | (9,720.8) |
| Price ($) | 75.9 | 70.2 | 121.9 | 65.7 | 57.9 | 95.0 |
| | (84.2) | (79.1) | (107.1) | (59.0) | (46.5) | (85.7) |
| Uses FBA/WFS | 0.89 | 1.00 | 0.00 | 0.79 | 1.00 | 0.00 |
| | (0.31) | (0.00) | (0.00) | (0.41) | (0.00) | (0.00) |
| Shipping weight (lb) | 12.7 | 11.7 | 21.3 | 11.9 | 10.5 | 17.2 |
| | (15.2) | (13.8) | (21.5) | (14.2) | (12.1) | (19.5) |
| FBA/WFS fee ($) | 13.1 | 12.3 | 19.0 | 9.6 | 11.8 | 16.4 |
| | (10.6) | (9.5) | (15.6) | (6.3) | (8.1) | (14.1) |
| Platform own retail | 0.27 | 0.31 | 0.00 | 0.39 | 0.50 | 0.00 |
| | (0.45) | (0.46) | (0.00) | (0.49) | (0.50) | (0.00) |
| Observations | 4,529 | 4,032 | 497 | 1,458 | 1,151 | 307 |

*Notes.* This table shows means with standard deviations in parentheses over product-month observations in the kitchen appliance panel, April–June 2025. The panel is built from weekly snapshots of approximately 140 bestsellers on Amazon and 40 on Walmart in 13 subcategories, aggregated to the product–month level; observations include the platforms' own retail listings. Columns split the listings on each platform by the logistics service used: the platform's integrated service (FBA on Amazon, WFS on Walmart) or independent logistics. Sales are monthly sales estimates from JungleScout (Amazon) and Bizmetrica (Walmart), which follow rank-sales conversion as shown in Appendix Section B. "Uses FBA/WFS" equals one if the listing is fulfilled by the platform's integrated logistics; it is one or zero by construction in the FBA, WFS, and Independent columns. "FBA/WFS fee" is the platform's posted logistics fee for the product's size tier and shipping weight, reported whether or not the seller uses the service. "Platform own retail" equals one if the product is sold by the platform as a first-party retailer.

Table 2 summarizes the kitchen appliance panel across Amazon and Walmart by logistics choice. These data include key variables for model estimation and counterfactual analyses, such as sales proxies, logistics choices across both platforms, prices, and logistics fees. The summary statistics

[34] There are 13 subcategories in the kitchen appliance category. These are coffee makers, bread machines, air fryers, blenders, electric skillets, food processors, kettles, waffle irons, hand mixers, electric pressure cookers, food steamers, microwave ovens, and electric grill.

[35] JungleScout uses its own conversion formula to attain the sales estimates. Yu (2024) uses sales estimates from JungleScout and verifies them with the actual sales records of a seller. The relationship between Bizmetrica's sales estimates and bestseller ranks shows a similar rank-to-sales conversion. In the Appendix figure A4, I examine the validity of the sales estimates as a proxy for actual sales.

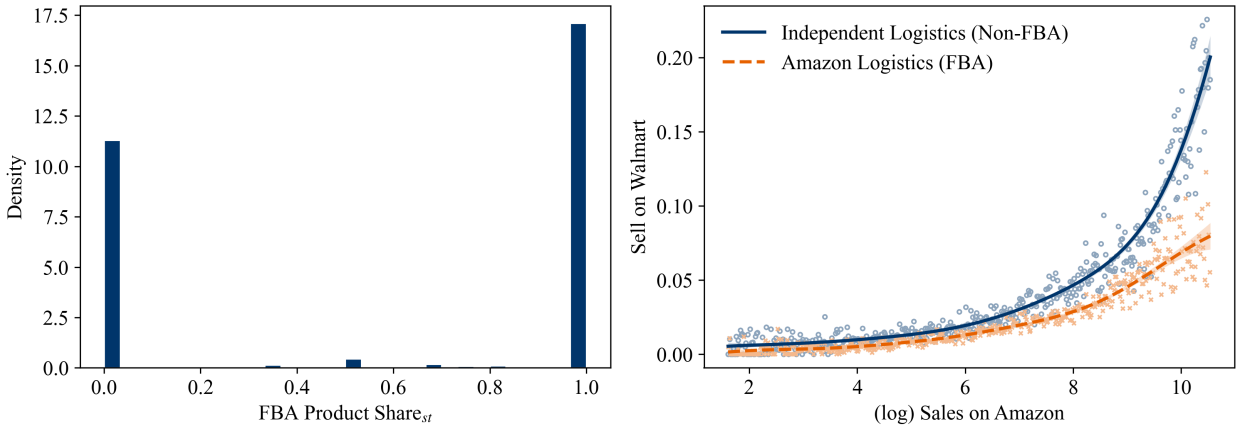
show that platform-integrated logistics account for the vast majority of listings on both marketplaces: 89% on Amazon (FBA) and 79% on Walmart (WFS). Across both platforms, products fulfilled by integrated logistics are lighter than those fulfilled by independent logistics. Platform own retail accounts for 27% of products on Amazon and 39% on Walmart.

**Numerator Consumer Panel** I additionally use the Numerator consumer panel to obtain information about cross-platform shopping behavior of U.S. households. Numerator collects receipts from a panel of about 200,000 U.S. households. Each panelist carries a household weight that aligns the panel to Census targets and a national projection factor that scales the panel to the U.S. household population. The data contain household demographics, including income and subscription status for Amazon Prime and Walmart+, as well as cross-platform shopping behavior, which are key inputs to the demand analysis.

## 3.2 Descriptive Analysis

In this subsection, I establish stylized facts and their implications, focusing on FBA usage, multihoming and their relationship, which is motivated by the institutional feature described in Section 2: the use of integrated logistics hinders cross-platform selling.

Figure 1: Logistics Usage, Multihoming, and Scale on Amazon




(a) Distribution of FBA Product Share

(b) Multihoming and Scale on Amazon

*Notes.* Panel (a) shows the distribution of the share of seller $s$'s FBA products in month $t$. The distribution is bimodal, with mass concentrated at 0 and 1, indicating that most Amazon sellers use FBA for either all or none of their products. Panel (b) shows the relationship between multihoming and seller scale by the seller's logistics choice. "Sells on Walmart" equals one if seller $s$ lists at least one product on Walmart in month $t$. Seller scale is measured by the seller's total estimated unit sales on Amazon in month $t$. Lines show local quadratic fits with a Gaussian kernel, and shaded areas denote their 95% confidence bands. Dots are means of the outcome within 500 equal-width bins of log sales.

The first fact is that Amazon sellers mostly use FBA for either all or none of their products. Panel (a) in Figure 1 shows the distribution of the FBA product share across seller-months: it

is bimodal, with mass concentrated near zero and one.[36] This pattern is consistent with sellers preferring to use a single logistics service to reduce operating costs. The bimodality also implies that the logistics decision is made at the seller level despite product-level fee heterogeneity.[37]

Second, larger sellers are more likely to multihome. I classify a seller as multihoming in a given month if it also appears on Walmart, and I measure seller size by total product sales. The binned scatter in Panel (b) of Figure 1 shows that the likelihood of appearing on Walmart rises monotonically with seller size. This positive correlation between size and multihoming is stronger among independent logistics sellers.[38] This pattern suggests that adopting FBA is negatively associated with multihoming. The cost-side impacts of inventory fragmentation are consistent with this pattern. FBA sellers have to store inventory in Amazon's logistics facilities, raising the cost of selling on Walmart. In contrast, an independent logistics seller, holding channel-flexible inventory, faces no comparable constraint. However, because FBA adoption is correlated with many confounding factors, causality cannot be established from this descriptive exercise alone.

### 3.3 Regression Discontinuity Analysis

In this subsection, I investigate the effects of adopting Amazon logistics (FBA) on market outcomes by using local discontinuities in the Amazon logistics fee schedule. Appendix D provides additional details and results for the analysis below.

#### 3.3.1 Product Categorization of Amazon Logistics Fee Policy

Under Amazon's logistics fee policy, the per-unit logistics fee (i.e., shipping fee) is the main fee that sellers pay to Amazon. The fee is calculated in two steps. First, Amazon assigns each product to a size tier based on its size and shipping weight. Second, it applies that tier's fee schedule, under which the fee increases with shipping weight.[39] These fees matter for market outcomes because they account for a large share of prices. For products fulfilled by FBA, the fee averages 26.8% of the price.[40]

I focus on three size tiers, to which most Amazon products are assigned: Small, Large, and Bulky.[41] Table A4 reports the size-tier thresholds. A product is assigned to a tier based on its shipping weight and size, measured by length, height, and width. If any attribute exceeds its

[36] Panel (a) of Figure A5 in the Appendix shows the distribution of FBA product share, restricting to sellers with more than five products. This shows that the bimodality is not driven by sellers who have a small number of products.

[37] I exploit this bimodality when modeling sellers' entry and logistics decisions in Section 4.

[38] This pattern is robust to additional control variables. Panel (b) of Figure A5 in the Appendix further shows that the gap between independent logistics and FBA sellers persists after residualizing both the multihoming indicator and seller size. For the residualization, I use the average (log) weight and (log) volume of the seller's products and product-category fixed effects. In addition, I run a regression of the multihoming indicator on FBA usage with seller and month fixed effects. Table A3 in the Appendix reports the results

[39] Amazon also classifies products as apparel, dangerous goods, or other, and applies the two steps within each class.

[40] Appendix Figure A6 shows the distribution of this share.

[41] Amazon's official names for these tiers are small standard, large standard, and small bulky.

threshold for a tier, the product is assigned to the next larger tier.[42] Formally, let $T^L$ and $T^B$ be the Large and Bulky size-tier indicators. For each tier $\in \{T^L, T^B\}$, I measure the differences from the size and shipping-weight cutoffs as:

$$x_j^{\text{size,tier}} = \max\left\{l_j - c_l^{\text{tier}},\ h_j - c_h^{\text{tier}},\ w_j - c_w^{\text{tier}}\right\}, \qquad x_j^{\text{shwgt,tier}} = \text{shwgt}_j - c_{\text{shwgt}}^{\text{tier}}. \tag{1}$$

where $l_j, h_j, w_j$, and $\text{shwgt}_j$ are length, height, width, and shipping weight of product $j$ and $c_l^{\text{tier}}$, $c_h^{\text{tier}}$, $c_w^{\text{tier}}$, $c_{\text{shwgt}}^{\text{tier}}$ are the corresponding cutoff values for given tier. If any of these differences is positive, the product is assigned to the larger tier and charged a higher fee.

The key source of variation is that Amazon logistics fees jump by approximately 30–50% upon crossing any threshold. Figure 2 displays the average logistics fee for products binned by their distance from the cutoff in both product size and shipping weight. In both Panels (a) and (b), FBA fees jump sharply at each cutoff.[43] The two tier transitions, each with a size and a shipping-weight threshold, give four cutoffs. I take the large size-tier cutoff as the main specification and use the small shipping-weight cutoff as corroborating evidence, rather than the large shipping-weight or the small size-tier cutoffs. At the large shipping-weight cutoff, Walmart's WFS schedule increases fees at the same threshold, so the discontinuity is not unique to Amazon.[44] At the small size-tier cutoff, the running variable has narrow support below the cutoff.[45]

### 3.3.2 Regression Model

The size-tier schedule generates discrete fee jumps at weight-size cutoffs, as shown in Figure 2, providing variation for an RD design. Specifically, I estimate the following regression model: for product $j$ in month $t$,

$$y_{jt} = \beta_0 + \beta_1 \mathbf{1}(x_j \geq 0) + m(x_j) + \tau_{year(t)} + u_{jt}, \tag{2}$$

where $x_j$ is the (normalized) running variable for product $j$, as introduced in (1).[46] I choose three main outcomes for $y_{jt}$: whether product $j$ has an FBA seller, prices on Amazon, and the number of multihoming sellers. These outcomes capture the channels through which FBA affects the market. I approximate $m(\cdot)$ using a flexible local-linear specification with MSE-optimal bandwidth selection

[42] For example, according to Table A4, a product weighing under one pound but exceeding 15 inches in length is classified as Large rather than Small.

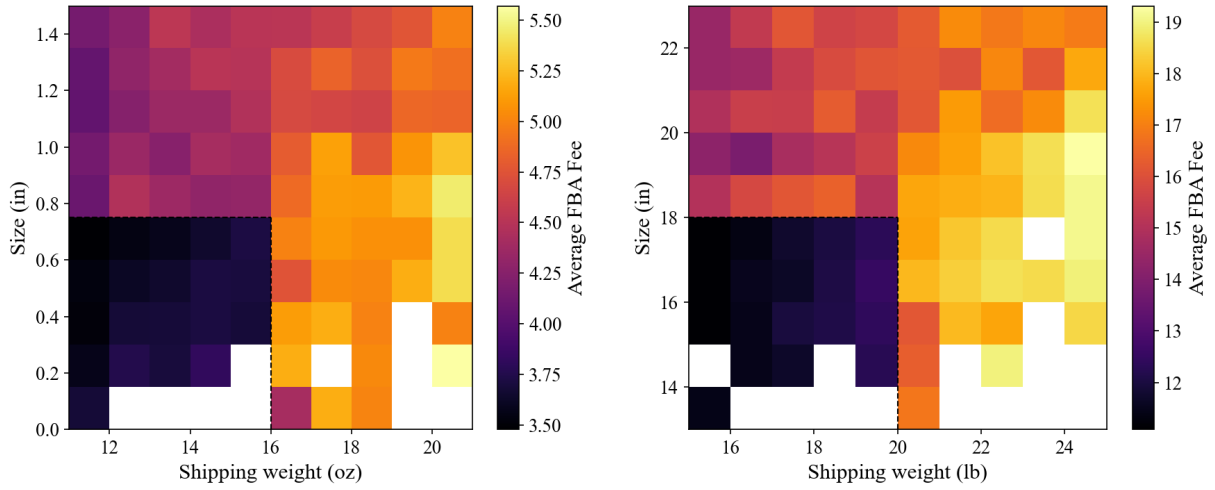
[43] Products in the Small tier are thin: the maximum width is 0.75 inches.

[44] In addition, only 105 distinct products lie above this cutoff in the entire panel. The MSE-optimal bandwidth for the FBA fee contains only 19 distinct products above the cutoff.

[45] It extends only 0.711 inches and contains 172 distinct values. The MSE-optimal bandwidth for the FBA fee is 0.017 inches, which is too narrow for estimation.

[46] Keepa provides length and weight in millimeters and grams rather than inches and pounds. Therefore, additional conversion of metrics is necessary to match Amazon's size-tier criteria, which may induce rounding errors. To account for this, I exclude observations within 0.1 inch or 0.1 pound of a cutoff.

Figure 2: Distributions of FBA Logistics Fees Around Product Size-Tier Cutoffs




(a) Small to Large Tier

(b) Large to Bulky Tier

*Notes.* This figure shows the heatmaps of Amazon logistics (FBA) fees around size-tier cutoffs defined by product size and shipping weight. Amazon classifies products into different tiers by size and weight, and a product is assigned to the larger size tier if any criterion exceeds its threshold. Panels (a) and (b) display the distributions around the cutoffs from the Small tier to the Large tier and from the Large tier to the Bulky tier, respectively. In each panel, the black dashed lines indicate cutoffs based on either product size measures or shipping weight measures. Each cell reports the average FBA logistics fee for products in the corresponding size–weight bin, averaged over time as well. The total number of products in Panels (a) and (b) is 1,620 and 3,356, respectively.

(Calonico *et al.*, 2014), allowing the bandwidth to be chosen in a data-driven way.[47] $\tau_{year(t)}$ denotes year fixed effects. $u_{jt}$ is a regression error term. In specification (2), the main parameter of interest is $\beta_1$, which captures the effect of the fee increase on the outcome $y_{jt}$. This parameter reflects equilibrium responses of both consumers and sellers through competition in the relevant platform markets.[48] For each running-variable measure and tier, I restrict the sample to products for which the other measure meets the criterion for the smaller size tier. This ensures that variation in the primary running variable alone determines tier assignment.

The interpretation of $\beta_1$ depends on whether there is sorting (manipulation) around the cutoff. In the absence of such sorting, $\beta_1$ can be interpreted as the local effect of the fee increase, holding the composition of products around the cutoff fixed. However, if sellers sort into the cheaper FBA fee regime, then $\beta_1$ also absorbs the effects of endogenous adjustment of product characteristics (e.g., packaging) and selection into treatment status. To assess the plausibility of sorting, I test for discontinuities in the density of the running variable at the cutoffs. Let $f_X(\cdot)$ denote the density

[47] It is documented that RD estimators are sensitive to global high-order polynomial approximations (Gelman and Imbens, 2019). Following their recommendations, I use low-order local-linear regressions to approximate $m(\cdot)$.

[48] Because potential outcomes would depend on the outcomes of competing products (i.e., interference), the estimand in this paper should not be interpreted as the standard average treatment effect at the cutoff targeted by textbook RD designs (e.g., Lee and Lemieux, 2010).

of the running variable. The key hypotheses are $H_0 : \lim_{x\to 0^-} f_X(x) = \lim_{x\to 0^+} f_X(x)$ vs. $H_1 : \lim_{x\to 0^-} f_X(x) \neq \lim_{x\to 0^+} f_X(x)$. Figure A7 presents the density estimates. I find limited evidence of sorting around the cutoffs, consistent with formal density-discontinuity tests (Cattaneo *et al.*, 2018, 2020).[49]

### 3.3.3 RD Estimation Results

Figure 3 visualizes the RD estimates and Table A6 reports the corresponding point estimates in details. I focus on the large-to-bulky cutoff; the estimates at the small-to-large cutoff are qualitatively the same. Panel (a) shows that Amazon logistics (FBA) fees jump at the size-tier cutoff by \$3.24, or approximately 50%. The remaining panels reveal how sellers and prices respond to this fee increase. In Panel (b), the likelihood of having an Amazon logistics seller drops sharply at the cutoff, by 0.202. In Panel (c), the consumer prices on Amazon rise discontinuously, by \$8.36, consistent with sellers passing the fee increase through to consumers. Panel (d) shows that the number of multihoming sellers increases at the cutoff by about 0.10.

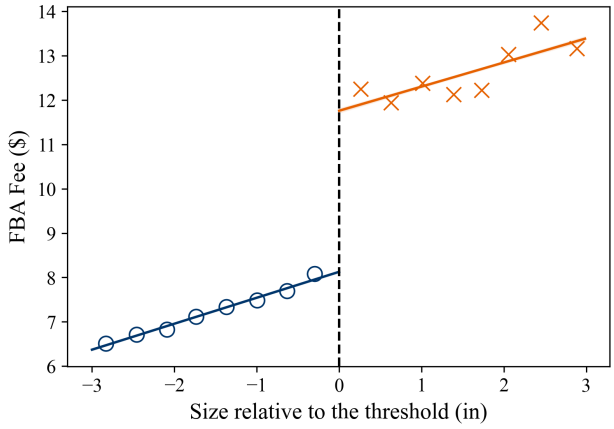
These results imply that Amazon logistics affects sellers' cross-platform behavior and consumer prices. When FBA fees rise, sellers substitute away from FBA. Some of these sellers switch to independent logistics providers, which can serve multiple platforms from a single inventory pool. This substitution is reflected in the simultaneous decline in FBA adoption in Panel (b) and increase in multihoming in Panel (d): the same cost increase that pushes sellers off FBA also reduces their dependence on Amazon as a sales channel. The consumer price increase in Panel (c) confirms that FBA fees are not simply absorbed by sellers but are passed through to consumers, ultimately affecting consumer welfare.

## 4 Empirical Model of Integrated Platform Economy

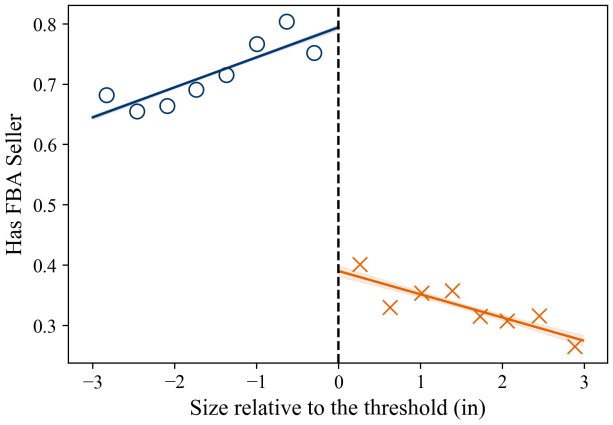
Motivated by evidence on the impacts of using FBA on consumer prices and discouraging multihoming, I develop an equilibrium model to quantify platforms' pricing incentives and the welfare effects of logistics integration and policy interventions. On the demand side, the goal is to capture consumer preferences over product characteristics with a focus on price sensitivity and heterogeneous valuations of logistics and platforms. On the supply side, the model characterizes variable and fixed costs of sellers as functions of their logistics choices. Given the demand function, the cost structure pins down how sellers trade off demand- and cost-side (dis)advantages from using integrated logistics. The central challenge is to accurately recover these primitives while keeping the model tractable for counterfactual analyses.

[49] Table A5 reports the density tests. I find no statistically significant discontinuity in the density of the running variable at either the large-to-bulky size cutoff ($T$-stat $= -1.31$, $p$-value $= 0.19$) or the small-to-large shipping-weight cutoff ($T$-stat $= -0.39$, $p$-value $= 0.70$).

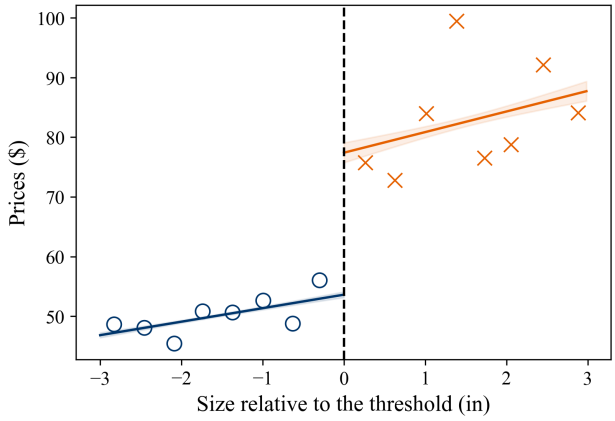
Figure 3: Regression Discontinuity Analyses



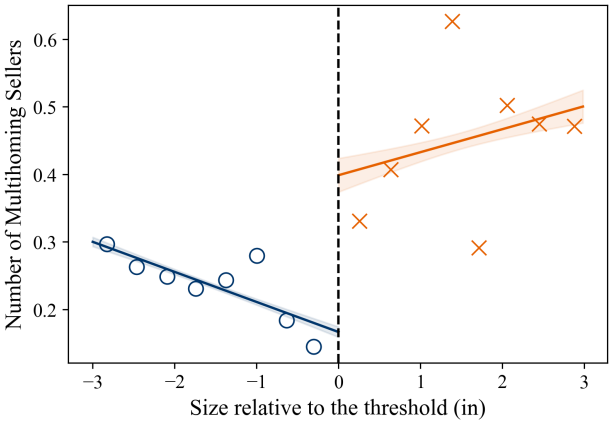

(a) FBA Fees




(b) Has FBA Seller




(c) Consumer Prices on Amazon

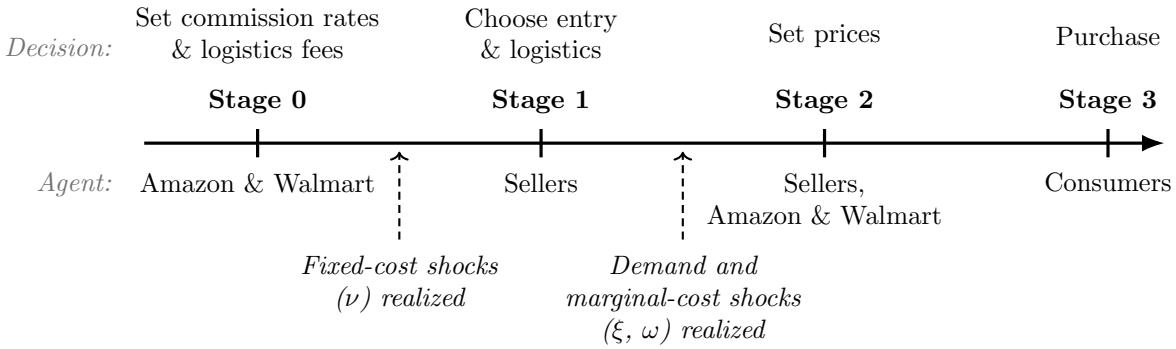



(d) Multihoming

*Notes.* This figure presents the RD estimates from (2) around the large size-tier cutoff, using the product-size running variable. Table A6 shows the point estimates in details. In each panel, the horizontal axis is the normalized distance to the cutoff: positive values correspond to products in the larger size tier, which are charged higher FBA logistics fees, and negative values correspond to products in the smaller tier. Dots are averages of the outcome within bins of the running variable; and the solid lines are local-linear fits estimated separately on each side of the cutoff; and the shaded regions denote 95% confidence intervals. Panel (a) plots the FBA logistics fee and confirms the discontinuous fee increase at the cutoff. Panel (b) plots the dummy variable indicating having an FBA seller, Panel (c) plots prices on Amazon, and Panel (d) plots the number of multihoming sellers.

**Model Overview** A market is defined as a subcategory–month pair $(c, t)$. On the demand side, consumers with heterogeneous preferences choose among differentiated products listed on Amazon and Walmart, or an outside option. On the supply side, sellers make decisions in two stages. In the first stage, sellers simultaneously choose which platforms to enter and which logistics service to use on each platform. After entry and logistics decisions are made, demand and marginal-cost shocks are realized. In the second stage, sellers observe these shocks and set prices. Commission rates and logistics fees are taken as given by consumers and sellers; they are set by Amazon and Walmart before sellers make any decisions. The two platforms aggregate profits across the markets and set rates and fees. Figure 4 summarizes the timeline, including when the structural errors are realized.

Figure 4: Model Timeline




*Notes.* This figure shows the timeline of the model. Platforms set commission rates and logistics fees before sellers make any decision. Sellers observe their fixed-cost shocks and simultaneously choose which platforms to enter and which logistics service to use on each platform. Demand and marginal-cost shocks are then realized, and sellers and the platforms set prices simultaneously. Consumers purchase in the last stage.

The timing assumption is based on institutional details introduced in Section 2. Commission rates are stable across time on both platforms and the two platforms announce logistics fee structures on a periodic basis. For example, Amazon typically updates FBA fee schedules annually. Sellers observe these schedules and make decisions more frequently in response. Each schedule is a publicly observable menu and sellers behave accordingly. Unlike fee schedules, retail prices change much more frequently, so a simultaneous game structure is more suitable than a sequential one when it comes to own-product retail pricing decisions.

## 4.1 Consumer Demand

I characterize consumers by their income level, platform memberships, and idiosyncratic random tastes. Consumers are first divided into four income groups based on their annual income level: under \$50k, \$50k–100k, \$100k–150k, and over \$150k. I further distinguish consumers by their Amazon Prime and Walmart+ membership subscription status. Therefore, there are 16 $(= 4 \times 2 \times 2)$

consumer types.[50] To capture cross-platform substitution patterns, I introduce a nest structure as in McFadden (1977) in which each platform forms a nest.[51] Figure A9 visualizes the nest structure. To accommodate the heterogeneous preferences over product and platform characteristics, I also include demographic interactions and random coefficients with taste shocks (Berry *et al.*, 1995; Petrin, 2002).

**Consumer Utility** Consumer $i$ gets indirect utility from purchasing product $j$ on platform $f$ in month $t$ as:

$$\begin{aligned} U_{ijft} = {} & \beta_i^p \log(p_{jft}) + \beta_i^{\text{FBA}} \text{FBA}_{jft} + \beta^{\text{WFS}} \text{WFS}_{jft} + \beta_i^{\text{Wmt}} \text{Wmt}_{jft} \\ & + X'_{jft}\beta^x + \xi_{b(j)} + \xi_{jft} + \epsilon_{ijft} \end{aligned} \tag{3}$$

The first line captures the key variables. The coefficient $\beta_i^p$ on log prices ($\log(p_{jft})$) governs price sensitivity that varies with income level, membership status, and a taste shock. Specifically, the coefficient is modeled as:

$$\beta_i^p = \beta_0^p + \sum_{k \in \{1,2,3\}} \beta_k^p 1\{i \in \text{Inc}_k\} + \beta_{\text{Prime}}^p \text{Prime}_i + \beta_{\text{W}^+}^p \text{W}_i^+ + \sigma^p u_i^p \tag{4}$$

where $\text{Inc}_1$, $\text{Inc}_2$, and $\text{Inc}_3$ denote income groups whose annual income level is between 50k to 100k, 100k to 150k, and over 150k. Also, $\text{Prime}_i = 1\{i \text{ subscribes to Amazon Prime}\}$ and $\text{W}_i^+ = 1\{i \text{ subscribes to Walmart +}\}$. This specification allows price sensitivity to differ by a consumer's income level and membership status. Additionally, $u_i^p$ is an i.i.d. random preference shock drawn from a standard normal distribution. This shock captures residual unobserved heterogeneity in price sensitivity not explained by a consumer's observed income group or membership status. $\sigma^p$ is an additional parameter to be estimated that governs the dispersion of price sensitivity.

The indicators $\text{FBA}_{jft}$ and $\text{WFS}_{jft}$ equal one when the product is fulfilled by the platform's integrated logistics on Amazon and Walmart, respectively. Their coefficients $\beta_i^{\text{FBA}}$ and $\beta^{\text{WFS}}$ capture the utility consumers derive from the associated service quality. To capture cross-platform substitution patterns by membership status, I allow for the utility from FBA products to differ by Prime subscription, because the high-quality service—such as fast shipping and reliability—is

[50] I also consider the presence of nearby Walmart offline stores as an additional demographic feature of consumers. This is mainly to capture diversion to Walmart offline stores. However, almost 90% of U.S. households live within 10 miles of Walmart offline stores (Walmart, 2022), so the geographic variation of distances is not informative for identifying the diversion.

[51] An alternative model of consumer choice is that consumers have a specific product they wish to buy and compare the same product across the two platforms. The main reason for choosing my specification over this alternative is to capture the cross-product substitution within a platform. Such shopping behavior is supported by experimental evidence in Farronato *et al.* (2025). Specifically, they show that when they randomly hide Amazon products on a search page, consumers tend to shop within Amazon, rather than visiting a competing platform such as Walmart.

exclusive to Prime subscribers. Specifically,

$$\beta_i^{\text{FBA}} = \beta_0^{\text{FBA}} + \beta_{\text{Prime}}^{\text{FBA}} \text{Prime}_i \tag{5}$$

I incorporate Walmart+ membership program in a different fashion to account for the program details. Walmart provides 2-day shipping on WFS items to *every* customer regardless of their membership status.[52] Instead, the membership is associated with other Walmart services such as grocery delivery and savings on fuel charges. As this associated perk can generate cross-platform substitution, I include the interaction between the platform dummy and Walmart+ membership as:

$$\beta_i^{\text{Wmt}} = \beta_0^{\text{Wmt}} + \beta_{\text{W}^+}^{\text{Wmt}} \text{W}_i^+ \tag{6}$$

where $\text{Wmt}_{jft}$ is a dummy variable indicating that the product is listed on Walmart.com.

I control for organic search ranks in $X_{jft}$ to separate the demand effect of fulfillment quality from algorithmic prominence that FBA products receive in search results. Specifically, I include search rank tier indicators constructed by grouping every ten position ranks.[53] The remaining observed characteristics in $X_{jft}$ include star ratings, number of reviews, log of product shipping weight, material dummies, and whether the product is sold by the platform directly. $\xi_{b(j)}$ are brand fixed effects. $\xi_{jft}$ is an unobserved demand shock at the product–platform–month level.

The error structure reflects the platform-level nesting in Figure A9. Consumers who have chosen to shop on Amazon substitute more readily among Amazon products than they do toward Walmart products or toward the outside option, and symmetrically for Walmart. I capture this pattern by defining a nest as a platform, so that $\mathcal{F} = \{\text{Amz}, \text{Wmt}\}$ indexes the inside nests and the outside option $j = 0$ sits in its own nest. Specifically, the error term $\epsilon_{ijft}$ can be decomposed as:

$$\epsilon_{ijft} = \bar{\epsilon}_{ift} + (1 - \rho)\bar{\epsilon}_{ijft} \tag{7}$$

where $\bar{\epsilon}_{ijft}$ is an i.i.d. Type I extreme value shock and $\bar{\epsilon}_{ift}$ is the unique random variable such that the sum $\bar{\epsilon}_{ift} + (1 - \rho)\bar{\epsilon}_{ijft}$ is itself Type I extreme value (Cardell, 1997). The parameter $\rho \in [0, 1)$ governs the within-platform correlation of unobserved idiosyncratic tastes: higher $\rho$ means consumers substitute more readily within a platform than across platforms, and $\rho = 0$ collapses the model to a standard random-coefficients logit.

**Market Shares** Consumers choose the product that maximizes their utility among products in the choice set. Let $\text{sh}_{ijft}$ be consumer $i$'s choice probability. Aggregate market shares, $\text{sh}_{jft}$, are

[52] Walmart offers free shipping on WFS items, provided consumers meet a $35 basket minimum. Walmart+ does not unlock a new tier of shipping speed for WFS items; rather, it waives the $35 minimum for the entire platform.

[53] The demand estimates are robust to alternative specifications that use either a linear or a log search-rank term. The main specification is to accommodate the fact that consumer search costs can vary discretely with whether the consumer searches an additional page, a feature that a linear-in-rank specification may not capture.

derived by integrating over the joint distribution of observable demographics and the unobserved taste shocks $u_i^p$:

$$\text{sh}_{jft} = \sum_{\tau \in \mathcal{G}} w_{\tau t} \int \text{sh}_{ijft}(\tau, u)\, f(u)\, du, \tag{8}$$

where $\mathcal{G}$ indexes the 16 observable consumer groups (income group $\times$ Prime $\times$ Walmart+), $w_{\tau t}$ are the household weights for each type in month $t$, and $f(\cdot)$ is the standard normal density. Derivation of market share is provided in appendix E.2 in detail.

## 4.2 Seller Supply

Considering the demand, each seller $s$ makes decisions in two stages for given month $t$. In the first stage, sellers simultaneously choose which platforms to enter and which logistics service to adopt, incurring fixed costs. After entry and logistics decisions are made, demand and marginal-cost shocks are realized. In the second stage, after observing these shocks, sellers then set prices. The marginal cost depends on the seller's logistics choice and economies of scale, which interact because integrated logistics fragments inventory across platforms. Sellers take platform policies and product portfolios as given for every decision.

**Seller Profit** The variable profit from selling product $j$ on platform $f$ in month $t$ is

$$\pi_{jft} = \underbrace{M_{c(j)t}\, \text{sh}_{jft}}_{\equiv Q_{jft}} \left[(1 - \phi_f)\, p_{jft} - AVC_{jft}\right] \tag{9}$$

where $M_{c(j)t}$ is the market size of category $c(j)$ in month $t$, $\text{sh}_{jft}$ is the market share function defined in the consumer demand model, $\phi_f$ is the commission rate charged by platform $f$ on product $j$, and $AVC_{jft}$ is the average variable cost. Revenue net of commission is $(1 - \phi_f)\, p_{jft}$ per unit.[54]

The total profit of seller $s$ aggregates variable profits across all products and platforms. Entering into platform $f \in \mathcal{F} = \{\text{Amz}, \text{Wmt}\}$ incurs fixed costs, so the total profit is:

$$\Pi_{st} = \sum_{f \in \mathcal{F}} \sum_{j \in \mathcal{J}_{sf}} (\pi_{jft} \times e_{sft} - FC_{jt}) \tag{10}$$

where $\mathcal{J}_{sf}$ is the set of products offered by seller $s$ on platform $f$, and $e_{sft} \in \{0, 1\}$ is the entry indicator. $FC_{jt}$ is the incurred fixed cost and is a function of entry and logistics choices that are introduced in section 4.2.2 below.

[54] The market size is defined to be the number of households who purchased any kitchen appliance among all the subcategories, $N_t^H$, times Google Trend search interest of the subcategory that lies within 0 to 1, $\text{GoogleTrend}_{c(j)t}$. That is, $M_{c(j)t} = N_t^H \times \text{GoogleTrend}_{c(j)t}$. See section E.3 in Appendix for further detail.

### 4.2.1 Seller Pricing and Marginal Costs

I solve the model backward, stage-by-stage. In the second stage, sellers observe demand and marginal-cost shocks and set prices to maximize variable profit, taking entry and logistics decisions from the first stage as given. Each seller solves a Bertrand–Nash pricing problem. I use both the demand and pricing models to recover the marginal costs, which are a function of logistics fees, shipping weight, and the seller's effective scale in addition to other cost shifters.

**Pricing** In the second stage, each seller $s$ chooses prices for its products across platforms to maximize variable profit, taking entry and logistics decisions and rival prices as given. This means that the seller solves a Bertrand–Nash pricing problem as:

$$\mathbf{p}_s^* = \arg\max_{\mathbf{p}_s} \sum_{f \in \mathcal{F}} \sum_{j \in \mathcal{J}_{sf}} \pi_{jf}(\mathbf{p}_s, \mathbf{p}_{-s}), \tag{11}$$

where $\mathbf{p}_s$ is the vector of prices set by seller $s$ and $\mathbf{p}_{-s}$ collects the prices of all other sellers.

The first-order conditions from (11) yield a system that can be inverted to recover marginal costs. In vector form, for the products of seller $s$:

$$\mathbf{p}_s = \tilde{\mathbf{mc}}_s - \mathbf{\Omega}_s^{-1}\, \mathbf{s}_s \tag{12}$$

where $\tilde{\mathbf{mc}}_s$ is the vector of commission rate absorbed marginal costs (i.e. $\tilde{mc}_{jft} = \frac{mc_{jft}}{1-\phi_f}$) and $\mathbf{\Omega}_s$ is the matrix of own- and cross-price demand derivatives for seller $s$'s products and $\mathbf{s}_s$ is the vector of market shares.[55] I recover $mc_{jft} = (1 - \phi_f)\tilde{mc}_{jft}$ and parameterize independent seller's marginal costs in the next section. The marginal costs of platform's own-retail are also recovered in the same fashion while $\phi_f = 0$ holds. I do not specify the function of own-retail marginal costs and use them as they are recovered throughout the analyses.

Aside from the commission rate, this is the standard markup equation from differentiated-product oligopoly, adapted to the multi-platform setting. The key feature is that, for a multihoming seller who lists products on both Amazon and Walmart, the matrix $\mathbf{\Omega}_s$ captures cross-platform demand derivatives. A multihoming seller internalizes the fact that raising its price on Amazon may divert some demand to its own listing on Walmart, and vice versa. Introducing the product in a competing platform induces cannibalization, discouraging multihoming.[56]

**Marginal Cost of Independent Seller** In the marginal-cost function of an independent seller, there are three main terms that determine a seller's per-unit cost in addition to standard cost

[55] Each element of $\mathbf{\Omega}_s$ is $\Omega_{jk} = \partial s_{jf}/\partial p_{kf'}$ for products $j, k \in J_s$, where products may be listed on different platforms $f, f'$. The matrix is non-zero off the diagonal when the same seller lists products on multiple platforms and a price change on one platform affects demand on the other.

[56] Appendix Section A discusses more details about sellers' pricing and multihoming.

shifters: the fee paid to the platform's logistics service (if used), the cost of independent logistics (if not), and economies of scale.

I specify the marginal cost of product $j$ on platform $f$ in month $t$ as:

$$mc_{jft} = \underbrace{\gamma^{\ell}\,\text{fee}^{\ell}_{jf}\,\ell_{jft}}_{\text{platform logistics fee}} + \underbrace{\gamma^{\text{UPS}}\,\text{fee}^{\text{UPS}}_{j}\,(1-\ell_{jft})}_{\text{independent logistics fee}} + \underbrace{\gamma^{Q}\bar{Q}_{s(j)ft}}_{\text{scale economies}} + \mathbf{W}'_{jf}\gamma^{W} + \omega_{c(j)} + \omega_{t} + \omega_{jft} \tag{13}$$

where $\ell_{jft}$ is an indicator for whether the product uses the platform's integrated logistics (FBA on Amazon, WFS on Walmart), $\text{fee}^{\ell}_{jf}$ is the platform logistics fee, $\text{fee}^{\text{UPS}}_{j}$ is the posted UPS shipping rate, and $\bar{Q}_{s(j)ft}$ is the effective scale of the seller. The term $\mathbf{W}_{jf}$ is a vector of cost-shifters (material dummies, major brand indicators, third-order polynomial in shipping weights and an Amazon-market dummy), while $\omega_{c(j)}$ and $\omega_t$ are category and month fixed effects, respectively. Finally, $\omega_{jft}$ is an i.i.d. marginal cost shock.

The coefficient on the platform logistics fee is restricted to $\gamma^{\ell} = 1$. This normalization reflects that the logistics fee is a per-unit cost that the seller pays directly to the platform. When a seller does not use the platform's logistics ($\ell_{jft} = 0$), it ships through an independent provider. Because I do not observe exact independent logistics fees, I instead proxy for them with the posted UPS rate $\text{fee}^{\text{UPS}}_{j}$. UPS and FedEx are the two largest private independent parcel carriers, and their posted schedules are nearly identical, so UPS is representative of the independent logistics option. The coefficient $\gamma^{\text{UPS}}$ therefore captures the average fraction of the posted rate actually paid.[57]

Scale economies arise through volume discounts on storage, shipping, and wholesale procurement, operating through $\bar{Q}_{s(j)ft}$. When a seller uses a common independent logistics provider across platforms, it can pool inventory and aggregate volume. When a seller uses a platform's integrated logistics on any platform, inventory is fragmented and scale is limited to single-platform volume. Formally,

$$\bar{Q}_{sft} = \begin{cases} Q_{sft} & \text{if } \ell_s = 1 \\ \sum_{f' \in \mathcal{F}} Q_{sf't} & \text{if } \ell_s = 0, \end{cases} \tag{14}$$

where $\ell_s = \max_{f \in \mathcal{F}}\{\ell_{sf}\}$ indicates whether the seller uses integrated logistics on any platform, and $Q_{sft} = \sum_{j \in \mathcal{J}_{sf}} Q_{jft}$ is the seller's total volume on platform $f$. The resulting fragmentation might reduce the seller's effective scale on each platform, raising per-unit costs through the scale economy channel. Therefore, the choice of logistics creates scale spillovers; because the seller's effective scale is the function of whether to use integrated logistics or not, marginal costs across *both* platforms ($mc_{sf}$ and $mc_{sf'}$) shift.

[57] This interpretation is supported by an interview with an industry expert. Typically, sellers (or the independent logistics provider) pay a discounted rate off UPS's posted schedule, negotiated between the seller and UPS.

### 4.2.2 Entry and Logistics Choice with Fixed Costs

In the first stage, each independent seller $s$ simultaneously chooses which platforms to enter and which logistics arrangement to use on each platform. Omitting the month subscript $t$, the seller's decision vector is:

$$\mathbf{d}_s = (e_{s\text{Amz}},\ e_{s\text{Wmt}},\ \ell_{s\text{Amz}},\ \ell_{s\text{Wmt}}) \in \mathcal{D}$$

where $e_{sf} \in \{0,1\}$ indicates entry into platform $f$, $\ell_{s\text{Amz}} \equiv \text{FBA}_s \in \{0,1\}$ indicates the use of FBA (relevant only if $e_{s\text{Amz}} = 1$), and $\ell_{s\text{Wmt}} \equiv \text{WFS}_s \in \{0,1\}$ indicates the use of WFS (relevant only if $e_{s\text{Wmt}} = 1$). Modeling the logistics decision at the seller level rather than the product level is motivated by the bimodal pattern of FBA adoption documented in Panel (a) of Figure 1—sellers store either all or none of their products at Amazon's facility. By omitting the infeasible case (e.g., using FBA to Amazon orders without entering into Amazon marketplace), sellers can choose among 8 decisions.

Sellers choose their decision vectors to maximize expected total profit, taking their own product portfolio and the decisions of all other sellers as given:

$$\mathbf{d}_s^*(\mathbf{d}_{-s}) = \arg\max_{\mathbf{d}_s \in \mathcal{D}} \mathbb{E}_{(\xi,\omega)} \left[\Pi_s(\mathbf{d}_s, \mathbf{d}_{-s})\right], \tag{15}$$

where the expectation is over the demand and marginal-cost shocks.

**Fixed Cost** The fixed cost of seller $s$ introducing product $j$ at month $t$ according to the decision $\mathbf{d}_s$ is

$$FC_{jt}(\mathbf{d}_s) := \overline{FC}(\mathbf{d}_s) + \nu_{jt}(\mathbf{d}_s) \tag{16}$$

where $\overline{FC}$ is the mean fixed cost for decision $\mathbf{d}$, and $\nu_j(\mathbf{d})$ is a decision-specific, mean-zero stochastic fixed cost shock (i.e., $\mathbb{E}\left[\nu_j(\mathbf{d})\right] = 0$). Here, $\overline{FC}(\mathbf{d}_s)$ is the parameter of interest to be estimated. The fixed costs include both the upfront setup costs of an e-commerce business, such as initial stocking, listing, branding, and marketing, and the opportunity cost of each decision vector, which are not absorbed in the marginal costs (i.e., variable costs).

## 4.3 Platform Duopoly

At the beginning (Stage 0) of the game, platforms set commission rates ($\phi_f$) and logistics fees ($\text{fee}_{jf}$) to maximize the total profit. Platforms also set retail prices of their own products, i.e., first-party retail, in the second stage, simultaneously with independent sellers.[58] The profits of platform $f$ can

[58] Platforms do not participate in the first-stage entry and logistics choice. Their entry and logistics choices are fixed as observed.

be written as:

$$\begin{aligned}
\Pi_f &= \Pi_f^{\text{1P}} + \Pi_f^{\ell} + \Pi_f^{\text{Com}} \\
\Pi_f^{\text{1P}} &= \sum_{t\in\mathcal{T}} \sum_{j\in\mathcal{J}_{ft}^{\text{1P}}} \left(p_{jft} - mc_{jft}^{\text{1P}}\right) Q_{jft} \\
\Pi_f^{\ell} &= \sum_{s\in\mathcal{S}^{\text{Ind}}} \sum_{t\in\mathcal{T}} \sum_{j\in\mathcal{J}_{sft}} \left(\text{fee}_{jf}^{\ell} - mc_{jf}^{\ell}\right) \ell_{sft} e_{sft} Q_{jft} \\
\Pi_f^{\text{Com}} &= \sum_{s\in\mathcal{S}^{\text{Ind}}} \sum_{t\in\mathcal{T}} \sum_{j\in\mathcal{J}_{sft}} \phi_f \, p_{jft} \, e_{sft} Q_{jft}
\end{aligned} \tag{17}$$

where $\mathcal{T}$ is the set of months, $\mathcal{J}_f^{\text{1P}}$ is the set of products offered by platform $f$, and $mc_{jft}^{\text{1P}}$ is the marginal cost of own-product retail. In the second stage in which every structural shock is realized, platform $f$ maximizes the total profit through retail pricing as:

$$\mathbf{p}_f^{\text{1P}*}(\mathbf{p}_{-f}) = \arg\max_{\mathbf{p}_f^{\text{1P}}} \Pi_f(\mathbf{p}_f^{\text{1P}}, \mathbf{p}_{-f}) \tag{18}$$

where $\mathbf{p}_f^{\text{1P}}$ is the vector of own-product retail prices and $\mathbf{p}_{-f}$ is the vector of prices of all the other products, including prices charged by the competing platform and independent sellers on every platform. In the initial stage 0, platform $f$ maximizes the expected total profit through fee charges and commission rates as:

$$\left(\mathbf{fee}_f^{\ell*}(\mathbf{fee}_{-f}^{\ell}, \phi_{-f}), \phi_f^*(\mathbf{fee}_{-f}^{\ell}, \phi_{-f})\right) = \arg\max_{(\mathbf{fee}_f^{\ell}, \phi_f)} \mathbb{E}\left[\Pi_f\left(\mathbf{fee}_f^{\ell}, \phi_f, \mathbf{fee}_{-f}^{\ell}, \phi_{-f}\right)\right] \tag{19}$$

where $\mathbf{fee}_f^{\ell}$ is the vector of every logistics fee on platform $f$ and $-f$ denotes the rival platform. The platform takes expectation over distributions of structural shocks that include demand ($\xi$), marginal cost ($\omega$) and fixed cost shocks ($\nu$). To simplify the platform pricing model in (19), I parameterize both the fee schedule and the logistics marginal cost as proportional to the observed schedule. Let $\overline{\text{fee}}_{jf}^{\ell}$ denote the observed logistics fee of product $j$ on platform $f$ (the FBA schedule for Amazon and the WFS schedule for Walmart), and let

$$\text{fee}_{jf}^{\ell} = \lambda_f \, \overline{\text{fee}}_{jf}^{\ell}, \qquad mc_{jf}^{\ell} = \kappa_f^{\ell} \, \overline{\text{fee}}_{jf}^{\ell}, \tag{20}$$

where $\lambda_f > 0$ scales every product fee proportionally and $\kappa_f^{\ell} > 0$ is the platform's logistic cost per dollar of the baseline schedule. The observed schedule corresponds to $\lambda_f = 1$, and the platform earns the per-unit logistic margin $\left(\lambda_f - \kappa_f^{\ell}\right) \overline{\text{fee}}_{jf}$ on every fulfilled unit. To state the first-order conditions, I first define the platform's logistic revenue at the baseline schedule and its commission

base as

$$R_f^{\ell} \equiv \sum_{s \in \mathcal{S}^{\text{Ind}}} \sum_{t \in \mathcal{T}} \sum_{j \in \mathcal{J}_{sft}} \overline{\text{fee}}_{jf}\, \ell_{sft}\, e_{sft}\, Q_{jft} \qquad R_f^{\text{Com}} \equiv \sum_{s \in \mathcal{S}^{\text{Ind}}} \sum_{t \in \mathcal{T}} \sum_{j \in \mathcal{J}_{sft}} p_{jft}\, e_{sft}\, Q_{jft} \tag{21}$$

Under (20), the platform profit can be written as $\Pi_f = \Pi_f^{\text{1P}} + \left(\lambda_f - \kappa_f^{\ell}\right) R_f^{\ell} + \phi_f\, R_f^{\text{Com}}$. Then, the first-order conditions of (19) are:

$$\lambda_f = \underbrace{\kappa_f^{\ell}}_{\text{logistics cost}} + \underbrace{\frac{\mathbb{E}\left[R_f^{\ell}\right]}{-\mathbb{E}\left[dR_f^{\ell}/d\lambda_f\right]}}_{\text{logistics markup (+)}} + \underbrace{\phi_f \frac{\mathbb{E}\left[dR_f^{\text{Com}}/d\lambda_f\right]}{-\mathbb{E}\left[dR_f^{\ell}/d\lambda_f\right]}}_{\text{commission-base erosion (-)}} + \underbrace{\frac{\mathbb{E}\left[d\Pi_f^{\text{1P}}/d\lambda_f\right]}{-\mathbb{E}\left[dR_f^{\ell}/d\lambda_f\right]}}_{\text{1P demand diversion (+)}} \tag{22}$$

$$\phi_f = \underbrace{\frac{\mathbb{E}\left[R_f^{\text{Com}}\right]}{-\mathbb{E}\left[dR_f^{\text{Com}}/d\phi_f\right]}}_{\text{commission markup (+)}} + \underbrace{\left(\lambda_f - \kappa_f^{\ell}\right) \frac{\mathbb{E}\left[dR_f^{\ell}/d\phi_f\right]}{-\mathbb{E}\left[dR_f^{\text{Com}}/d\phi_f\right]}}_{\text{logistics-margin erosion (-)}} + \underbrace{\frac{\mathbb{E}\left[d\Pi_f^{\text{1P}}/d\phi_f\right]}{-\mathbb{E}\left[dR_f^{\text{Com}}/d\phi_f\right]}}_{\text{1P demand diversion (+)}} \tag{23}$$

These first-order conditions decompose the observed rates into interpretable components: the fee condition splits the observed logistics fees into the logistics cost rate and three adjustment terms, while the commission condition consists of the three adjustment terms alone. The adjustments formalize the central tension of integrated platform pricing: market power versus commission incentives.

The markup terms capture market power—incentives to increase fees and rates above the costs. They are pinned down by the (expected) inverse of semi-elasticity of revenue bases with respect to corresponding fee and rate charges. The logistics revenue base in (21) has both logistics choice and entry margins, therefore, platforms consider sellers' incentive to join the platform together. In contrast, the commission base does not directly incorporate the logistics choice margin. Competition with the rival platform disciplines how sensitive sellers are to logistics fees and commission rates, as they can substitute toward the rival.

The erosion terms capture the commission incentive that disciplines this market power. The two bases are complementary: a higher logistics fee raises sellers' costs and retail prices and thus shrinks the commission base ($dR_f^{\text{Com}}/d\lambda_f < 0$), while a higher commission depresses sales and thus fulfilled volume ($dR_f^{\ell}/d\phi_f < 0$). Because the platform internalizes the complementary margin—the commission rate $\phi_f$ in (22) and the logistic margin $(\lambda_f - \kappa_f^{\ell})$ in (23)—each erosion term is negative and pulls the rate below its stand-alone level.

The 1P demand-diversion terms reflect the platform's third role as a retail competitor to its own sellers. Given platforms' and independent sellers' products are substitutes, a higher fee or commission raises independent sellers' prices and diverts demand toward the platform's own listings, so both diversion terms are positive—a classic force of raising rivals' costs (Salop and Scheffman,

1983) within the marketplace. Each observed rate therefore balances two upward forces, markup and diversion, against one downward force, erosion.

The decomposition also clarifies what logistics separation changes. If logistics is divested to an independent provider with pricing power, the provider sets $\lambda_f$ without earning commissions or first-party retail profit, so the commission-base-erosion and diversion terms drop out of (22); the platform continues to set $\phi_f$ but no longer holds the logistic margin, so the logistics-margin-erosion term drops out of (23). The commission loses only a downward term and therefore tends to rise. The logistics fee loses one downward term (erosion) and one upward term (diversion), so its direction is theoretically ambiguous.

### 4.4 Closing the Model

**Market Equilibrium** The equilibrium concept is a subgame perfect Nash equilibrium (SPNE) which consists of a Nash equilibrium in every subgame. No platform or seller can increase (expected) profit by unilaterally deviating from their strategy. I assume the existence of a pure-strategy SPNE, yet do not assume the uniqueness of it.

**Discussion** Several limitations are worth noting. First, independent logistics providers do not participate in the pricing competition in my model, which may restrict their responses to counterfactual interventions. An alternative would be to treat them as a single strategic player in the stage-0 pricing, but I follow what data indicates. An analysis of the fee-schedule dynamics of Amazon and other logistics providers (UPS and FedEx) in Appendix I shows that these providers respond little to Amazon's fee changes, suggesting that they do not engage in Bertrand-style competition over their publicly posted fees.[59]

In addition, the model restricts competition to the online channel, omitting Walmart's extensive brick-and-mortar stores. The model also abstracts away other online and offline retail channels available to independent sellers beyond the Amazon and Walmart platforms, such as Shopify stores. Although I do not explicitly model these channels, I show in Appendix A.3 that substitution toward them is absorbed into the marginal cost recovered from retail pricing. Therefore, counterfactual results can be interpreted as the changes while these incentives related to outside channels being fixed.

Lastly, the model abstracts from the platforms' intertemporal, forward-looking strategies, which matter for analyzing the dynamic equilibrium effects of logistics integration. The model does not entirely exclude such incentives, since the cost rates ($c_f^\ell$) can absorb investment incentives.[60] What

[59] This remains a limitation, however, because these providers can offer negotiated discounts off their posted schedules. One extension would let a representative independent logistics provider choose $\gamma^{\text{UPS}}$ in the stage-0, which would require modeling that provider's pricing and recovering their logistics costs.

[60] For instance, the cost rates include the Pareto weights that platforms allocate to consumer and producer surpluses, which are explicitly modeled in Gutierrez (2022) and Yu (2024).

the static formulation cannot capture is how these incentives evolve over time, which remains as limitation. I instead regard the static equilibrium as the long-run equilibrium state of the economy. I believe this focus is adequate for the research questions of this paper, and incorporating the platforms' dynamic incentives is an important extension for future research.

# 5 Identification and Estimation

## 5.1 Consumer Demand

I estimate the demand parameters by using a combination of aggregate and micro-level moment conditions. First, I use the aggregate-level moment condition as:

$$\mathbb{E}\left[\xi_{jft} \mid \mathbf{Z}^{\mathrm{D}}_{jft}\right] = 0 \tag{24}$$

where $\xi_{jft}$ is the unobserved demand shock in (3) and $\mathbf{Z}^{\mathrm{D}}_{jft}$ are the demand-side instruments.

To address the endogeneity of prices, I use $\log(\text{fee}_{jf})$ as a cost shifter instrument.[61] Two arguments support the validity of this instrument. First, the instrument satisfies the exclusion restriction and is relevant by construction: the fee is excluded from consumer utility but enters the seller's marginal cost and therefore shifts equilibrium prices. Also, the institutional details of platform fee-setting corroborate exogeneity. Platforms announce their fee schedules generally at annual intervals and apply them uniformly across the marketplace. Therefore, it is unlikely that an idiosyncratic, unobserved demand shock is correlated with this predetermined fee structure, particularly conditional on a rich set of observed product characteristics. Non-price product characteristics are assumed to be exogenous and therefore instrument for themselves. I use additional instruments to identify the nesting parameter ($\rho$). I use the number of products in the choice set in each market to identify the nesting parameter. This gives variations in within-nest shares and is exogenous under the timing assumption.

I build micro moments to identify demographic interaction terms ($\beta^p_1, \beta^p_2, \beta^p_3, \beta^p_{\text{Prime}}, \beta^p_{\text{W+}}, \beta^{\text{FBA}}_{\text{Prime}}$, $\beta^{\text{Wmt}}_{\text{W+}}$) and standard deviation on price coefficients ($\sigma^p$). I compute average log of prices by income groups and membership subscriptions for pinning down interaction terms ($\beta^p_1, \beta^p_2, \beta^p_3, \beta^p_{\text{Prime}}, \beta^p_{\text{W+}}$). In addition, I calculate Amazon's inside share by Prime and Walmart+ subscription status. These moments convey information about how strongly subscription status tilts a consumer toward or away from Amazon, thereby identifying ($\beta^{\text{FBA}}_{\text{Prime}}$, $\beta^{\text{Wmt}}_{\text{W+}}$). Lastly, I compute the second moment of log prices, which, together with the income-group averages of log prices, pins down the standard deviation of the price coefficient ($\sigma^p$). Additional details of moment construction are provided in the appendix section E.4.

I estimate the demand parameters by minimizing the GMM objective function that stacks

[61] I take log to match the log price variable in the consumer demand model and to improve the first-stage fit.

moment conditions described above. I implement a two-step procedure to improve efficiency: I first estimate the demand parameters by using the introduced aggregate and micro moments and then use the estimates to approximate the optimal micro moments in the spirit of Chamberlain (1987).[62] Additional details of demand estimation are discussed in the appendix section E.

### 5.2 Marginal Costs

The key condition to identify the cost parameters in (13) is

$$\mathbb{E}\left[\omega_{jft} \mid \mathbf{Z}^{\text{MC}}_{jft}\right] = 0 \tag{25}$$

where $\mathbf{Z}^{\text{MC}}_{jft}$ is a vector of the instrumental variables.

The endogenous variable in (13) is the effective scale ($\bar{Q}$). As instruments, I consider demand shifters that are excluded from the marginal cost function (Berry and Haile, 2014), alongside the Google Trends search popularity measure. These excluded demand shifters, however, include the number of reviews and star ratings, which may be correlated with unobserved cost shocks. For instance, a negative realization of the cost shock lowers marginal cost, which raises sales and hence the number of reviews. To address this potential endogeneity issue, I use leave-one-out instruments that average each demand shifter across the other products sold by the same seller, and separately across products sold by rival sellers.[63] Instrumenting for the effective scale also mitigates the bias from the rank-to-sales conversion measurement error. This is because these instruments are correlated with the true sales quantity but orthogonal to the conversion error. I estimate the marginal cost function using two-stage least squares (2SLS).

### 5.3 Fixed Costs

I use the revealed preference rationale to identify the fixed costs in (16). In addition, I restrict the support of differences in fixed costs to account for selection bias, following Eizenberg (2014). I introduce the identification with more detail in section H.1.

Suppose a seller $s$ chooses $\mathbf{d}^*_s \in \mathcal{D}$ as an optimal decision as in (15). Holding others' decisions fixed as $\mathbf{d}_{-s}$, the revealed preference rationale with the decision rule in (15) and Nash equilibrium condition imply that for any deviation $\mathbf{d}'_s \neq \mathbf{d}^*_s$,

$$\mathbb{E}_{(\xi,\omega)}\left[\Pi_s(\mathbf{d}^*_s, \mathbf{d}_{-s})\right] \geq \mathbb{E}_{(\xi,\omega)}\left[\Pi_s(\mathbf{d}'_s, \mathbf{d}_{-s})\right] \tag{26}$$

where $\mathbb{E}_{(\xi,\omega)}$ means the expectation is taken with respect to the distribution of $(\xi, \omega)$. We can rearrange these inequalities to denote the difference between the expected variable profits from the

[62] I follow the estimation routines in Conlon and Gortmaker (2025)

[63] This remains valid under the weaker assumption that $\omega_{jft}$ and $x_{j'ft}$ are uncorrelated for $j' \in \mathcal{J}_{sft} \setminus \{j\}$ or $j' \in \mathcal{J}_{s'ft}$ where $s' \neq s$, and $x \in \{\text{Star Rating}, \text{Reviews}\}$.

observed optimal choice $\mathbf{d}_s^*$ and any deviation $\mathbf{d}_s'$ is larger than incurred fixed cost differences of them.

These inequalities are informative as an upper and lower bound of the differences in fixed costs, yet the *selection bias* hinders us from directly using the above inequality conditions. To circumvent this selection issue, I follow the approach by Eizenberg (2014) and assume that the support of differences in fixed costs ($\Delta FC_s$) falls within the range of the supremum and infimum of expected variable profit differences ($\bar{\pi}_s$):

$$\Delta FC_s(\mathbf{d}, \mathbf{d}') \subset \left[\inf_s \Delta\bar{\pi}_s(\mathbf{d}, \mathbf{d}'), \sup_s \Delta\bar{\pi}_s(\mathbf{d}, \mathbf{d}')\right] \text{ for any } s \in \mathcal{S} \tag{27}$$

This support restriction allows me to attain the lower and upper bounds for every decision, so I can take the *unconditional* expectation, resolving the selection bias.

The support restriction is corroborated by the highly skewed sales distributions on e-commerce platforms, often resembling a Pareto distribution (Chevalier and Goolsbee, 2003; Brynjolfsson *et al.*, 2003, 2025). This means that a small number of best-selling products account for the vast majority of sales, causing expected variable profits to vary widely across products and sellers. In addition, standardized practices, such as wholesaling, warehousing, and fulfillment, and the consequently stable cost structures of e-commerce operations further support the validity of this assumption. Stacking the moment inequalities, I estimate the set by the linear-programming method. Estimation details are in Appendix H.1.1.

# 6 Estimation Results

## 6.1 Demand Estimation Results

Table 3 reports the parameter estimates from the main demand specification. There are three key features that the model aims to capture: cross-platform substitution, heterogeneity in price sensitivity, and valuation of logistics quality.[64]

Consumers without any membership subscription prefer Amazon to Walmart. The mean coefficient on the Walmart-platform indicator is $-1.300$ and significant. The Walmart+ interaction of 2.070 more than offsets this baseline, so Walmart+ subscribers are substantially more likely to purchase on Walmart. Beyond subscriptions, the nesting parameter $\rho = 0.190$ (significant at the 1% level) indicates residual within-platform correlation in unobserved tastes, which is statistically significant yet economically modest.

Income-based heterogeneity in price sensitivity is statistically significant but not substantially different from each other: relative to the mean log-price coefficient of $-3.790$, the income-bin in-

[64] As a robustness check, I estimate the demand model excluding star ratings, the number of reviews, and search-rank tiers. The resulting parameter estimates are very close to their counterparts in Table 3.

Table 3: Demand Estimates

| Variable | Mean | Interactions with Demographics | | | | | Std. ($\sigma$) / Corr. ($\rho$) |
|---|---|---|---|---|---|---|---|
| | | 50–100k | 100–150k | 150k+ | Prime | Walmart+ | |
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) |
| log of Prices | −3.790*** | 0.020 | 0.043*** | 0.057*** | 0.354*** | 0.033** | 0.000 |
| | (0.367) | (0.014) | (0.015) | (0.015) | (0.039) | (0.014) | (0.005) |
| FBA | −0.873*** | | | | 1.310*** | | |
| | (0.237) | | | | (0.191) | | |
| WFS | −0.183 | | | | | | |
| | (0.180) | | | | | | |
| Walmart Platform | −1.300*** | | | | | 2.070*** | |
| | (0.242) | | | | | (0.099) | |
| Star Ratings | 0.141* | | | | | | |
| | (0.086) | | | | | | |
| Reviews (in 1K) | 0.024*** | | | | | | |
| | (0.003) | | | | | | |
| log of Shipping Weight | 1.840*** | | | | | | |
| | (0.201) | | | | | | |
| 1P Retailer | −0.177 | | | | | | |
| | (0.119) | | | | | | |
| Stainless steel | 0.276*** | | | | | | |
| | (0.071) | | | | | | |
| Aluminum | −0.226** | | | | | | |
| | (0.100) | | | | | | |
| Plastic | −0.050 | | | | | | |
| | (0.072) | | | | | | |
| Platform Nests | | | | | | | 0.190*** |
| | | | | | | | (0.066) |
| Brand Fixed Effects | Yes | | | | | | |
| Search Rank Tiers | Yes | | | | | | |
| GMM objective | 52.17 | | | | | | |
| Number of Observations | 5,585 | | | | | | |
| Number of Markets | 39 | | | | | | |
| Micro-moment Observations | 41,165 | | | | | | |
| Mean Own Price Elasticity | −4.2 | | | | | | |

*Notes.* This table reports parameter estimates from the demand model. Column (1) presents the mean coefficients; columns (2)–(6) present interactions of product characteristics with household demographics, and column (7) reports the standard deviation $\sigma$ of the random coefficients on log price and the nesting parameter $\rho$ for the Amazon/Walmart nest structure. Income interactions are indicators for the household income bins 50–100k, 100–150k, and 150k+, with "below 50k" as the omitted category. Prime and Walmart+ denote subscription status. Robust standard errors are in parentheses. ***, **, and * indicate the coefficient estimate is statistically significant at 1, 5, and 10% level, respectively.

teractions range from 0.020 to 0.057. Prime subscribers are notably less price sensitive, with an interaction of 0.354;[65] the Walmart+ interaction is substantially smaller. The standard deviation of the random coefficient on log price is close to zero, indicating little residual unobserved heterogeneity. The implied mean own-price elasticity of −4.2 is consistent with the literature (Farronato *et al.*, 2025; Yu, 2024); Figure A10 shows the full distribution.

Prime subscribers place high value on FBA's logistics quality. The mean FBA coefficient of

[65] This estimate is qualitatively similar to the interaction term in Farronato *et al.* (2025).

−0.873 combines with the Prime–FBA interaction of 1.310 to give Prime subscribers a net utility weight of 0.437 on FBA fulfillment, whereas non-Prime consumers face only the negative mean coefficient. Evaluated at the median income bin among Prime subscribers and an average price of $74, this implies a willingness to pay of approximately $9.40 per delivery, equivalent to a 13% price discount. The WFS coefficient is statistically insignificant. Its point estimate close to zero implies that non-Prime consumers value WFS above FBA. This is consistent with Walmart offering two-business-day WFS shipping on orders over $35 regardless of membership status, whereas standard shipping on FBA orders for non-Prime customers takes 5–8 business days. The remaining characteristics have the expected signs.

### 6.2 Marginal Cost Estimation Results

Table 4 reports estimates of the main variables in the marginal cost function (13). The coefficient on $\text{fee}^{\text{UPS}} \times (1 - \ell_{sf})$, which captures the fraction of the posted UPS rate that independent sellers actually pay, is estimated at 0.607 and statistically significant at the 1% level. The estimate is stable across alternative specifications, ranging between 0.598 and 0.609 (Appendix Table A8). The estimates imply that sellers shipping through an independent provider pay roughly 60% of the posted UPS Ground schedule—an average negotiated discount of about 40%.[66]

Table 4: Marginal Cost Function Estimates

| Variable | Parameter | Estimate |
|---|---|---|
| $\text{fee}^{\ell} \times \ell_{sf}$ | $\gamma^{\ell}$ | 1.000 |
| $\text{fee}^{\text{UPS}} \times (1 - \ell_{sf})$ | $\gamma^{\text{UPS}}$ | $0.607^{***}$ |
| | | (0.090) |
| Scale Economy ($\bar{Q}$) | $\gamma^{Q}$ | $-0.144^{***}$ |
| | | (0.047) |
| Control Variables | | Yes |
| Category FE | | Yes |
| Month FE | | Yes |
| Weak IV $F$-stat | | 50.6 |
| Observations | | 4,092 |

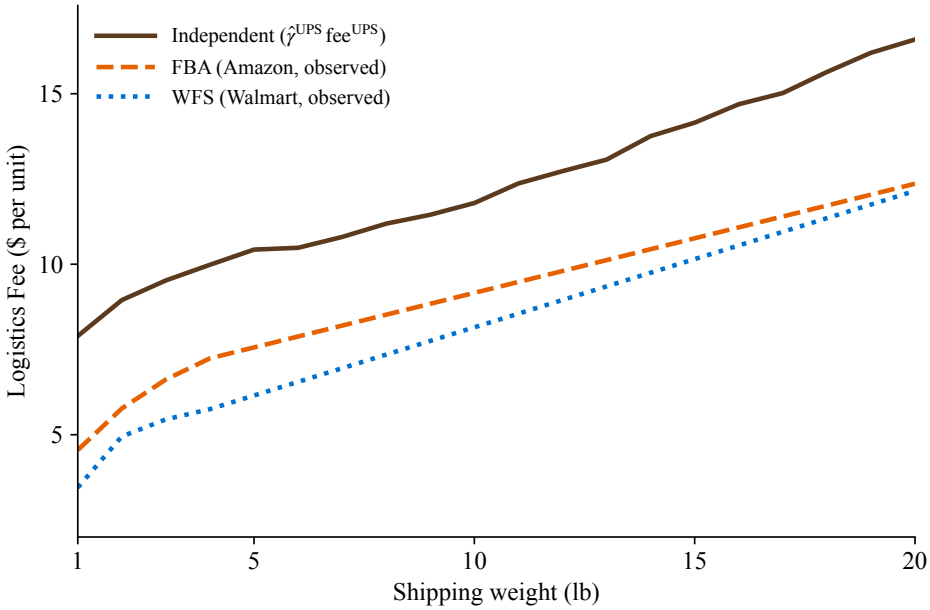
*Notes.* This table reports the estimates of main variables in the marginal cost function (13). The results correspond to column (8) of Table A8 in Appendix F, which reports coefficient estimates of all regressors, including controls, and alternative specifications. The coefficient on $\text{fee}^{\ell} \times \ell_{sf}$ is constrained to one. The independent logistics fee ($\text{fee}^{\text{UPS}}$) is the posted 2025 UPS Ground rate at the product's shipping weight. The unit of scale economy is 1,000 units. Heteroskedasticity-robust standard errors in parentheses. The weak IV $F$ is the rank-based Wald $F$-statistic of Kleibergen and Paap (2006). $^{***}$, $^{**}$, and $^{*}$ indicate the coefficient estimate is statistically significant at 1, 5, and 10% level, respectively.

The scale economy parameter $\gamma^{Q}$ is negative and statistically significant in the main specification

[66] This interpretation is consistent with an industry expert saying sellers negotiate substantial discounts off carriers' posted rate schedules.

and across almost every alternative specification in Appendix Table A8, indicating that sellers with larger effective volumes benefit from lower marginal costs.[67] The main specification is reported in Table 4 and carried into the counterfactual analysis; its estimate of −0.144 implies a reduction of about $0.14 per 1,000 units.

Figure 5: Logistics Fee Comparison




*Notes.* This figure compares the platforms' integrated logistics fee schedules with the estimated per-unit cost of independent logistics across shipping weights. FBA and WFS are the platforms' 2025 logistics fee schedules, shown for the large-standard size tier, to which most products in the sample are assigned. The brown line is the estimated per-unit fee of independent logistics, $\hat{\gamma}^{UPS}$fee$^{UPS}$, evaluated at the main-specification estimate $\hat{\gamma}^{UPS} = 0.607$ from Table 4. fee$^{UPS}$ is the posted 2025 UPS Ground rate.

The marginal cost estimates speak to the recent regulatory debate on whether FBA is a costly option for sellers. The FTC complaint describes FBA as a "costly" fulfillment service that has made selling on Amazon "substantially more expensive" for sellers (*FTC v. Amazon*, 2023), whereas Amazon characterizes FBA as a "cost-effective" option.[68] Figure 5 translates the estimates into a direct comparison among the logistics options.[69] Amazon's and Walmart's integrated logistics fees lie below the estimated per-unit cost of independent logistics at every shipping weight, with Walmart's fee the lowest. This supports the Amazon's claim.

The fee comparison alone, however, cannot settle the debate because of economies of scale. Integrated logistics forces sellers to split inventory by platform, forgoing the scale economies that a

[67] In the baseline linear specification of equation (13), reported in columns (2), (4), (6), and (8) of Appendix Table A8, each additional 1,000 units of effective monthly scale lowers a seller's marginal cost by roughly $0.06 to $0.32. The logarithmic alternative, reported in columns (1), (3), (5), and (7) of Appendix Table A8, delivers the same qualitative message. The weak IV $F$-statistics indicate that the instruments are relevant in every IV specification.

[68] Amazon's public response to the FTC complaint, September 2023 (aboutamazon.com).

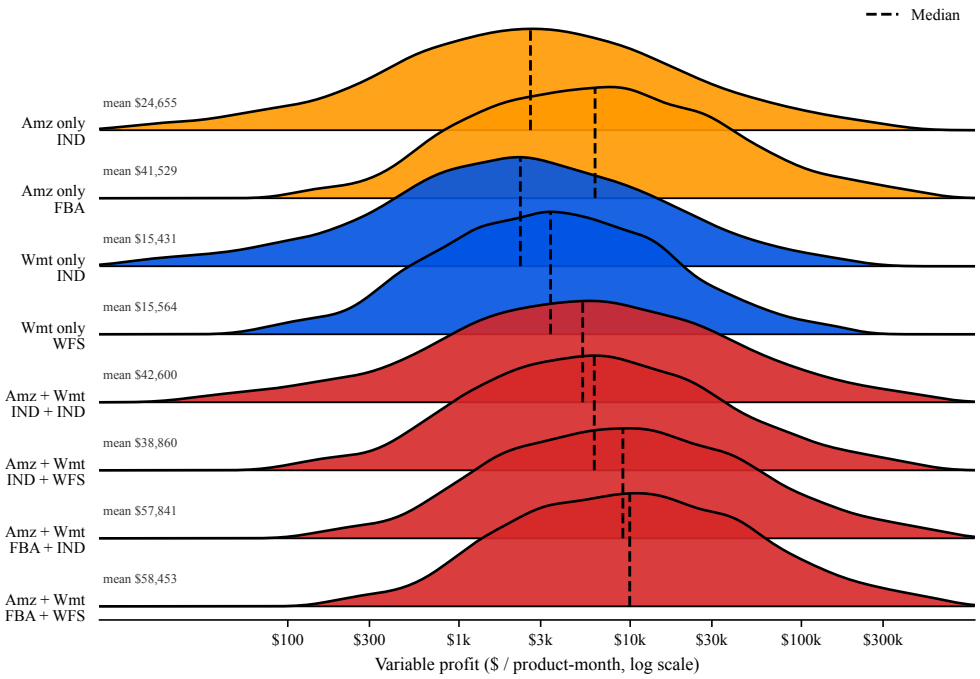
[69] Figure A12 shows the fee comparison for shipping weights up to 50 lbs. This figure also includes the posted fees of Amazon's multi-channel fulfillment (MCF) service, which apply to orders placed outside Amazon.

common independent provider would deliver. Setting $\gamma^Q = 0$ and recomputing marginal costs, I find that scale economies lower sellers' marginal costs by roughly 3.8% on average and by up to 20% for large sellers. Whether FBA is costly therefore depends on how a seller's gain from the lower posted fee compares with its loss from fragmented inventory. Answering this question requires accounting for sellers' fixed costs and for their equilibrium entry and logistics decisions, each of which embodies a trade-off including the marginal-cost side.

### 6.3 Estimated Variable Profits by Entry and Logistics Decisions

Using the demand and marginal cost estimates, I quantify the trade-off that sellers face in their entry and logistics choices. Specifically, I simulate each seller's variable profits under each $\mathbf{d} \in \mathcal{D}$ while holding the other sellers' decisions fixed; these profits are crucial inputs for estimating the fixed costs, because sellers directly compare the variable profit differences in order to choose optimal $\mathbf{d}$ as described in (15). The simulation details are explained in Appendix G.

Figure 6: Simulated Variable Profits by Entry and Logistics Decisions




*Notes.* This figure shows the estimated expected variable profits by entry and logistics decisions. Each ridge is a kernel density estimate of the distribution of simulated expected variable profit across product-months under one $\mathbf{d} \in \mathcal{D}$. The unit of observation is a product–month; variable profit is summed over the product's active platform listings. The horizontal axis is on a log scale (USD per product-month); each density is estimated on log variable profit trimmed to its 1st–99th percentiles. Dashed lines mark the median and "mean" reports the full-sample mean. Orange, blue and red denote Amazon-only, Walmart-only and multihoming entry decisions, respectively. IND denotes independent logistics.

Figure 6 plots the distributions of simulated expected variable profit per product-month under

each entry and logistics decision. Overall, selling on Amazon is more profitable than selling on Walmart. On Amazon, using FBA raises mean variable profit by about 70% (from \$24.7K to \$41.5K), reflecting both Prime subscribers' demand for FBA listings and FBA fees that are below the cost of independent logistics. This is not the case on Walmart: WFS and independent logistics yield similar mean variable profits, about \$15.5K.

Multihoming increases variable profits because it adds a sales channel and, with independent logistics, generates scale economies through inventory pooling. Multihoming with independent logistics on both platforms yields 6.3% more than the sum of the corresponding single-homing profits (Amazon-Ind and Walmart-Ind), whereas the multihoming decisions that use FBA, under which inventory cannot be pooled, exceed their sums by only 1.5–2.4%. Because FBA prevents inventory pooling, sellers face a trade-off: FBA raises sales on Amazon but forgoes the scale benefits of pooling.

### 6.4 Fixed Costs Estimation Results

Figure 7 reports the fixed-cost estimates of each entry and logistics decision. All estimates are reported relative to the Amazon–independent-logistics decision, $FC(1,0,0,0) \equiv 0$. The figure shows the identified-set bounds for each decision along with the best-fitting point of the identified set.[70].

There are several features worth noting. First, three of the four multihoming sets lie strictly above zero. The only multihoming decision whose identified set contains zero is the one with independent logistics on both platforms. Second, among the multihoming decisions, the fixed cost is lowest under independent logistics on both platforms and highest when FBA is combined with independent logistics on Walmart, consistent with the data that the majority (56%) of observed multihomers operate independent logistics on both platforms. That adding WFS lowers the multihoming cost conditional on FBA suggests that the two integrated services are partially complementary in resolving the inventory fragmentation that FBA creates.

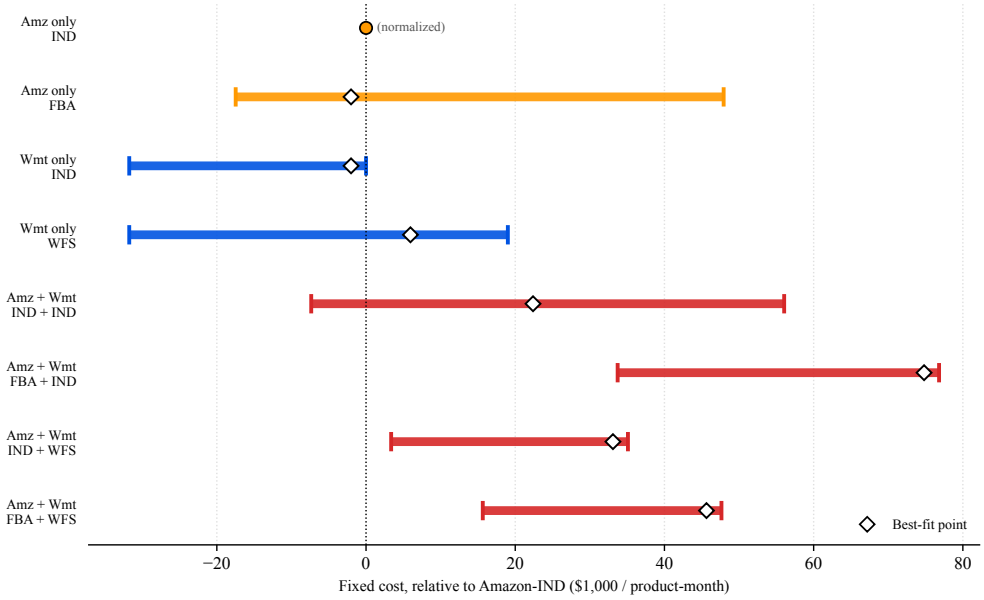
### 6.5 Decomposing the Platform Pricing

The full demand and cost estimates tell us how sensitive consumers and sellers are to prices, fees and commission rates, which further informs the platforms' pricing incentives. In this section, I estimate and decompose factors of platform's pricing that internalize both profits from logistics service provision and marketplace commissions.

I evaluate each term in the first-order conditions (22)–(23) and estimate $\kappa_f^\ell$. I obtain the derivatives with respect to each platform's logistics fee and commission rate by taking a central-difference, simulating entry-logistics decisions, and solving the estimated second-stage price equilibrium. Given

[70]This is the point that best predicts the observed distribution of seller choices while setting the unobserved fixed cost shocks to be zero as in Eizenberg (2014). Together with the simulated expected variable profits and decision rule (26), the best-fit point predicts 66% of observed decisions.

Figure 7: Fixed Cost Estimates by Entry and Logistics Decision




*Notes.* This figure shows fixed cost estimates of each decision, $\overline{FC}(\mathbf{d})$. IND denotes independent logistics. The fixed cost of Amazon-IND decision is normalized to be zero. Horizontal bars are the estimated sets; diamonds mark the best-fitting point of the estimated set, i.e., the point that best fits the observed distribution of seller choices when the unobserved fixed cost shocks are imputed to be zero. Orange, blue, and red denote Amazon-only, Walmart-only, and multihoming entry decisions, respectively. The number of sellers is 1,839. Exact values with 95% confidence intervals are reported in Appendix Table A10; details of the estimation are in Appendix Section H.1.1.

these derivatives, I solve for the unique $\kappa_f^\ell$ that rationalizes the observed schedule ($\lambda_f = 1$). This simultaneously pins down the 1P retail margins through the platforms' own retail pricing conditions.

Table 5 shows the decomposition results. I discuss the results mainly based on the best-fit point, but the implications are qualitatively consistent with more conservative partially identified results. In the fee condition, the logistics markup is 0.64 on Amazon and 0.43 on Walmart: the platforms can exert significant market power through the logistics chain. The implied logistics cost rate is $\kappa_f^\ell = 0.54$ on Amazon and 0.89 on Walmart, so the per-unit logistics margin $1 - \kappa_f^\ell$ is 0.46 and 0.11: Amazon earns a sizable margin on each fulfilled unit while Walmart prices its logistics service close to cost.

The logistics markup is largely offset by commission-base erosion ($-0.35$ on Amazon, $-0.41$ on Walmart—about half the markup on Amazon and nearly the full markup on Walmart). As an integrated platform internalizes that a higher logistics fee drives sellers off the marketplace and shrinks the commission base, the platform faces downward pressure on the logistics fee. This suggests that vertically separating the logistics operator can reduce consumer and independent sellers' welfare as the independent logistics operator would not internalize any of this erosion and therefore have incentives to raise fees.

Table 5: Decomposition of the Platforms' Fee and Commission Pricing

| | Amazon | Walmart |
|---|---|---|
| *Panel A: fee FOC* (22) | | |
| Logistics cost rate $\kappa_f$ | 0.543 | 0.895 |
| | [0.377, 1.374] | [0.367, 1.752] |
| Logistics markup | 0.636 | 0.431 |
| | [0.138, 0.784] | [0.070, 0.983] |
| Commission-base erosion | -0.355 | -0.412 |
| | [-0.693, -0.212] | [-0.839, 0.010] |
| 1P demand diversion | 0.176 | 0.086 |
| | [0.039, 0.217] | [0.014, 0.193] |
| *Panel B: commission FOC* (23) | | |
| Commission markup | 0.322 | 0.253 |
| | [0.058, 0.381] | [0.021, 0.356] |
| Logistics-margin erosion | -0.130 | -0.022 |
| | [-0.142, 0.070] | [-0.058, 0.126] |
| 1P demand diversion | 0.036 | 0.017 |
| | [0.007, 0.043] | [0.001, 0.024] |

*Notes.* This table shows the estimates of each term in the first-order conditions of the platform pricing model in (19). Brackets report the range of each term across 50 fixed-cost vectors sampled uniformly from the estimated set. I compute the derivatives with respect to each platform's logistics fee and commission rate by taking a central-difference, simulating entry-logistics decisions, and solving the second-stage pricing.

The commission rate is mainly determined by the commission markup (0.32 on Amazon and 0.25 on Walmart), with smaller contributions from the logistics-margin and 1P channels.[71] The 1P demand-diversion terms are small relative to the other components (0.18 and 0.09 in the fee condition, 0.04 and 0.02 in the commission condition). These numbers imply that the platforms' incentive to raise costs of independent sellers in order to steer demand toward their own 1P products is relatively weak on both Amazon and Walmart. The numbers, however, are strictly above zero, which means that there still exist anti-competitive incentives.

# 7 Counterfactuals

I use the estimated model to conduct counterfactual analyses. The main analysis separates the logistics service (FBA) from Amazon to measure the net welfare effects of Amazon's FBA integration.[72] I then conduct three additional analyses to further understand the interaction between integrated logistics and platform competition. The first is a policy counterfactual of interoperable logistics. Enforcing interoperability is often suggested as a regulatory tool to foster competition (Bourreau

[71] According to the decomposition, there might be a discrepancy between the model implied $\hat{\phi}_f$ that is the sum of three adjustment terms and the observed $\phi_f = 0.15$. To account for the potential bias from this discrepancy in the counterfactual analysis, I fully re-solve the status-quo integrated regime and use it as the benchmark for comparison.

[72] I focus on Amazon's logistics service because Amazon is the larger platform in terms of market share.

and Krämer, 2023; Scott Morton *et al.*, 2023; Colangelo and Martinez, 2025). I provide data-driven evidence of the impacts of making a platform's logistics service available to the competing platform at the same quality and fees. The second is an Amazon monopoly counterfactual, which quantifies the competitive pressure from Walmart in disciplining Amazon's pricing incentives under logistics integration. The last is a reduced FBA cost efficiency counterfactual, which measures whether the large cost efficiency gains from Amazon's logistics investment, documented in Houde *et al.* (2023), are indeed passed on to consumers and sellers.

To evaluate the welfare outcomes, I define consumer surplus (CS) as (Small and Rosen, 1981; Petrin, 2002):

$$\text{CS} = \sum_{ct} \left[ M_{ct} \sum_{i \in I_{ct}} w_{ict} \frac{\bar{p}_{ct}}{|\beta_i^p|} \log \left( 1 + \sum_{f \in \mathcal{F}} \exp V_{ifct} \right) \right], \quad V_{ifct} = (1-\rho) \log \sum_{j \in J_f} \exp \left( \frac{V_{ijct}}{1-\rho} \right) \tag{28}$$

in which $w_{ict}$ are the weights of the consumers, $V_{ijct} := \beta_i^p \log(p_{jft}) + \beta_i^{\text{FBA}} \text{FBA}_{jft} + \beta^{\text{WFS}} \text{WFS}_{jft} + \beta_i^{\text{Wmt}} \text{Wmt}_{jft} + X'_{jft} \beta^x + \xi_{b(j)} + \xi_{jft}$ in (3), and $V_{ifct}$ is the inclusive value of platform-nest $f$ with nesting parameter $\rho$. To convert consumer surplus into the money metric, I use the observed share-weighted market price $\bar{p}_{ct}$, which is held fixed across scenarios.

In addition, seller surplus (SS) aggregates variable profits net of fixed costs, as in (10), over each seller's equilibrium decision. Each platform's profit ($\Pi_f$) comprises commission revenue on independent sellers' sales, own retail profit, and logistics profit, as in (17). Total welfare is the sum of these components:

$$\text{W} = \text{CS} + \text{SS} + \Pi_{\text{Amz}} \ (+\Pi_{\text{FBA}}) + \Pi_{\text{Wmt}}. \tag{29}$$

In the separation counterfactual, $\Pi_{\text{Amz}}$ does not include the profit of the separated FBA operator, $\Pi_{\text{FBA}}$, which I report separately. In the others, $\Pi_{\text{Amz}}$ includes $\Pi_{\text{FBA}}$.

### 7.1 Measuring the Welfare Effects of Amazon's Logistics Integration

Separation of logistics services has both pro- and anti-competitive effects, so its welfare consequence is theoretically ambiguous. Amazon's two roles, as a marketplace intermediary and as a retailer of its own products, generate different effects. On the one hand, separation removes the exclusivity of the high-quality logistics service and the scale losses due to inventory fragmentation, and makes the fixed cost of joining the competing platform equal to that of using independent logistics.[73] Separation also removes the incentive to raise sellers' costs through high logistics fees in order to divert sales to Amazon's own retail business (Salop and Scheffman, 1983; Salinger, 1988).[74] On the

[73] I assume that the share of consumers holding Prime subscriptions, and hence with access to high-quality Amazon logistics, is unchanged under separation. In the counterfactual regime, those consumers also receive the same quality of Amazon logistics on their Walmart orders. This can be interpreted as consumers subscribing to a Prime membership program offered by the separated Amazon logistics operator.

[74] I let Amazon continue to use the separated Amazon logistics in the separation regime. This abstracts away from the contractual relationship between platforms and logistics providers in the counterfactual regime. A different

other hand, separation introduces double marginalization, and thus removes the downward pressure that integration places on both the commission rate and the logistics fee. In addition, separation also results in double marginalization between logistics operation and retail business (Spengler, 1950).

I re-solve the full game under the counterfactual separation regime: platforms set commissions and logistics fees, sellers choose entry and logistics, and prices adjust to equilibrium.[75] However, two obstacles make an exact solution infeasible. First, as in many static entry games analyses, the equilibrium may not be unique. Second, computing the set of all potential Nash equilibria is impractical.[76] Therefore, I instead employ a rather simplified algorithm. For given logistics fees and commission rates, I let sellers move sequentially in a randomly drawn order.[77] I still allow full strategic responses of Amazon, Walmart and the separated Amazon logistics and solve them by employing fixed point iterations.

Figure 8 contains the results. Panel (a) shows the impacts of separation on welfare. Separating FBA reduces consumer welfare by 19.2% and Amazon's profit without the FBA part by 18.2%. Aggregate seller profits increases by a small degree (+2.7%). The loss to Amazon excludes the profit of the separated FBA operator, which earns $117M, about 3.6 times the $33M logistics profit of the integrated FBA; Amazon and the operator together earn 11.1% more than the integrated Amazon. A clear winner is Walmart, whose profit rises by 50.4%. The gains of the separated operator and Walmart offset the losses of consumers and Amazon, so total welfare changes little (+1.6%). Separation therefore mainly redistributes surplus from consumers to the logistics operator and the competing platform.

Panel (b) of Figure 8 shows the impacts of separation on pricing and multihoming. The separated FBA charges a logistics fee 40.5% higher than the integrated FBA, and Amazon raises its commission rate by 16.1%. Walmart's best response is to raise its commission rate by a similar amount. The higher fees and rates are passed through to consumers: the average price rises from $73.00 to $91.14 (24.9%). This price increase dominates the pro-competitive effects of facilitating cross-platform competition, even though those effects are present: the multihoming rate nearly doubles, from 6.9% to 13.2%.

Panel (c) of Figure 8 decomposes platform profits by source. The separated FBA earns more than the integrated FBA mainly because it expands to the Walmart platform. Under separation, the

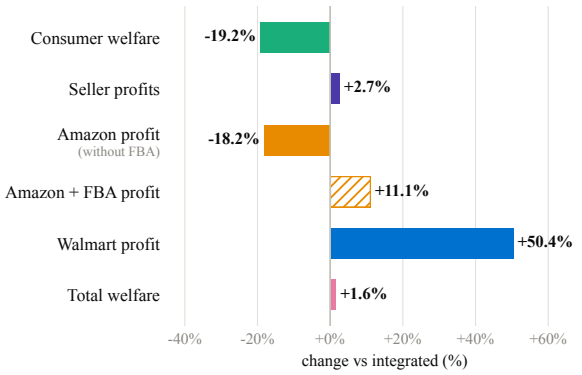
contractual relationship, such as two-part tariffs, could have different impacts on markets and welfare (Rey and Tirole, 1986; Bonanno and Vickers, 1988).

[75] As a benchmark, I also re-solve the full model under the status-quo integrated regime to ensure the model-based comparison.

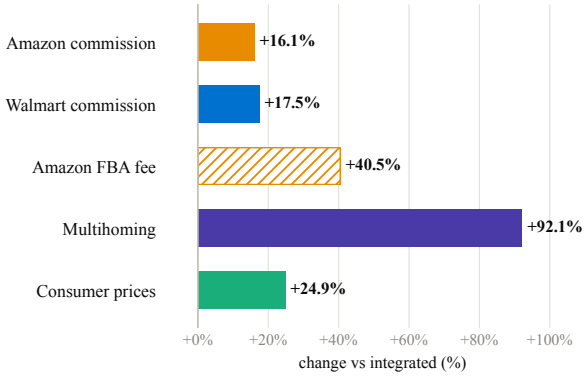
[76] The ideal procedure would evaluate every profile of seller decisions and check profitable deviations from each profile for a given logistics fees and commission rates charge, which is impossible to calculate. With 1,839 independent sellers each choosing among eight alternatives, there are $8^{1,839}$ profiles.

[77] For every counterfactual, I use the same order of sellers' moves. Another approach to make an order of the decision is to rank independent sellers in the order of profitability as in Berry (1992).

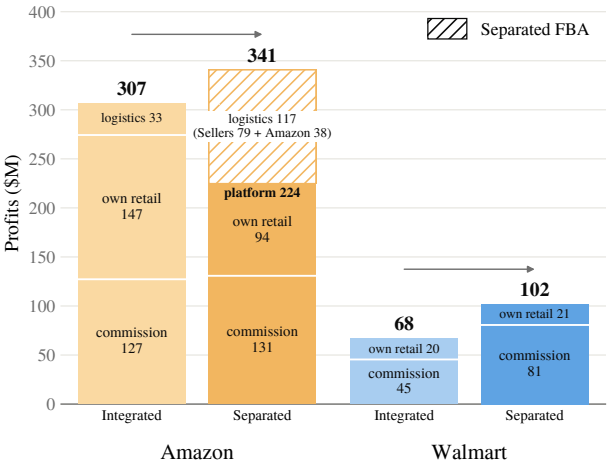
Figure 8: The Effects of Separating Logistics Service (FBA) from Amazon



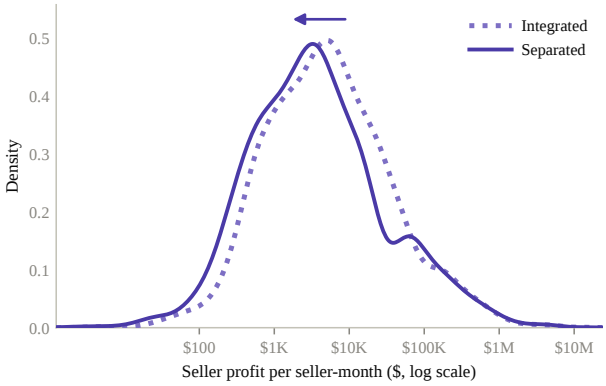

(a) Welfare




(b) Pricing and Multihoming




(c) Platform Profits by Parts




(d) Distribution of Seller Profits

*Notes.* This figure compares the integrated benchmark with the counterfactual that separates Amazon's logistics service (FBA) from its platform. Panels (a) and (b) show percentage changes relative to the integrated benchmark. In Panel (a), "Amazon profit (without FBA)" is the platform's profit excluding logistics, and the hatched bar, "Amazon + FBA profit," adds the profit of the separated FBA. Panel (b) shows changes in the commission rates, the FBA fee, the multihoming rate, and the average consumer price across both marketplaces, weighted by market shares. The change in the fee for Walmart's logistics service (WFS) is omitted because no seller adopts WFS in the market equilibrium under separation. Panel (c) decomposes platform profits ($M) into commissions, own retail, and logistics profits. The hatched block is the profit of the separated FBA; within it, "Sellers" and "Amazon" denote the profits earned from independent sellers' adoption and from Amazon's own retail, respectively. Panel (d) plots kernel density estimates of seller profit per seller-month; the horizontal axis is on a logarithmic scale.

operator earns $117M, of which $79M comes from independent sellers and $38M from Amazon's own retail. Amazon earns less from retail ($147M to $94M), primarily because double marginalization raises its retail cost, and slightly more from commissions ($127M to $131M) because the higher commission rate is applied to a smaller sales base. Walmart's commission profit rises by 77% ($45M to $81M), which accounts for nearly all of its gain; its retail profit is unchanged. Walmart's own logistics service, WFS, is no longer competitive once the separated FBA is available on Walmart, and no seller uses it.

Panel (d) of Figure 8 shows that separation increases earnings inequality among sellers. Although aggregate seller profits are nearly unchanged, the bulk of the distribution shifts to the left: the median seller-month profit falls by one third, from $4,836 to $3,220, while the share of profits earned by the top 10% of seller-months rises from 83% to 87%. Sellers in the top decile gain on average because they can absorb the higher commission rates and logistics fees by multihoming with the separated FBA. The use of the separated FBA allows inventory pooling across Amazon and Walmart, which further increases benefits from economies of scale.

Overall, logistics integration benefits consumers even though the pro-competitive effects of separation (multihoming, non-exclusive quality, and inventory pooling) are present. Separation raises logistics fees and commission rates by enough to more than offset these gains for consumers, and it substantially benefits Walmart, which raises its own commission rate in response. The gain in total welfare is small and accrues entirely to the separated FBA and to Walmart.

Why does fostering multihoming not offset the harm that separation does to consumers? To understand the effects of FBA separation further, I simulate a counterfactual in which FBA is separated from Amazon but remains exclusive to the Amazon marketplace. Figures A14 and A15 show the results. I find that multihoming with the separated FBA itself benefits consumers, but the pricing responses of the separated FBA and the two platforms allow them to capture these consumer welfare gains.[78] The separated FBA raises its logistics fees more as its reach extends beyond Amazon, as a monopolist upstream input provider.

Both platforms also raise their commission rates, not only because separation introduces double marginalization but also because it softens commission-rate competition between the platforms. In the separated-yet-exclusive counterfactual, both Amazon's commission rate and FBA fees rise by large amounts, reflecting substantial double marginalization, while Walmart's commission rate responds little to separation. Opening the separated FBA to Walmart sellers raises Amazon's commission rate by a further 3.1% and Walmart's by 17.5%. When there are more multihoming sellers, platforms likely have weaker incentives to cut commission rates to attract exclusive sellers.[79]

[78] Multihoming with the separated FBA has three benefits: it gives consumers greater access to their preferred logistics, introduces more variety through entry, and promotes price competition across platforms. The interoperability counterfactual in Section 7.2, which holds platforms' pricing responses fixed, provides evidence for these benefits.

[79] Theoretical predictions on how multihoming affects price competition between platforms are mixed and depend on which side multihomes (Bakos and Halaburda, 2020; Belleflamme and Peitz, 2019; Teh *et al.*, 2023). Even though

### 7.2 Interoperability of Integrated Logistics

Can a policy targeting the anti-competitive effects improve the welfare? This question is related to active discussions on enforcing interoperability of a platform service (Bourreau and Krämer, 2023; Colangelo and Martinez, 2025) as a policy tool to foster platform competition. In this subsection, I examine a policy that mandates full (or equitable) interoperability (Scott Morton *et al.*, 2023), which removes the exclusivity and discriminatory fees of integrated logistics. It is important to distinguish this counterfactual from Amazon's business practice, multi-channel fulfillment service (MCF). MCF can be regarded as a partial, instead of full, interoperable logistics service across platforms and Amazon charges higher fees on orders from competing platforms with slower shipping speed compared to within Amazon orders. Figure A12 shows the MCF fees.

The counterfactual experiment of equitable Amazon logistics is as follows. When a seller multihomes using FBA, I allow the seller to pool inventory across channels and to pay the same fixed cost as under independent logistics. The seller also pays the same publicly posted Amazon logistics fee. To remove the demand-side exclusive quality, I assume that Prime subscribers attain the same demand-side quality on the Walmart side as well. In this experiment I hold both platforms' logistics fees and commission rates at their benchmark levels, so the results capture the effects of removing exclusivity for given platform prices.[80]

Panel (b) in Figure 9 shows the results. Removing exclusivity suggests room for enhancing welfare. Social welfare increases by about 27.6%, together with both consumer and seller welfare. To examine the mechanism behind this result, I conduct two decomposition analyses: the first removes only cost-side frictions, and the second removes only demand-side exclusivity. The results are shown in panels (b) and (c) of Figure A16. Cost side gives larger consumer gains (+18.0% vs +14.0%); demand side gives larger seller (+20.0% vs +9.6%) and total-welfare (+14.8% vs +11.1%) gains.

I further conduct a counterfactual analysis of Amazon's MCF.[81] In this counterfactual, when a seller multihomes using Amazon logistics, the seller can pool inventory across platforms. However, the seller pays the MCF fee, which is publicly posted by Amazon, per Walmart-side unit.[82] Panel (a) in Figure A16 shows that Amazon's MCF has only a modest impact on welfare: consumer and social welfare increase by 5.5% and 4.0%, respectively. This suggests room to improve welfare by further removing exclusivity (or asymmetry) of Amazon's logistics service.

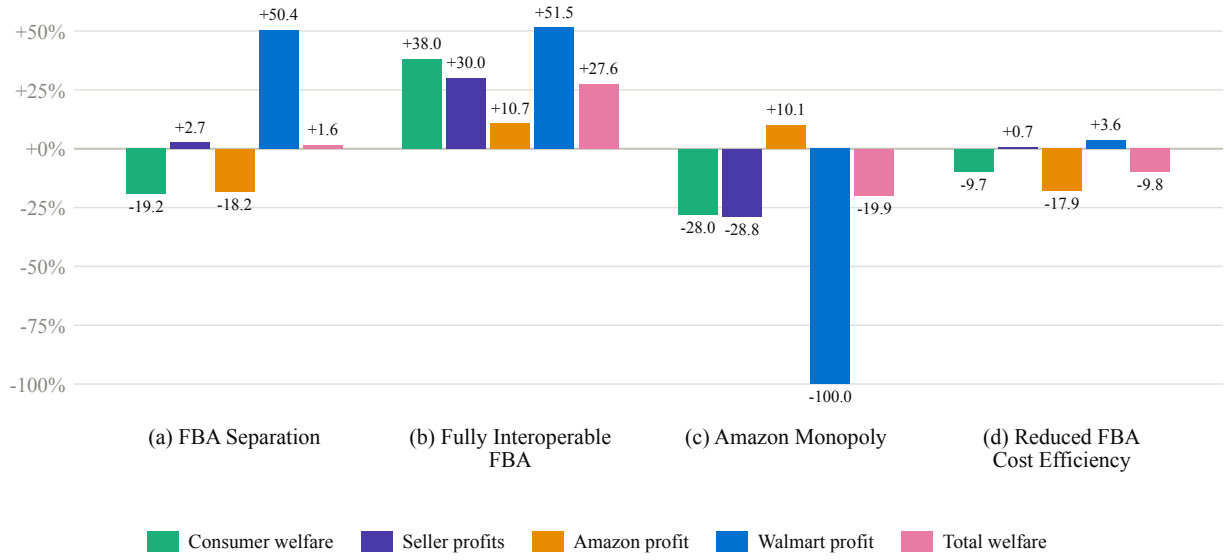
existing models do not consider the interaction of multihoming and vertically integrated services, my results are aligned with the prediction of Armstrong (2006) and Armstrong and Wright (2007): when sellers multihome, platforms compete less for them, and each platform instead charges them high fees for access to its own single-homing consumers.

[80] Appendix Figure A16 reports the decompositions under the same assumption.

[81] Walmart began allowing the use of the MCF on May 15th, 2025. Most of the data period, which ends on June 4th, 2025, precedes this intervention; hence my analysis throughout assumes that the MCF is not allowed. In this subsection, I instead examine the impact of introducing the MCF through counterfactual analyses.

[82] I assume the fixed costs of multihoming using Amazon logistics stay the same as estimated. Because Amazon's MCF provides two-business-day shipping as the fastest shipping speed, I set the demand-side coefficient of using Amazon logistics on the Walmart side equal to that of Walmart logistics.

Figure 9: Welfare Effects of the Counterfactuals




*Notes.* This figure shows the welfare outcomes of four counterfactuals as percent changes relative to the integrated benchmark. Total welfare is the sum of consumer welfare, seller profits, and platform profits. Panel (a) separates Amazon's logistics service (FBA) from Amazon: an independent operator runs FBA, sets its logistics fee, and serves both marketplaces. Amazon profit in Panel (a) excludes the profit of the separated FBA, which is included in total welfare. The result is identical to Panel (a) in Figure 8. Panel (b) makes FBA fully interoperable: a seller that multihomes using FBA can fulfill Walmart orders through FBA at the posted FBA fee, pool inventory across the two marketplaces, and pay the same fixed cost as under independent logistics, and Prime subscribers obtain the same benefit from FBA on the Walmart side as on the Amazon side. Panel (c) removes the Walmart marketplace and Walmart's logistics service, so Walmart's profit falls by 100%. Panel (d) sets Amazon's logistics cost equal to Walmart's as a benchmark.

### 7.3 Amazon Monopoly

To what extent does a competing platform discipline the pricing incentives of a dominant platform, and thereby enhance overall welfare? To address this question, I conduct a counterfactual simulation that removes the Walmart marketplace, together with Walmart's logistics service. Beyond e-commerce, the simulation has implications for a market with a similar structure: app stores. In *Epic v. Google* (2025), the court ordered Google to open the Android app market to rival app stores.[83] By simulating a counterfactual in which Amazon is a monopolist, I measure the welfare impact of the presence of a competing platform.[84]

Panel (c) of Figure 9 shows the welfare outcomes, and Figure A17 in the Appendix reports the pricing, platform profits, and seller profits. The results show that a rival constrains the dominant platform's commission, and removing it lowers total welfare by 19.9%. Consumer welfare falls by 28.0% and seller profits fall by 28.8%, while Amazon's profit rises by 10.1%. Without the competing platform, Amazon raises its commission rate by 23% and lowers its logistics fee by 5.9%. Nearly all sellers adopt Amazon's logistics (99.96%), so the marketplace consolidates into Amazon's own logistics. Amazon's gain comes primarily from commissions, which rise from \$127M to \$153M.

Although there are large increases in commission rates from Amazon's monopoly pricing, the losses of consumers and sellers come mainly from the removal of Walmart as a sales channel rather than from higher average prices. While the prices of Amazon listings rise by 5–6%, the number of units sold falls by 18% because Walmart's listings, including its own retail products, disappear. Sellers that multihomed see their variable profits fall by 42.6%; the variable profits of sellers that singlehome on Amazon fall by 14.6%.

### 7.4 Pass-Through of FBA Cost Efficiency

Amazon has attained substantial cost-side gains from its logistics integration and investment (Houde *et al.*, 2023). However, it is unclear to what extent Amazon passes these gains through to consumers and sellers because Amazon is the dominant platform and can exert market power. In this subsection, I conduct a counterfactual that lowers Amazon's logistics cost efficiency, and I measure how much this change affects consumer and seller welfare. Specifically, I set Amazon's logistics cost equal to the benchmark Walmart logistics cost and solve the full model.

Panel (d) of Figure 9 shows the results. Due to increased costs (about 65%), Amazon's profit drops by 17.9%. Consumers are worse off, by 9.7%, yet the impact on seller profits is relatively negligible. Due to Amazon's reduced logistics efficiency, Walmart's e-commerce profits also increase by 3.6%, which is the biggest welfare gain among market agents. Overall, this result suggests that Amazon's cost-side efficiency gains increase consumer welfare modestly.

[83] https://www.nytimes.com/2024/10/07/technology/google-app-store-epic-games-anticompetitive.html
[84] I follow the procedure of the logistics separation counterfactual.

# 8 Conclusion

This paper investigates the welfare effects of a platform's vertical integration of complementary services, focusing on e-commerce competition between Amazon and Walmart in the U.S. I develop an equilibrium model and quantify the welfare effects of Amazon's logistics integration. Estimating the model with the novel cross-platform data, I find that Amazon's integration benefits consumers and small sellers.

However, I also find anti-competitive effects of integration: the RD estimates and the seller cost estimates show that using FBA reduces sellers' multihoming, which is detrimental to platform competition. I propose policies that target these anti-competitive effects, such as making integrated logistics interoperable, to improve welfare. Overall, promoting platform competition should be a policy priority: as I show, removing competition from a rival platform substantially reduces consumer and seller welfare.

I conclude the paper with directions for future research. Digital platforms extend their role beyond intermediation by integrating complementary services, and this practice calls for regulatory scrutiny. Google and Apple have faced antitrust litigation for tying their payment systems to their app stores (*Epic v. Google*, 2025; *Epic v. Apple*, 2023), and Google for linking its advertising exchange to its ad tools (*United States v. Google*, 2025). These markets are important for the welfare of numerous consumers and businesses that rely on them, and the framework developed here provides a foundation for studying platform integration beyond e-commerce.

In addition, relatively little is known about the dynamics of platform industries, which are often described as a winner-take-all trajectory. Foreclosure through integration at an early stage of industry evolution can have substantial dynamic equilibrium effects and may be a key driver of such trajectories. Understanding the role of integration in shaping dynamic competition between platforms is therefore a first-order issue and could inform the design of appropriate regulation.

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# Appendix

## A Stylized Model

In this section, I present a stylized model of e-commerce markets with integrated logistics to aggregate and delineate the economic forces introduced in the institutional background. The model highlights two main forces. On the seller side, the use of integrated logistics affects not only demand through fulfillment quality, but also costs by hindering sellers' ability to aggregate scale across platforms. This can generate cross-platform competition effects through economies of scale. On the platform side, because logistics are integrated with own platform retail and platform marketplace, the platform sets prices, commissions, and logistics fees while internalizing all of these profits. This internalization generates both pro- and anti-competitive effects, motivating the empirical analysis.

**Economic Environment** There are two platforms, indexed by $f$, with $-f$ denoting the rival platform, and a set of single-product independent sellers indexed by $s \in \mathcal{S}^{\text{Ind}}$.[85] Each seller makes three decisions: whether to enter platform $f$ ($e_{sf} \in \{0,1\}$), whether to use platform $f$'s integrated logistics ($\ell_{sf} \in \{0,1\}$), and what retail price to charge $p_{sf} \in \mathbb{R}_+$. In addition to intermediating transactions for independent sellers, each platform also sells its own product and operates its own logistics service. Accordingly, platform $f$ has three decision variables: an own retail price ($p_f^{\text{1P}} \in \mathbb{R}_+$), an ad valorem commission rate ($\phi_f \in [0,1]$), and a per-unit logistics fee ($\text{fee}_f^{\ell} \in \mathbb{R}_+$).[86] There is also an independent logistics provider, indexed by $o$, which sets only a per-unit logistics fee $\text{fee}_o^{\ell}$.

### A.1 Seller Trade-offs, Logistics Choice, and Economies of Scale

I begin with the seller side of the model. Integrated logistics affects seller decisions through both demand and cost. On the demand side, a platform's logistics service provides high fulfillment quality. On the cost side, logistics choices affect per-unit logistics fees, the ability to aggregate scale across platforms, and the fixed costs of serving multiple marketplaces.

The total profit of seller $s$ across platforms $f$ is

$$\Pi_s = \sum_f \big( \underbrace{p_{sf} \times Q_{sf}}_{\text{Revenue}} - \underbrace{TC_{sf}}_{\text{Total Costs}} \big) \times \underbrace{e_{sf}}_{\text{Entry into } f} \tag{A.1}$$

[85] In the stylized model, I assume each seller offers only a single product, so the product index $j$ of the empirical model coincides with the seller index $s$, and I suppress the time index $t$. This single-product assumption is intended to illustrate key economic forces and will be relaxed to allow for multi-product sellers in the empirical model in Section 4.

[86] Although I assume platform-charged commission rates and logistics fees are uniform across products for illustrative purposes, in practice, these charges are uniform only across sellers but vary across products. Commission rates differ by product category, and fees are determined by a function mapping product size, weight, and category to shipping and storage costs. I introduce heterogeneity in the empirical model, where the product-specific fee is written $\text{fee}_{jf}^{\ell}$, and exploit these variations for the empirical analyses.

where $Q_{sf}$ is the sales quantity of seller $s$ on platform $f$. $TC_{sf}$ represents the total costs, which include both variable and fixed costs.

The total costs consist of variable and fixed costs:

$$TC_{sf}(\ell_{sf}, \bar{Q}_{sf}, W_s, Q_{sf}) = \underbrace{AVC_{sf}(\ell_{sf}, \bar{Q}_{sf}, W_s) \times Q_{sf}(\ell_{sf})}_{\text{Variable Costs } (VC_{sf})} + \underbrace{FC_{sf}(\ell_{sf})}_{\text{Fixed Costs}} \tag{A.2}$$

where $AVC_{sf}$ is the average (or per-unit) variable cost and $FC_{sf}$ is the fixed cost of joining platform $f$. Average variable cost depends on three objects. First, the logistics choice ($\ell_{sf}$) affects variable costs because the logistics service directly charges per-unit logistics fees. The second argument comprises cost shifters, $W_s$, such as wholesale costs. Third, it depends on the seller's effective scale $\bar{Q}_{sf}$, reflecting the fact that both procurement terms and logistics contracts often improve with volume.[87]

The use of integrated logistics affects whether a seller can *aggregate* scale across platforms. I define the seller's effective scale on platform $f$ as

$$\bar{Q}_{sf} := \begin{cases} Q_{sf} & \text{if } \ell_s = 1, \\ Q_{sf} + Q_{s,-f} & \text{if } \ell_s = 0, \end{cases} \tag{A.3}$$

where $\ell_s = \max\{\ell_{sf}, \ell_{s,-f}\}$ indicates whether seller $s$ uses any platform's integrated logistics service. The idea is that once a seller adopts a platform-exclusive fulfillment system, inventory and operations become fragmented across platforms, preventing the seller from fully pooling scale. By contrast, when the seller relies on non-exclusive logistics, sales across platforms can be aggregated.[88]

Integrated logistics also affects fixed costs. On the focal platform, using the platform's own logistics service may reduce setup and operating burdens, lowering the fixed cost of participation. At the same time, because integrated logistics is platform-specific, it may require additional costs to expand to rival platforms relative to using an independent logistics provider; independent providers allow sellers to share fulfillment infrastructure across platforms more easily. Integrated logistics therefore creates a fixed-cost trade-off: it may facilitate participation on one platform while making multihoming more costly.

I investigate the channels that affect sellers' decisions and the trade-offs created by integrated logistics. First, I rewrite the total profit in (A.1) by substituting the total cost equation (A.2) as:

$$\Pi_s = \sum_f (p_{sf} Q_{sf} - VC_{sf} - FC_{sf}) \times e_{sf}$$

[87] For instance, sellers can leverage their sales volume to negotiate better deals with both logistics providers and manufacturers

[88] In practice, integrated logistics need not be fully exclusive. For example, Amazon provides Multi-Channel Fulfillment (MCF) for orders placed outside Amazon. Yet, because Walmart had not accepted MCF for Walmart orders for most of the data period, I assume exclusivity in this paper.

Suppose seller $s$ multihomes ($e_{sf} = e_{s,-f} = 1$) and profit functions are differentiable; we can then obtain the following first-order condition with respect to prices:

$$\frac{\partial \Pi_s}{\partial p_{sf}} = \underbrace{Q_{sf}}_{\text{Direct}} + \underbrace{p_{sf}\frac{\partial Q_{sf}}{\partial p_{sf}}}_{\text{Within-Platform Demand Loss}} + \underbrace{p_{s,-f}\frac{\partial Q_{s,-f}}{\partial p_{sf}}}_{\text{Cross-Platform Substitution}} - \underbrace{\frac{\partial VC_{sf}}{\partial Q_{sf}}\frac{\partial Q_{sf}}{\partial p_{sf}}}_{\text{Within-Platform Cost Effects}} - \underbrace{\frac{\partial VC_{s,-f}}{\partial Q_{s,-f}}\frac{\partial Q_{s,-f}}{\partial p_{sf}}}_{\text{Cross-Platform Cost Spillover}} = 0 \tag{A.4}$$

The first-order condition with respect to $p_{sf}$ shows that pricing is not separable across platforms. Sellers not only consider standard within-platform factors but also internalize cross-platform interactions. Sellers consider business stealing effects across platforms and further take cost-side effects into account, which are linked via economies of scale.

By rearranging the terms in (A.4) and denoting $mc_{sf} = \frac{\partial VC_{sf}}{\partial Q_{sf}}$, we obtain the following pricing equations:

$$\begin{pmatrix} p_{sf} \\ p_{s,-f} \end{pmatrix} = \begin{pmatrix} mc_{sf} \\ mc_{s,-f} \end{pmatrix} - \underbrace{\begin{pmatrix} \frac{\partial Q_{sf}}{\partial p_{sf}} & \frac{\partial Q_{s,-f}}{\partial p_{sf}} \\ \frac{\partial Q_{sf}}{\partial p_{s,-f}} & \frac{\partial Q_{s,-f}}{\partial p_{s,-f}} \end{pmatrix}^{-1}}_{\Omega^{-1}} \begin{pmatrix} Q_{sf} \\ Q_{s,-f} \end{pmatrix} \tag{A.5}$$

A seller's adoption of a platform's integrated logistics ($\ell_{sf} = 1$) introduces four primary economic effects. First, a fee pass-through effect occurs, as the per-unit logistics fee $\text{fee}_f^{\ell}$ directly alters the seller's marginal cost, $mc_{sf}$. Second, adopting integrated logistics triggers distinct demand effects by potentially boosting sales on the focal platform ($Q_{sf} \uparrow$) while simultaneously reducing sales on the rival platform ($Q_{s,-f} \downarrow$), which consequently changes the demand substitution matrix, $\Omega$. Third, the choice of logistics creates scale spillovers; because the seller's effective scale is modified, marginal costs across both platforms ($mc_{sf}$ and $mc_{s,-f}$) may shift. Fourth, by altering both variable and fixed profits, it affects entry and multihoming decisions, which then feed back into equilibrium competition. Logistics choice therefore influences seller outcomes not only mechanically through fees, but also strategically through pricing, market participation, and scale aggregation.

## A.2 Logistics Integration and Platform Pricing

This section discusses how a platform's integration of complementary services changes pricing incentives when the platform simultaneously operates retail, intermediation, and logistics with more details.[89]

I first present the model under logistics integration and then compare it to a benchmark model of logistics separation. In both regimes, platform $f$ performs two roles in the product market: it acts as an own product retailer and intermediates transactions for independent sellers. The difference is whether logistics service is operated inside the platform or by an independent logistics provider.

### A.2.1 Under Logistics Integration

Following the decomposition of platform profit in the empirical model, the profit function of platform $f$, ($\Pi_f$), splits into 1P retail profit ($\Pi_f^{\text{1P}}$), logistics profit ($\Pi_f^{\ell}$), and commission profit ($\Pi_f^{\text{Com}}$):

$$\Pi_f = \Pi_f^{\text{1P}} + \underbrace{\Pi_f^{\ell} + \Pi_f^{\text{Com}}}_{\Pi_f^{\text{Ind}}} \tag{A.6}$$

$$\Pi_f^{\text{1P}} = \left(p_f^{\text{1P}} - mc_f^{\text{1P}}(mc_f^{\ell})\right) Q_f^{\text{1P}} \tag{A.7}$$

$$\Pi_f^{\ell} = \sum_{s \in \mathcal{S}^{\text{Ind}}} \left(\text{fee}_f^{\ell} - mc_f^{\ell}\right) \ell_{sf}\, e_{sf}\, Q_{sf} = \left(\text{fee}_f^{\ell} - mc_f^{\ell}\right) Q_f^{\ell} \tag{A.8}$$

$$\Pi_f^{\text{Com}} = \sum_{s \in \mathcal{S}^{\text{Ind}}} \phi_f\, p_{sf}\, e_{sf}\, Q_{sf} = \phi_f R_f^{\text{Com}} \tag{A.9}$$

where $p_f^{\text{1P}}$, $mc_f^{\text{1P}}$, and $Q_f^{\text{1P}}$ denote the 1P price, retail marginal cost, and sales quantity in platform marketplace $f$; $Q_f^{\ell} \equiv \sum_s \ell_{sf} e_{sf} Q_{sf}$ is fulfilled volume; and $R_f^{\text{Com}} \equiv \sum_s p_{sf} e_{sf} Q_{sf}$ is the commission base.[90] When it comes to the 1P profit, the platform sets its own retail price and earns a margin equal to the difference between that price and its retail marginal cost. Because the platform fulfills its own retail orders using its integrated logistics service, its 1P retail cost ($mc_f^{\text{1P}}$) depends directly on the underlying logistics cost ($mc_f^{\ell}$).

The profit earned from independent sellers, $\Pi_f^{\text{Ind}}$, has two components: logistics profits and commission revenues. In the logistics market chain, platforms charge a unit fee ($\text{fee}_f^{\ell}$) to a seller and fulfillment incurs $mc_f^{\ell}$ to the platform. Whenever a seller enters the platform and adopts the integrated logistics service, the platform fulfills that seller's orders and earns a logistics margin on the resulting volume. In addition, the platform charges an ad valorem commission on independent sellers' transactions, so commission revenue is proportional to independent seller revenue.

[89] Gutierrez (2022) also consider Amazon's problem of price, commission and fee charges, yet my model has distinctions and extensions. I explicitly consider the effects of integration on platform's pricing compared to the separation, and, importantly, competition between platforms.

[90] For simplicity, I assume the platform is a single-product seller in a stylized model, which will be relaxed in the empirical model. Because the fee is uniform here, the empirical model's fulfillment revenue base satisfies $R_f^{\ell} = \overline{\text{fee}}_f^{\ell} Q_f^{\ell}$; the stylized model works directly with the volume $Q_f^{\ell}$.

Following the decision timing in Figure 4, I take partial differentiation with respect to 1P retail prices and total differentiation with respect to the logistics fee and commission rate as:[91]

$$\frac{\partial \Pi_f}{\partial p_f^{1P}} = Q_f^{1P} + \left(p_f^{1P} - mc_f^{1P}(mc_f^{\ell})\right)\frac{\partial Q_f^{1P}}{\partial p_f^{1P}} + \left(\text{fee}_f^{\ell} - mc_f^{\ell}\right)\frac{\partial Q_f^{\ell}}{\partial p_f^{1P}} + \phi_f \frac{\partial R_f^{\text{Com}}}{\partial p_f^{1P}} = 0 \quad \text{(A.10)}$$

$$\frac{\mathrm{d}\Pi_f}{\mathrm{d}\text{fee}_f^{\ell}} = \left(p_f^{1P} - mc_f^{1P}(mc_f^{\ell})\right)\frac{\mathrm{d}Q_f^{1P}}{\mathrm{d}\text{fee}_f^{\ell}} + Q_f^{\ell} + \left(\text{fee}_f^{\ell} - mc_f^{\ell}\right)\frac{\mathrm{d}Q_f^{\ell}}{\mathrm{d}\text{fee}_f^{\ell}} + \phi_f \frac{\mathrm{d}R_f^{\text{Com}}}{\mathrm{d}\text{fee}_f^{\ell}} = 0 \quad \text{(A.11)}$$

$$\frac{\mathrm{d}\Pi_f}{\mathrm{d}\phi_f} = \left(p_f^{1P} - mc_f^{1P}(mc_f^{\ell})\right)\frac{\mathrm{d}Q_f^{1P}}{\mathrm{d}\phi_f} + \left(\text{fee}_f^{\ell} - mc_f^{\ell}\right)\frac{\mathrm{d}Q_f^{\ell}}{\mathrm{d}\phi_f} + R_f^{\text{Com}} + \phi_f \frac{\mathrm{d}R_f^{\text{Com}}}{\mathrm{d}\phi_f} = 0 \quad \text{(A.12)}$$

To understand pricing incentives along the supply chain, I derive the platform's pricing equations by rearranging the first-order conditions:

$$p_f^{1P} = mc_f^{1P}(mc_f^{\ell}) + (\sigma_{f,p}^{1P})^{-1} + \overbrace{\underbrace{\phi_f D_{f,p}^{R,\text{Com}}}_{(+)} + \underbrace{\mu_f^{\ell} D_{f,p}^{\ell}}_{(+)}}^{\text{Cross-chain Internalization}} \quad \text{(A.13)}$$

$$\text{fee}_f^{\ell} = mc_f^{\ell} + (\sigma_{f,\text{fee}}^{\ell})^{-1} + \overbrace{\underbrace{\mu_f^{1P} D_{f,\text{fee}}^{1P}}_{(+)} + \underbrace{\phi_f D_{f,\text{fee}}^{R,\text{Com}}}_{(-)}}^{\text{Cross-chain Internalization}} \quad \text{(A.14)}$$

$$\phi_f = (\sigma_{f,\phi}^{R,\text{Com}})^{-1} + \overbrace{\underbrace{\mu_f^{1P} D_{f,\phi}^{1P}}_{(+)} + \underbrace{\mu_f^{\ell} D_{f,\phi}^{\ell}}_{(-)}}^{\text{Cross-chain Internalization}} \quad \text{(A.15)}$$

where the definitions and expected signs of economic objects are introduced in Table A1. Each pricing rule consists of three components: marginal cost ($mc$), the standard markup term determined by demand elasticities ($\sigma$), and additional internalization terms, profit margin ($\mu$) and diversion ratios ($D$), reflecting how the platform accounts for the effects of pricing decisions across different supply chains.

Because of logistics integration, the platform internalizes how prices, logistics fees, and commission rates affect outcomes across the different supply chains. Platform pricing therefore reflects a key trade-off. On the one hand, the platform competes with independent sellers as a 1P retailer and therefore has an incentive to steal business from them. On the other hand, the platform earns commissions and logistics fees from independent sellers' transactions, which creates an incentive to promote their sales.

Logistics integration generates different pricing incentives for 1P retail prices and commission rates. Consider first the 1P retail pricing in (A.13). When the platform raises its own retail price, some demand is diverted toward independent sellers. Under integration, part of that diverted de-

[91] Note that the terms $\frac{\partial \Pi_f}{\partial p_f^{1P}}\frac{\mathrm{d}p_f^{1P}}{\mathrm{d}\text{fee}_f^{\ell}}$ and $\frac{\partial \Pi_f}{\partial p_f^{1P}}\frac{\mathrm{d}p_f^{1P}}{\mathrm{d}\phi_f}$ are zero by the envelope theorem (i.e., $\frac{\partial \Pi_f}{\partial p_f^{1P}} = 0$ in the pricing stage) and hence omitted.

Table A1: Definitions of Margin, (Semi-)Elasticity and Diversion Ratio Terms

| Measure | Notation | Definition | Expected Sign |
|---|---|---|---|
| 1P Margin | $\mu_f^{1P}$ | $p_f^{1P} - mc_f^{1P}$ | $(+)$ |
| 1P Price Semi-elasticity | $\sigma_{f,p}^{1P}$ | $-\frac{\partial Q_f^{1P}/\partial p_f^{1P}}{Q_f^{1P}}$ | $(+)$ |
| 1P to Commission-Base Diversion | $D_{f,p}^{R,\text{Com}}$ | $-\frac{\partial R_f^{\text{Com}}/\partial p_f^{1P}}{\partial Q_f^{1P}/\partial p_f^{1P}}$ | $(+)$ |
| 1P to Logistics Diversion | $D_{f,p}^{\ell}$ | $-\frac{\partial Q_f^{\ell}/\partial p_f^{1P}}{\partial Q_f^{1P}/\partial p_f^{1P}}$ | $(+)$ |
| Logistics Margin | $\mu_f^{\ell}$ | $\text{fee}_f^{\ell} - mc_f^{\ell}$ | $(+)$ |
| Fee Semi-elasticity | $\sigma_{f,\text{fee}}^{\ell}$ | $-\frac{\mathrm{d}Q_f^{\ell}/\,\mathrm{d}\text{fee}_f^{\ell}}{Q_f^{\ell}}$ | $(+)$ |
| Logistics to 1P Diversion | $D_{f,\text{fee}}^{1P}$ | $-\frac{\mathrm{d}Q_f^{1P}/\,\mathrm{d}\text{fee}_f^{\ell}}{\mathrm{d}Q_f^{\ell}/\,\mathrm{d}\text{fee}_f^{\ell}}$ | $(+)$ |
| Logistics to Commission-Base Diversion | $D_{f,\text{fee}}^{R,\text{Com}}$ | $-\frac{\mathrm{d}R_f^{\text{Com}}/\,\mathrm{d}\text{fee}_f^{\ell}}{\mathrm{d}Q_f^{\ell}/\,\mathrm{d}\text{fee}_f^{\ell}}$ | $(-)$ |
| Commission Semi-elasticity of the Commission Base | $\sigma_{f,\phi}^{R,\text{Com}}$ | $-\frac{\mathrm{d}R_f^{\text{Com}}/\,\mathrm{d}\phi_f}{R_f^{\text{Com}}}$ | $(+)$ |
| Commission-Base to 1P Diversion | $D_{f,\phi}^{1P}$ | $-\frac{\mathrm{d}Q_f^{1P}/\,\mathrm{d}\phi_f}{\mathrm{d}R_f^{\text{Com}}/\,\mathrm{d}\phi_f}$ | $(+)$ |
| Commission-Base to Logistics Diversion | $D_{f,\phi}^{\ell}$ | $-\frac{\mathrm{d}Q_f^{\ell}/\,\mathrm{d}\phi_f}{\mathrm{d}R_f^{\text{Com}}/\,\mathrm{d}\phi_f}$ | $(-)$ |

*Notes.* This table shows the economic measure, notation, definition and expected sign of terms in (A.13), (A.14), and (A.15). The commission base is $R_f^{\text{Com}}$, defined in (A.9). Margins measure per-unit profits in each supply chain, semi-elasticities determine the standard markup components, and diversion ratios capture cross-chain spillovers that arise because the platform simultaneously operates retail, marketplace, and logistics services. Under the standard monopolistic pricing assumption, $\mathrm{d}R_f^{\text{Com}}/\,\mathrm{d}\phi_f < 0$ holds as shown in Appendix A.2.2.

mand generates logistics profits in addition to commission revenue. This creates an additional incentive to soften 1P price competition.

In contrast, as shown in (A.15), increasing the commission rate raises the marginal costs of independent sellers. Sellers respond by increasing prices, which reduces their sales and the associated demand for logistics services. Because the platform internalizes this loss in logistics profit, integration creates downward pressure on commission rates. Both effects also enter the logistics fee pricing rule, generating offsetting pricing pressures on logistics fees.

### A.2.2 Under Logistics Separation

I next consider a benchmark regime with logistics separation. The profit functions of the logistics-segregated platform $f$ and the independent logistics operator $o$ are

$$\begin{aligned}
\Pi_f^{\text{1P}} &= \left(p_f^{\text{1P}} - mc_f^{\text{1P}}(\text{fee}_o^\ell)\right) Q_f^{\text{1P}} \\
\Pi_f^{\text{Com}} &= \phi_f \sum_{s \in \mathcal{S}^{\text{Ind}}} p_{sf}\, e_{sf}\, Q_{sf} = \phi_f R_f^{\text{Com}} \\
\Pi_o^\ell &= \left(\text{fee}_o^\ell - mc_o^\ell\right) \sum_f \sum_{s \in \mathcal{S}^{\text{Ind}}} e_{sf}\, Q_{sf} = \left(\text{fee}_o^\ell - mc_o^\ell\right) \sum_f Q_{of}^\ell = \left(\text{fee}_o^\ell - mc_o^\ell\right) Q_o^\ell
\end{aligned}$$

There are three main changes from the profit functions of the vertically integrated case. First, the platform does not internalize the profit from the logistics market. Second, the platform's 1P retail cost directly depends on the fee charged by the independent logistics operator, $\text{fee}_o^\ell$, instead of the logistics cost $mc_o^\ell$. Lastly, the logistics operator internalizes cross-platform sales, as its service is not exclusive to platform $f$.

As in the integrated case, I use and arrange the first-order conditions to attain the following pricing equations:

$$p_f^{\text{1P}} = mc_f^{\text{1P}}(\text{fee}_o^\ell) + (\sigma_{f,p}^{\text{1P}})^{-1} + \phi_f D_{f,p}^{R,\text{Com}} \tag{A.16}$$

$$\phi_f = (\sigma_{f,\phi}^{R,\text{Com}})^{-1} + \mu_f^{\text{1P}} D_{f,\phi}^{\text{1P}} \tag{A.17}$$

$$\text{fee}_o^\ell = mc_o^\ell + (\sigma_{o,\text{fee}}^\ell)^{-1} \tag{A.18}$$

Under separation, the platform no longer internalizes spillovers into logistics profits, while the logistics operator sets fees to maximize its own profit and internalizes logistics demand across platforms.

Logistics integration reduces double marginalization in 1P retail. This can be seen by comparing the retail-cost terms in the integrated and separated pricing equations, (A.13) and (A.16). Under logistics integration, as shown in (A.13), the platform's retail cost depends directly on the underlying logistics cost. By contrast, under logistics separation in (A.16), the platform's retail cost depends on the logistics fee charged by an independent logistics operator. As a result, separation introduces

an additional markup through the logistics provider, which logistics integration eliminates.

Under separation, the independent logistics provider internalizes demand substitution across platforms: if a fee increase raises consumer prices on one platform, some demand may shift to a rival platform that the same logistics provider also serves. This dampens the pricing pressure coming from the demand response to fee increases and can result in higher markups.

**Commission Rate ($\phi$) and the Commission Base** Let $R_f^{\text{Com}}$ denote the commission base on platform $f$, that is, total independent-seller revenue. Suppose sellers choose prices $p$, and demand depends on prices. The commission base can be written as

$$R_f^{\text{Com}} = p \cdot Q(p).$$

When the platform increases the commission rate $\phi_f$, sellers face higher marginal costs (as the commission acts like an ad valorem tax). Sellers adjust prices $p$, which in turn changes demand. Therefore $R_f^{\text{Com}}$ changes through both price and quantity responses. Differentiating the base with respect to the commission rate yields

$$\frac{\mathrm{d}R^{\text{Com}}}{\mathrm{d}\phi} = \frac{\mathrm{d}p}{\mathrm{d}\phi}Q + p\frac{\mathrm{d}Q}{\mathrm{d}p}\frac{\mathrm{d}p}{\mathrm{d}\phi}.$$

Factoring out $\frac{\mathrm{d}p}{\mathrm{d}\phi}Q$ gives

$$\frac{\mathrm{d}R^{\text{Com}}}{\mathrm{d}\phi} = \frac{\mathrm{d}p}{\mathrm{d}\phi}Q\left(1 + \frac{p}{Q}\frac{\mathrm{d}Q}{\mathrm{d}p}\right),$$

where $\frac{p}{Q}\frac{\mathrm{d}Q}{\mathrm{d}p} = -\varepsilon$ and $\varepsilon$ is the price elasticity of demand. Substituting this expression yields

$$\frac{\mathrm{d}R^{\text{Com}}}{\mathrm{d}\phi} = \frac{\mathrm{d}p}{\mathrm{d}\phi}Q(1-\varepsilon).$$

Therefore, the elasticity of demand ($\varepsilon$) determines the sign of the commission-base response to commission rate increases. When demand is elastic, $1-\varepsilon < 0$, higher commissions increase seller prices and reduce quantity substantially, implying

$$\frac{\mathrm{d}R^{\text{Com}}}{\mathrm{d}\phi} < 0.$$

On the other hand, when demand is inelastic, $1-\varepsilon > 0$, the price increase dominates the quantity reduction, implying

$$\frac{\mathrm{d}R^{\text{Com}}}{\mathrm{d}\phi} > 0.$$

Under the monopolistic pricing assumption, profit maximization is achieved on the elastic part of demand. This implies that the expected sign of $\frac{\mathrm{d}R^{\text{Com}}}{\mathrm{d}\phi}$ is negative.

## A.3 Online and Offline Retails and the Recovery of Marginal Costs

The baseline model focuses on competition between the online marketplaces. Walmart, however, operates an extensive brick-and-mortar network and prices its online catalog while internalizing how online prices divert demand to its own stores. This subsection extends the stylized model of Section A.2 to let Walmart operate an offline retail channel, and shows precisely how the marginal cost in the original model absorbs this cross-channel substitution.

**Walmart's Multi-Channel Profit** Suppose $f = \text{Wmt}$ and add a brick-and-mortar retail channel alongside online 1P retail, the online marketplace for independent sellers, and logistics. Let $\mu^{1P}_{\text{Wmt}} = p^{1P}_{\text{Wmt}} - mc^{1P}_{\text{Wmt}}$ and $\mu^{B}_{\text{Wmt}} = p^{B}_{\text{Wmt}} - mc^{B}_{\text{Wmt}}$ denote the online and offline retail margins, where the superscript $B$ indexes the offline (brick-and-mortar) channel. Walmart's profit is

$$\Pi_{\text{Wmt}} = \underbrace{\mu^{1P}_{\text{Wmt}} Q^{1P}_{\text{Wmt}}}_{\text{online 1P}} + \underbrace{\mu^{B}_{\text{Wmt}} Q^{B}_{\text{Wmt}}}_{\text{offline 1P}} + \underbrace{\mu^{\ell}_{\text{Wmt}} Q^{\ell}_{\text{Wmt}} + \phi_{\text{Wmt}} R^{\text{Com}}_{\text{Wmt}}}_{\text{independent sellers}}. \tag{A.19}$$

The two retail channels are substitutes, so an increase in the online price weakly raises offline demand, $\partial Q^{B}_{\text{Wmt}} / \partial p^{1P}_{\text{Wmt}} \geq 0$.

**Online Pricing under Multi-Channel Retail** Differentiating (A.19) with respect to the online 1P price—the price inverted on the supply side—yields one additional term relative to the single-channel case:

$$Q^{1P}_{\text{Wmt}} + \mu^{1P}_{\text{Wmt}} \frac{\partial Q^{1P}_{\text{Wmt}}}{\partial p^{1P}_{\text{Wmt}}} + \underbrace{\mu^{B}_{\text{Wmt}} \frac{\partial Q^{B}_{\text{Wmt}}}{\partial p^{1P}_{\text{Wmt}}}}_{\text{offline recapture}} + \mu^{\ell}_{\text{Wmt}} \frac{\partial Q^{\ell}_{\text{Wmt}}}{\partial p^{1P}_{\text{Wmt}}} + \phi_{\text{Wmt}} \frac{\partial R^{\text{Com}}_{\text{Wmt}}}{\partial p^{1P}_{\text{Wmt}}} = 0. \tag{A.20}$$

Define the diversion ratio from online 1P retail to the offline channel, following the convention of Table A1,

$$D^{B}_{\text{Wmt},p} \equiv -\frac{\partial Q^{B}_{\text{Wmt}} / \partial p^{1P}_{\text{Wmt}}}{\partial Q^{1P}_{\text{Wmt}} / \partial p^{1P}_{\text{Wmt}}} \geq 0. \tag{A.21}$$

Rearranging (A.20) gives the online pricing equation

$$p^{1P}_{\text{Wmt}} = mc^{1P}_{\text{Wmt}} + (\sigma^{1P}_{\text{Wmt},p})^{-1} + \overbrace{\underbrace{\phi_{\text{Wmt}} D^{R,\text{Com}}_{\text{Wmt},p}}_{(+)} + \underbrace{\mu^{\ell}_{\text{Wmt}} D^{\ell}_{\text{Wmt},p}}_{(+)}}^{\text{intra-platform internalization}} + \overbrace{\underbrace{\mu^{B}_{\text{Wmt}} D^{B}_{\text{Wmt},p}}_{(+)}}^{\text{offline recapture}}. \tag{A.22}$$

The only difference from the integrated 1P pricing rule (A.13) is the final, nonnegative term. It reflects that Walmart prices online above the single-channel optimum because a fraction of the demand lost to an online price increase is recaptured in its own stores at margin $\mu^{B}_{\text{Wmt}}$.

**Implication for Marginal-Cost Recovery** Suppose the estimated model omits the offline channel and therefore recovers the online marginal cost, $\widehat{mc}^{1P}_{\text{Wmt}}$, from the pricing rule (A.13) rather

than (A.22):

$$\widehat{mc}^{\text{1P}}_{\text{Wmt}} = p^{\text{1P}}_{\text{Wmt}} - (\sigma^{\text{1P}}_{\text{Wmt},p})^{-1} - \phi_{\text{Wmt}} D^{R,\text{Com}}_{\text{Wmt},p} - \mu^{\ell}_{\text{Wmt}} D^{\ell}_{\text{Wmt},p}. \tag{A.23}$$

Substituting the true price (A.22) into (A.23), and holding the demand derivatives at their true values, the recovered marginal cost satisfies

$$\widehat{mc}^{\text{1P}}_{\text{Wmt}} = mc^{\text{1P}}_{\text{Wmt}} + \mu^{B}_{\text{Wmt}}\, D^{B}_{\text{Wmt},p} \;\geq\; mc^{\text{1P}}_{\text{Wmt}}. \tag{A.24}$$

The recovered online marginal cost equals the true marginal cost plus the cross-channel internalization term $\mu^{B}_{\text{Wmt}} D^{B}_{\text{Wmt},p}$—the per-unit offline margin times the share of marginally lost online demand recaptured in stores.

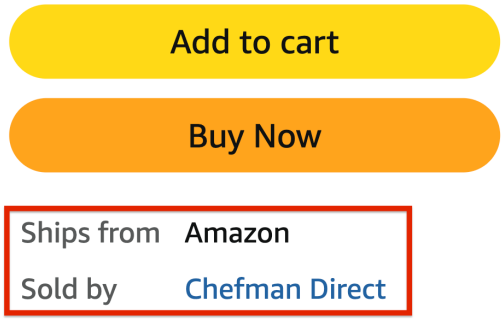
Note that this analysis extends to independent sellers' pricing incentives when they have an alternative retail channel. By the same logic, the marginal costs recovered from the demand estimates and Bertrand-Nash pricing conduct would also absorb the pricing pressure originating from the alternative channel.

# B Data Construction

## B.1 Sources of Data

The key variables for this research are whether a seller uses the platform's integrated logistics service for a given product, and whether a seller is present on Amazon, on Walmart, or on both. Both are collected from public web pages. Figure A1 shows where logistics usage appears on a product page, and Figure A2 shows the seller identifiers I use to link sellers across the two platforms. Table A2 summarizes sources of data and their usage and platform coverage.

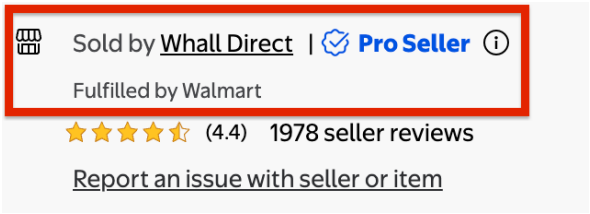
Figure A1: Logistics Service Usage on Product Web Pages



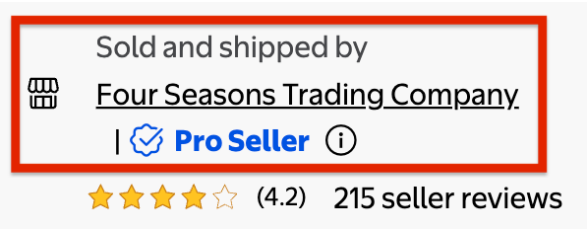

(a) Amazon (FBA)




(b) Amazon (Non-FBA)



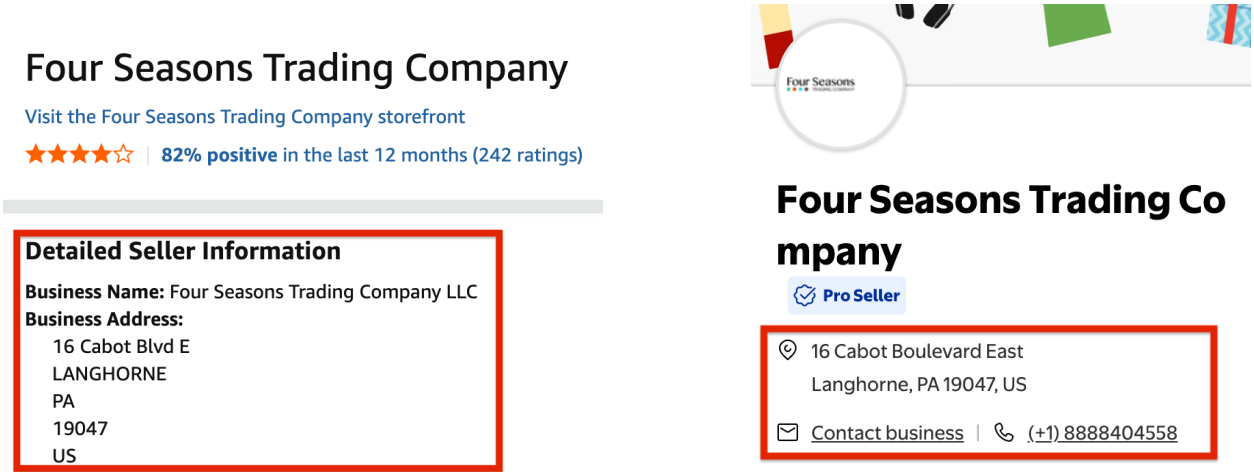

(c) Walmart (WFS)




(d) Walmart (Non-WFS)

*Notes.* This figure shows parts of product web pages on Amazon and Walmart. Panels (a) and (b) show sellers using Amazon logistics (FBA) and independent logistics (Non-FBA), respectively. Panels (c) and (d) show sellers using Walmart logistics (WFS) and independent logistics (Non-WFS), respectively. The boxes mark the text indicating whether the product is fulfilled through the platform's integrated logistics service.

Figure A2: Seller Identifiers




(a) Amazon

(b) Walmart

*Notes.* This figure shows part of a seller web page on Amazon (Panel (a)) and on Walmart (Panel (b)). The boxes mark the business name, address, and phone number, which I use to match sellers across platforms.

Table A2: Summary of Data Sources

| Source | Platform | Content and use |
|---|---|---|
| *Panel A: Full-category panel (January 2021–August 2025)* | | |
| Keepa API | Amazon | Historical prices, offers, and sales ranks; FBA use and Prime eligibility; product characteristics and logistics fees; seller ratings and contact information |
| DataSpark API | Walmart | Walmart listings matched by UPC; historical product and seller data (available through April 2024) |
| BlueCart API | Walmart | Seller business name, address, and phone number, used to match sellers across platforms |
| *Panel B: Kitchen appliance panel (April–June 2025)* | | |
| Web scraping | Both | Weekly snapshots of bestseller and featured listings in 13 subcategories: logistics service, prices, search ranks, and product characteristics |
| Jungle Scout | Amazon | Monthly sales estimates |
| Bizmetrica | Walmart | Monthly sales estimates |
| *Panel C: Consumer panel and other sources* | | |
| Numerator | Multi-retailer | Receipt panel of about 200,000 U.S. households: purchases by retailer, income, and Amazon Prime and Walmart+ membership; used for micro moments and market size |
| Google Trends | – | Monthly search interest by subcategory within the kitchen appliance category, used to scale the market size (Appendix E.3) |
| Posted fee schedules | Both | Amazon FBA and Walmart WFS logistics fees by size tier and shipping weight; UPS and FedEx Ground rates as the independent-logistics benchmark, 2022–2025 (Appendix I) |

*Notes.* This table shows the sources of the data. The full-category panel covers Amazon bestsellers in nine categories and their Walmart matches; the kitchen appliance panel covers both platforms. "Both" denotes Amazon and Walmart; "Multi-retailer" denotes purchases at Amazon, Walmart, and other retailers.

## B.2 Full-category Panel

The primary source of the full-category panel is Keepa.com, whose API grants access to historical data on products, sellers, and their offers within the Amazon marketplace.[92] For instance, it enables us to track the historical price of a product by each seller and identify which offers Amazon has featured over time. Notably, it includes information on whether a seller has opted for Amazon's fulfillment service for shipping and if an offer is Prime eligible. Additionally, it provides the seller's address, historical product and seller ratings, shipping costs, historical sales rank, product characteristics, and fulfillment costs.

Utilizing the Keepa API, I have initially retrieved a list of bestsellers from 9 product categories.[93] Subsequently, I have requested product and seller information for these bestsellers through the API. Then, I match listings on Amazon with those on other retail channels by using the Universal Product Code (UPC) of a product. A UPC is a unique code assigned to each product or variation thereof, meaning that if two products share the same UPC, they are identical. I used both BlueCart API and DataSpark API in order to search for product info through UPCs.

Furthermore, the DataSpark API provides historical information from Walmart side, which is similar to data from Keepa API about Amazon side. I use BlueCart API service that scrapes seller information from Walmart marketplace that contains the address and contact information.

[92] Data from this source have been utilized in previous research, such as Chen and Tsai (2024) and Reimers and Waldfogel (2021).

[93] I follow the category definition in Amazon and categories are Appliances, Electronics, Office Products, Tools & Home Improvement, Home & Kitchen, Sports & Outdoors, Pet Supplies, Beauty & Personal Care, Clothing, Shoes & Jewelry

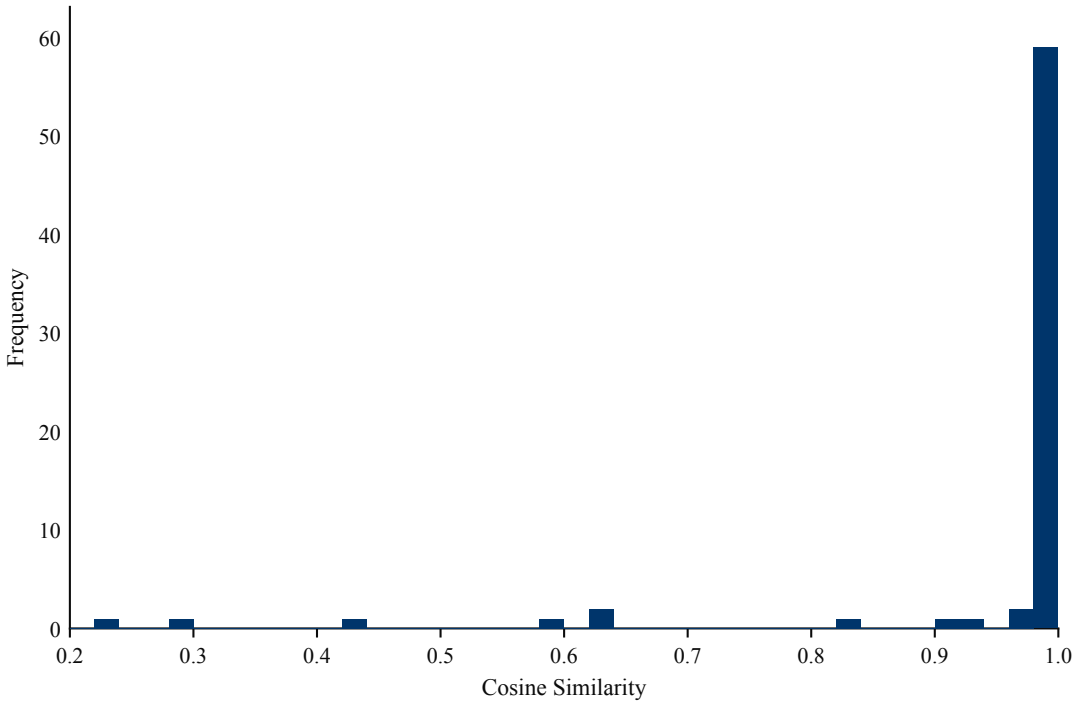
## B.3 Matching Sellers

In both Amazon and Walmart marketplaces, seller information such as name, phone number, email, and physical address is displayed. A key institutional fact is that this contact information is used for business and tax verification purposes in both marketplaces. Therefore, this information is useful to identify the same sellers across these platforms.

I establish a match between an Amazon seller and a Walmart seller if they share the exact same phone number, or if their registered business names or physical addresses are identical. To account for variations in text and quantify the closeness of these matches, I employ a semantic similarity measure. Specifically, I utilize a pre-trained Large Language Model, Sentence-BERT (Reimers and Gurevych, 2019), to generate embeddings of the seller profiles and calculate the cosine similarity between them.

I validate this approach by examining the cosine similarity distribution specifically for the subset of sellers matched via phone numbers, which serves as a ground-truth baseline. The results are shown in Figure A3. I verify that almost every phone-number-matched seller can be matched via the embedding-based method.

Figure A3: Distribution of Cosine Similarity of Phone-number Matched Sellers




*Notes.* This figure shows the histogram of the cosine similarity between Amazon and Walmart sellers who are pairwise matched by their phone numbers. For each of the 70 matched pairs, the similarity is the maximum over the name, business name, and address embeddings.

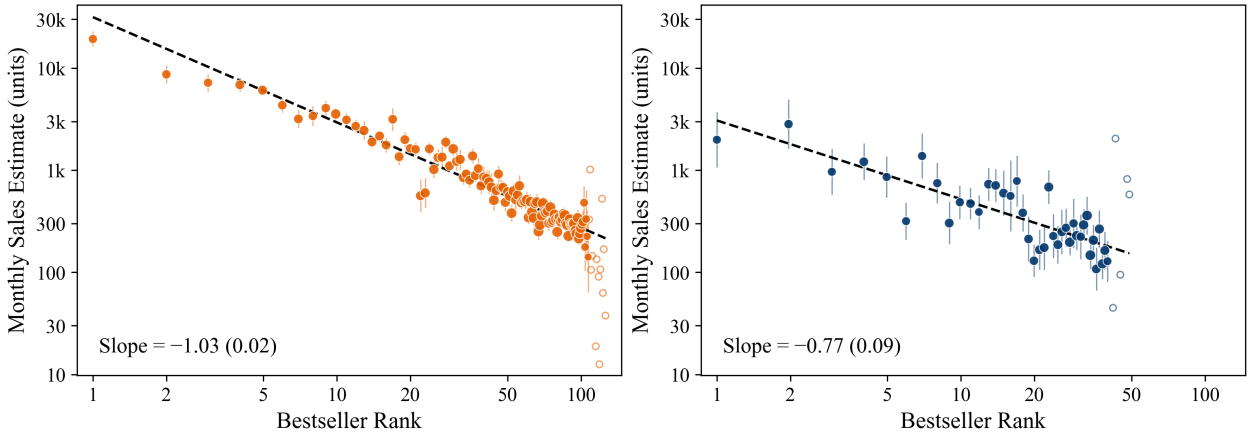
## B.4 Kitchen Appliance Panel

I web scrape bestsellers of each subcategory (e.g., air fryer) on both Amazon and Walmart and use API extensions to gather further details by weekly. I use keyword search on Amazon and Walmart by giving each subcategory as a keyword and scrape appeared products. I sort products by bestsellers and featured and scrape them both.

I use Jungle Scout sales estimates for the designated category to obtain proxy sales data. Jungle Scout is an intelligent service provider for Amazon sellers that uses actual sales records to convert best seller ranks into sales estimates. This service is also used by Yu (2024). For the Walmart-side sales proxy, I similarly use sales estimates provided by Bizmetrica.

To check the validity of the sales estimates obtained through the services, I inspect the relationship between the sales estimates and best seller ranks. A product's best seller rank indicates its relative position when products are sorted by sales volume on either platform. Figure A4 shows these relationships. There is a strong negative relationship on both platforms, although the Walmart side is noisier. This verifies that the sales estimates generally align with best seller ranks across each subcategory.

Figure A4: Bestseller Ranks and Sales Estimates

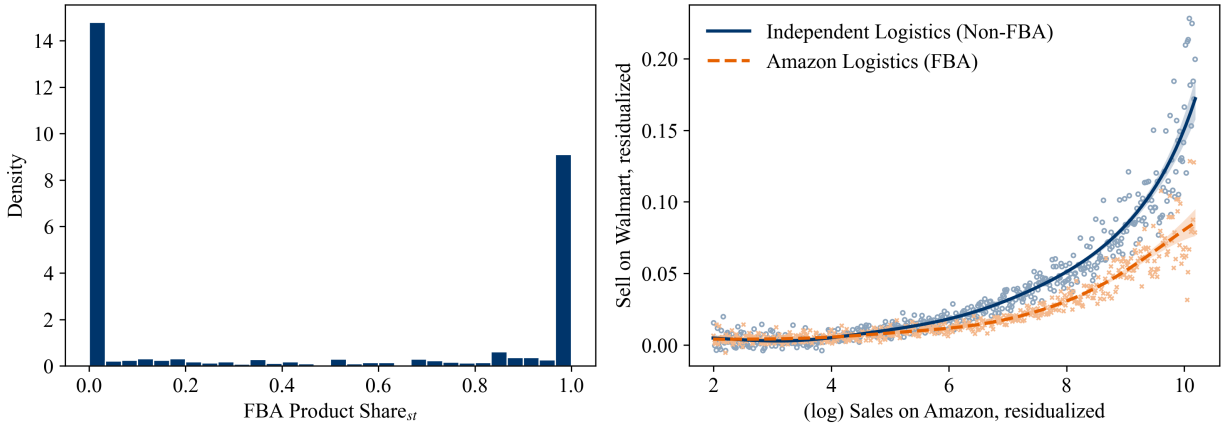



(a) Amazon (b) Walmart

*Notes.* This figure shows the relationship between monthly sales estimates and bestseller ranks on Amazon and Walmart, both on a log scale. Monthly sales estimates are obtained through the JungleScout and Bizmetrica APIs for the Amazon and Walmart sides, respectively. Each marker is the mean of log sales among products at a given rank across subcategories, with sales adjusted for subcategory fixed effects; marker size is proportional to the number of products at that rank, and hollow markers denote ranks with fewer than five products. Vertical bars are standard errors of the bin mean, and the dashed line is the OLS fit of log sales on log rank at the product level, with the slope and its standard error reported in each panel.

# C Additional Descriptive Analyses

Figure A5: Logistics Usage, Multihoming, and Scale on Amazon (Robustness)




(a) FBA Product Share, More Than Five Products

(b) Multihoming and Scale on Amazon, Residualized

*Notes.* Panel (a) shows the distribution of the share of seller $s$'s FBA products in month $t$, restricted to seller-months in which the seller offers more than five products on Amazon. The distribution is bimodal. Panel (b) repeats panel (b) of Figure 1 after residualizing both the "Sells on Walmart" indicator and log seller scale on the average log shipping weight and average log volume of the seller's products and on fixed effects for the seller's main category. Sample means are added back, and residualized log seller scale is trimmed at its 0.5th and 99.5th percentiles. Lines show local quadratic fits with a Gaussian kernel, and shaded areas denote their 95% confidence bands.

## C.1 Regression Analysis of Multihoming on FBA

I regress the following model:

$$\text{Multihome}_{st} = \alpha^{FBA}\text{FBA}_{st} + \alpha_t + \alpha_s + u_{st}$$

where $\text{Multihome}_{st}$ is the dummy indicating Amazon seller $s$ sells on Walmart at month $t$. $\text{FBA}_{st}$ is the dummy indicating seller $s$ uses FBA at month $t$. $\alpha_s$ and $\alpha_t$ are seller and month fixed effects, respectively. Table A3 shows the negative correlation between FBA and multihoming, controlling for the fixed effects.

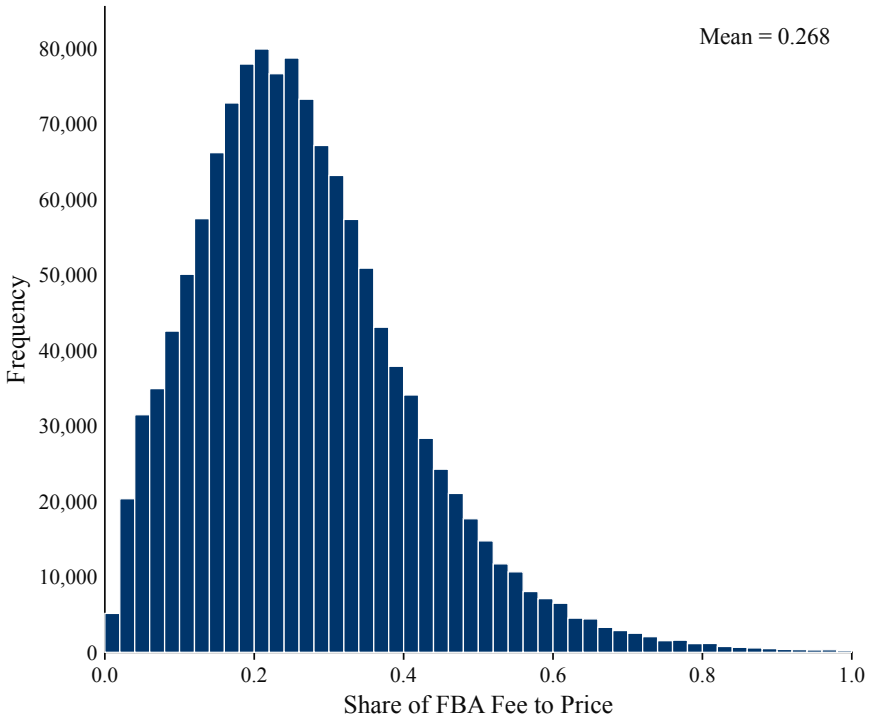
Table A3: Regressing Multihoming on FBA

| | OLS | | | WLS | | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Variables | $\text{Multihome}_{st}$ | | | | | |
| $\text{FBA}_{st}$ | -0.0097*** | -0.0110*** | -0.0020** | -0.1602*** | -0.1660*** | -0.0110** |
| | (0.001) | (0.001) | (0.001) | (0.021) | (0.022) | (0.005) |
| Intercept | 0.0330*** | | | 0.2214*** | | |
| | (0.001) | | | (0.021) | | |
| R-squared | 0.001 | 0.002 | 0.600 | 0.041 | 0.050 | 0.809 |
| Time FE | No | Yes | Yes | No | Yes | Yes |
| Seller FE | No | No | Yes | No | No | Yes |
| Weighted | No | No | No | Yes | Yes | Yes |
| Observations | 882,210 | 882,210 | 882,210 | 882,210 | 882,210 | 882,210 |

*Notes.* This table shows the estimates from the regression analysis of multihoming on FBA with fixed effects. Multihoming is defined as an indicator that seller $s$ appears on Walmart at month $t$. The number of product offerings on Amazon is used for running weighted least squares estimation. The simple and weighted averages of the dependent variable are 0.028 and 0.112. Standard errors clustered at seller-level are in parentheses. ***, **, and * indicate the coefficient estimate is statistically significant at 1, 5, and 10% level, respectively.

# D More on Regression Discontinuity Analysis

Figure A6: Distribution of FBA Fee to Price



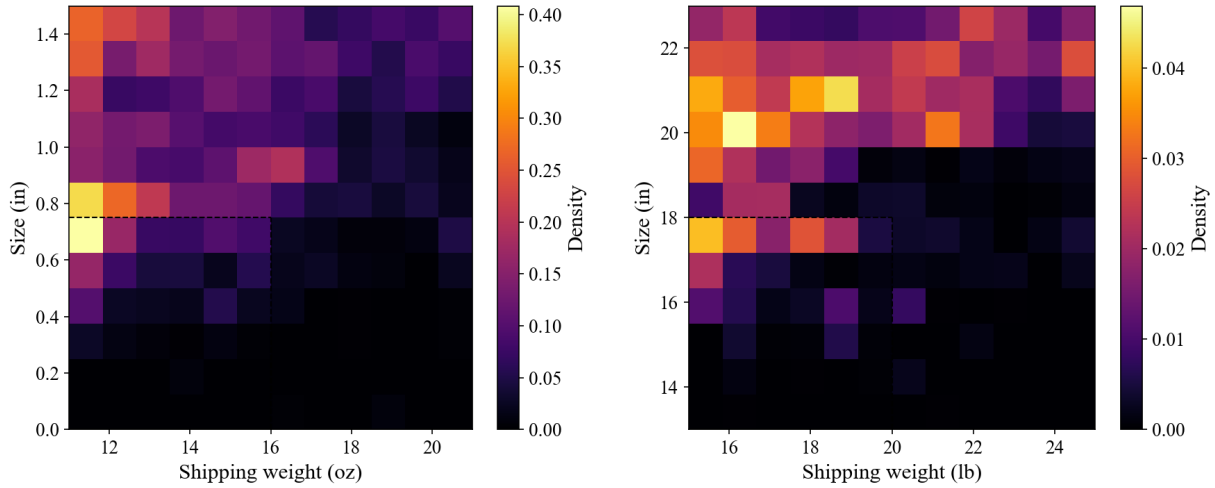

*Notes.* This figure shows the histogram of the FBA fee as a share of the consumer price in the full-category panel, with bins of width 0.02. The sample consists of product-seller pairs at the monthly level, conditional on using FBA. The dashed line marks the mean. FBA fees take a significant portion, on average 26.8%, of overall prices. The number of observations in this figure is 1,300,162.

Table A4: FBA Product Size-Tier

| | | Size | | |
|---|---|---|---|---|
| Size tier | Shipping weight (lb) | Length (in) | Height (in) | Width (in) |
| Small ($T^S$) | $\leq 1$ | $\leq 15$ | $\leq 12$ | $\leq 0.75$ |
| Large ($T^L$) | $\leq 20$ | $\leq 18$ | $\leq 14$ | $\leq 8$ |
| Bulky ($T^B$) | $\leq 50$ | $\leq 59$ | $\leq 33$ | $\leq 33$ |

*Notes.* This table reports the thresholds that define the three FBA product size tiers used in the RD analysis: Small ($T^S$), Large ($T^L$), and Bulky ($T^B$), to which most Amazon products are assigned. Amazon assigns each product to a size tier based on its shipping weight and its size, measured by length, height, and width, and then applies the tier's fee schedule, under which fees monotonically increase along with shipping weight. If any criterion exceeds a threshold, the product is assigned to the next larger tier. For example, a product weighing under one pound but exceeding 15 inches in length is classified as Large rather than Small. The thresholds correspond to Amazon's 2025 fee schedule and Amazon's official names for the tiers are small standard, large standard, and large bulky.

Figure A7: Distributions of Density Estimates Around Product Size-Tier Cutoffs




(a) Small to Large

(b) Large to Bulky

*Notes.* This figure shows the heatmaps of density estimates around size-tier cutoffs defined by product size and shipping weight. Panels (a) and (b) display the distributions around the cutoffs from the small standard tier to the large standard tier and from the large standard tier to the bulky tier, respectively. The black dashed lines trace the boundary of the smaller tier, at 0.75 inches and 16 ounces in Panel (a) and at 18 inches and 20 pounds in Panel (b); a product belongs to the smaller tier only if it lies below both cutoffs. Each cell reports the average kernel density estimates for products in the corresponding size–weight bin, averaged over time as well. The total number of products in Panels (a) and (b) is 1,620 and 3,356, respectively.

Table A5: Density-Discontinuity Tests at the Amazon Logistics Fee Cutoffs

| Cutoff<br>Running variable | Large to Bulky tier<br>Product size (in.) | Small to Large tier<br>Shipping weight (oz.) |
|---|---|---|
| $T$-statistic | −1.31 | −0.39 |
| $p$-value | (0.191) | (0.699) |
| Bandwidths | [0.67, 0.55] | [2.73, 2.73] |
| Effective observations below / above cutoff | 35,437 / 5,691 | 1,840 / 472 |

*Notes.* This table reports tests for a discontinuity in the density of the running variable at each cutoff, using the local polynomial density estimator of Cattaneo *et al.* (2020). Estimates use a local quadratic fit with robust bias correction and a triangular kernel. Bandwidths below and above the cutoff are selected separately by the MSE-optimal selector. Each cell reports the $T$-statistic for the difference in the estimated density above and below the cutoff, the two-sided $p$-value for $H_0 : \lim_{x\to 0^-} f_X(x) = \lim_{x\to 0^+} f_X(x)$ in parentheses, and the selected bandwidths below and above the cutoff in square brackets. Standard errors are computed from a bootstrap that resamples products (200 replications), holding the bandwidths fixed. The last row reports the number of product–month observations below and above the cutoff within the selected bandwidths.

Table A6: Regression Discontinuity Estimates at the Amazon Logistics Fee Cutoffs

| Cutoff<br>Running variable | Large to Bulky tier<br>Product size (in.) | Small to Large tier<br>Shipping weight (oz.) |
|---|---|---|
| FBA fee ($) | 3.243*** | 1.133*** |
| | (0.032) | (0.016) |
| | [0.96, 1.19] | [1.90, 3.14] |
| Has FBA seller | −0.202*** | −0.188*** |
| | (0.014) | (0.025) |
| | [0.22, 3.36] | [5.81, 5.96] |
| Price on Amazon ($) | 8.357*** | 6.379*** |
| | (1.209) | (1.334) |
| | [0.39, 1.71] | [2.63, 3.11] |
| Number of multihoming sellers | 0.100*** | 0.773***† |
| | (0.019) | (0.284) |
| | [0.64, 2.08] | [0.64, 8.24] |
| Observations below / above cutoff | 57,573 / 14,571 | 1,545 / 491 |

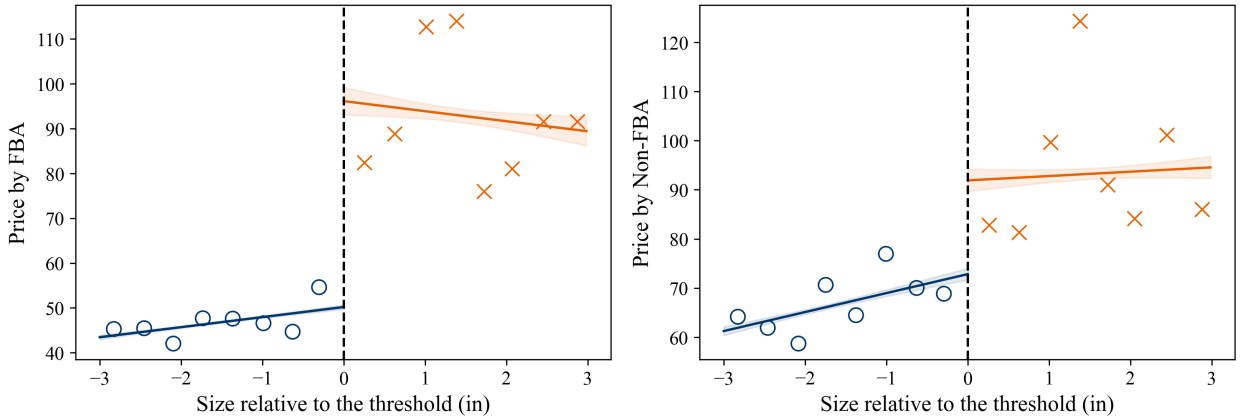
*Notes.* This table reports RD estimates of the discontinuity $\beta_1$ in each outcome at the cutoff from specification (2), estimated by local-linear regressions with a triangular kernel and year fixed effects. Bandwidths below and above the cutoff are selected separately for each outcome and cutoff by the MSE-optimal selector of Calonico *et al.* (2014). Each cell reports the point estimate, its standard error in parentheses, and the selected bandwidths below and above the cutoff in square brackets. "Has FBA seller" is an indicator equal to one if at least one seller of the product uses FBA. † marks an estimate whose bandwidth contains fewer than 10 distinct values of the running variable on one side of the cutoff. The last row reports the number of observations below and above the cutoff within the bandwidths selected for the FBA fee; Table A7 reports them for every outcome, running variable, and cutoff. ***, **, and * indicate statistical significance at the 1, 5, and 10% levels, respectively.

Table A7: Selected Bandwidths and Effective Samples for the RD Estimates

| | Bandwidth $h$ | | Observations within $h$ | |
|---|---|---|---|---|
| Outcome | Below cutoff | Above cutoff | Below cutoff | Above cutoff |
| *Panel A: Large size tier cutoff, product size running variable (in.)* | | | | |
| FBA fee ($) | 0.96 | 1.19 | 57,573 | 14,571 |
| Has FBA seller | 0.22 | 3.36 | 8,153 | 37,117 |
| Price on Amazon ($) | 0.39 | 1.71 | 18,613 | 20,598 |
| Number of multihoming sellers | 0.64 | 2.08 | 17,009 | 9,512 |
| *Panel B: Small size tier cutoff, shipping weight running variable (oz.)* | | | | |
| FBA fee ($) | 1.90 | 3.14 | 1,545 | 491 |
| Has FBA seller | 5.81 | 5.96 | 10,870 | 780 |
| Price on Amazon ($) | 2.63 | 3.11 | 2,309 | 491 |
| Number of multihoming sellers | 0.64 | 8.24 | 58 | 170$^{\dagger}$ |

*Notes.* This table shows the bandwidths selected by the two-sided MSE-optimal selector of Calonico *et al.* (2014) for each outcome and cutoff in Table A6, in the units of the running variable given in each panel heading, and the numbers of observations below and above the cutoff within the selected bandwidths. $^{\dagger}$ marks a bandwidth containing fewer than 10 distinct values of the running variable on one side of the cutoff.

Figure A8: RD Estimates (Prices by Logistics)



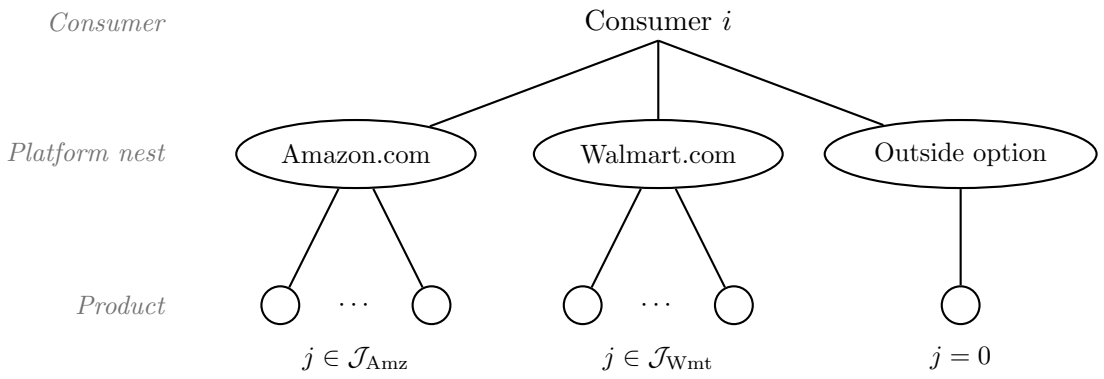

(a) Price by Seller using Amazon Logistics (FBA) (b) Price by Seller using Independent Logistics (Non-FBA)

*Notes.* This figure presents the RD estimates from (2) around the large size-tier cutoff, using the product-size running variable. In each panel, the horizontal axis is the normalized distance to the cutoff: positive values correspond to products in the larger size tier, which are charged higher FBA logistics fees, and negative values correspond to products in the smaller tier. Dots are averages of the outcome within bins of the running variable, and the solid lines are local-linear fits estimated separately and the shaded regions denote 95% confidence intervals. Panel (a) and (b) plot the consumer prices by sellers using FBA and independent logistics, respectively.

# E More on Consumer Demand

## E.1 Nesting Structure

Figure A9: Nest Structure of Consumer Choice




*Notes.* Consumer $i$ chooses among the products listed on Amazon and Walmart or the outside option. I characterize consumers by their income level, platform membership subscription, and idiosyncratic random tastes. Consumers are first divided into four income groups based on their annual income level: under 50K, 50k to 100k, 100k to 150k, and over 150k. I further categorize consumers by their Amazon Prime and Walmart+ membership subscription status. Each platform forms a nest, and the outside option ($j = 0$) is a nest of its own. $\mathcal{J}_f$ denotes the set of products available on platform $f$ for each market. The nesting parameter $\rho$ governs the correlation of unobserved tastes among products within a platform.

## E.2 Deriving and Computing Market Shares

Let $\delta_{jft}$ denote the mean utility and $\mu_{ijft}$ collect the consumer-specific deviations from mean utility generated by the demographic interactions and random coefficients. Then, we can rewrite the utility model in (3) as

$$U_{ijft} = \delta_{jft} + \mu_{ijft} + \bar{\epsilon}_{ift} + (1-\rho)\bar{\epsilon}_{ijft}. \tag{E.1}$$

Define the inclusive value of platform $f$ for consumer $i$,

$$IV_{ift} = (1-\rho)\log \sum_{k \in \mathcal{J}_{ft}} \exp\left(\frac{\delta_{kft} + \mu_{ikft}}{1-\rho}\right),$$

where $\mathcal{J}_{ft}$ is the set of products offered on platform $f$ in month $t$. Given the nested-logit error structure, the probability that consumer $i$ chooses product $j$ on platform $f$ decomposes into the within-platform conditional probability and the probability of selecting platform $f$:

$$\text{sh}_{ijft} = \text{sh}_{ijft|f} \cdot \text{sh}_{ift} = \frac{\exp\big[(\delta_{jft} + \mu_{ijft})/(1-\rho)\big]}{\exp\big[IV_{ift}/(1-\rho)\big]} \cdot \frac{\exp IV_{ift}}{1 + \sum_{f' \in \mathcal{F}} \exp IV_{if't}}. \tag{E.2}$$

The one in the denominator is the inclusive value of the outside option, which is normalized to zero ($\delta_{0t} = \mu_{i0t} = 0$).

I use the Halton sequence method (Halton, 1960) to simulate the random draws, $\nu$, from the standard normal density. This improves the coverage of the integration domain relative to pseudo-random Monte Carlo, yielding a faster convergence rate of $O(R^{-1}(\log aR))$ rather than $O(R^{`1/2})$ where $R$ is the number of draws (Train, 2009). The integration with respect to the normal density in the market share function (8) is approximated as

$$\text{sh}_{jft} \approx \sum_{\tau \in \mathcal{T}} w_{\tau t} \cdot \frac{1}{R} \sum_{r=1}^{R} \text{sh}_{ijft}(\tau, u_r), \tag{E.3}$$

where $u_r = \Phi^{-1}(\eta_r)$ and $\{\eta_r\}_{r=1}^{R} \subset [0,1]$ is a 1-dimensional Halton sequence with coprime prime base. I set $R = 100$.

### E.3 Defining Market Size

The market size is defined as the number of households that purchased any kitchen appliance across all subcategories, $N_t^H$, times the Google Trends search interest for the subcategory, $\text{GoogleTrends}_{ct} \in [0, 1]$. That is,

$$M_{ct} = N_t^H \times \text{GoogleTrends}_{ct} \tag{E.4}$$

To compute $N_t^H$, I project the number of purchasing households in the Numerator dataset for a given month $t$ to represent the total U.S. household population using demographic weights ("DEMO_WEIGHTS") and national factors ("NATIONAL_FACTOR").

To construct $\text{GoogleTrends}_{c(j)t}$, I first obtain daily search popularity data from the Google Trends for each subcategory keyword (e.g., air fryers) and aggregate them into monthly averages.

### E.4 Micro Moments

Following Petrin (2002), I match each micro moment with its model-predicted counterpart. The sample analogues are constructed from the Numerator consumer panel. Let $N_t^H$ be the number of households who purchased any kitchen appliance in month $t$, let $y_{ijft} \in \{0,1\}$ indicate that household $i$ purchases product $j$ on platform $f$ in month $t$, and let $\mathcal{J}_{ft}$ denote the set of products available on platform $f$ at time $t$. For example, the sample analogue of the average log price paid by households in income group $g$ is

$$\hat{\mathbb{E}}[\log p_{jft} \,|\, \text{Inc}_g = 1] = \frac{\sum_t \sum_{i=1}^{N_t^H} \sum_f \sum_{j \in \mathcal{J}_{ft}} y_{ijft} \log p_{jft} \cdot \mathbf{1}\{\text{Inc}_{g,it} = 1\}}{\sum_t \sum_{i=1}^{N_t^H} \mathbf{1}\{\text{Inc}_{g,it} = 1\}}.$$

The remaining moments—average log price conditional on Prime and Walmart+ subscription status, platform choice probabilities conditional on subscription status, and the unconditional second moment of log price—are constructed analogously.

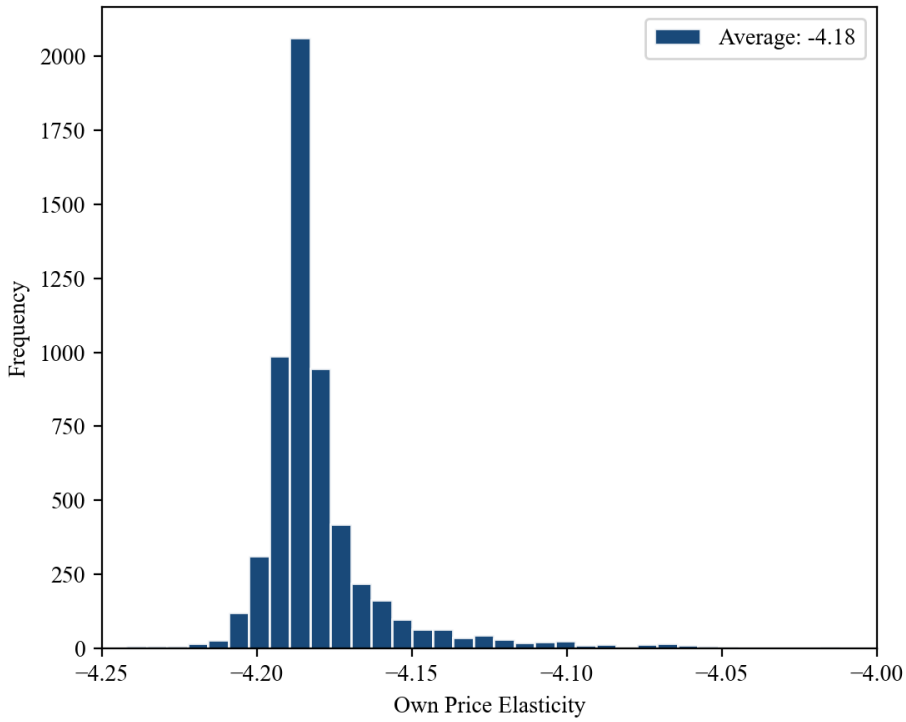
The model-implied counterparts replace observed purchases with predicted choice probabilities. Let $I_t$ denote the set of simulated consumer types in month $t$, $w_{it}$ the integration weight for type $i$, and $s_{ijft}(\boldsymbol{\theta})$ the model-predicted probability that type $i$ purchases product $j$ on platform $f$ in month $t$. The counterpart to the moment above is

$$\mathbb{E}[\log p_{jft} \,|\, \text{Inc}_g = 1] = \frac{\sum_t \sum_{i \in I_t} w_{it} \sum_f \sum_{j \in \mathcal{J}_{ft}} s_{ijft}(\boldsymbol{\theta}) \log p_{jft} \cdot \mathbf{1}\{\text{Inc}_{g,it} = 1\}}{\sum_t \sum_{i \in I_t} w_{it} \sum_f \sum_{j \in \mathcal{J}_{ft}} s_{ijft}(\boldsymbol{\theta}) \cdot \mathbf{1}\{\text{Inc}_{g,it} = 1\}},$$

with the other model-predicted moments constructed in the same manner. I implement these micro moments using the framework of Conlon and Gortmaker (2025), to which I refer the reader for the general construction.

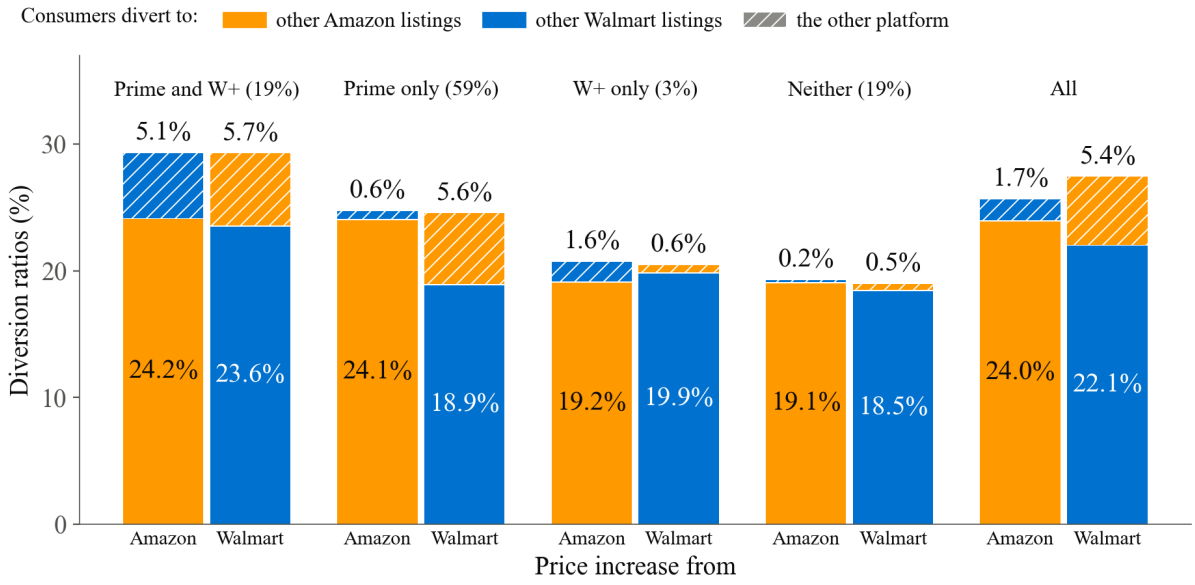
## E.5 Additional Demand Estimation Results

Figure A10: Distribution of Own Price Elasticity




*Notes.* This figure shows the distribution of own-price elasticities implied by the estimated demand model of (3). Each observation is a listing in a market. There are 5,855 product-month observations across 39 subcategory-month markets. The own-price elasticity is evaluated at observed prices. The average is −4.18.

Figure A11: Diversion Ratios by Consumer Membership Status




*Notes.* This figure shows (average) diversion ratios by consumer membership status. Each pair of bars corresponds to a group of consumers defined by their Amazon Prime and Walmart+ subscriptions (population share in parentheses). The last pair pools all consumers. Within a pair, the left bar is a price increase on an Amazon listing and the right bar a price increase on a Walmart listing. The solid segment is the diversion ratio to other listings on the same platform, and the hatched segment is the diversion ratio to listings on the other platform, colored by the destination platform (Amazon orange and Walmart blue). The remainder of each listing's lost sales goes to the outside option. Diversion ratios are computed from the estimated demand model in (3). Income groups are pooled.

# F More on Variable and Marginal Costs

## F.1 Recovering Variable Cost from Marginal Cost Function

Suppose the marginal cost function has the following form

$$MC(Q) = C_0 + h_0(Q) \tag{F.1}$$

By the definition of the variable and marginal cost,

$$VC(Q) = \int_0^Q MC(q)\ \mathrm{d}q = \int_0^Q \left(C_0 + h_0(q)\right) \mathrm{d}q = C_0 Q + \int_0^Q h_0(q)\ \mathrm{d}q$$

Therefore, the average variable cost is

$$AVC(Q) = \frac{VC(Q)}{Q} = C_0 + \frac{\int_0^Q h_0(q)\ \mathrm{d}q}{Q}$$

If the marginal cost function is linear in $Q$ (i.e., $h_0(Q) \equiv \beta_0 Q$), $AVC(Q)$ is also the linear function of $Q$ as:

$$AVC(Q) = C_0 + \frac{\int_0^Q \beta_0 q\ \mathrm{d}q}{Q} = C_0 + \frac{\beta_0}{2} Q$$

This means that the sign of $\beta_0$ determines also the sign of the economies of scale in variable costs.

Suppose the marginal cost function is linear in $\log(Q)$ (i.e., $h_0(Q) \equiv \gamma_0 \log(Q)$). The variable cost is then

$$VC(Q) = \int_0^Q MC(q)\ \mathrm{d}q = \int_0^Q \left(C_0 + \gamma_0 \log q\right) \mathrm{d}q = C_0 Q + \gamma_0 \int_0^Q \log q\ \mathrm{d}q.$$

By the integration by parts:

$$\int_0^Q \log q\ \mathrm{d}q = \Big[q \log q\Big]_0^Q - \int_0^Q q \cdot \tfrac{1}{q}\ \mathrm{d}q = Q \log Q - Q.$$

Therefore,

$$VC(Q) = C_0 Q + \gamma_0 Q(\log Q - 1) = (C_0 - \gamma_0) Q + \gamma_0 Q \log Q,$$

and dividing by $Q$ yields the average variable cost

$$AVC(Q) = \frac{VC(Q)}{Q} = (C_0 - \gamma_0) + \gamma_0 \log Q.$$

The functional form of $AVC$ is therefore identical to that of $MC$ with the same slope coefficient $\gamma_0$.

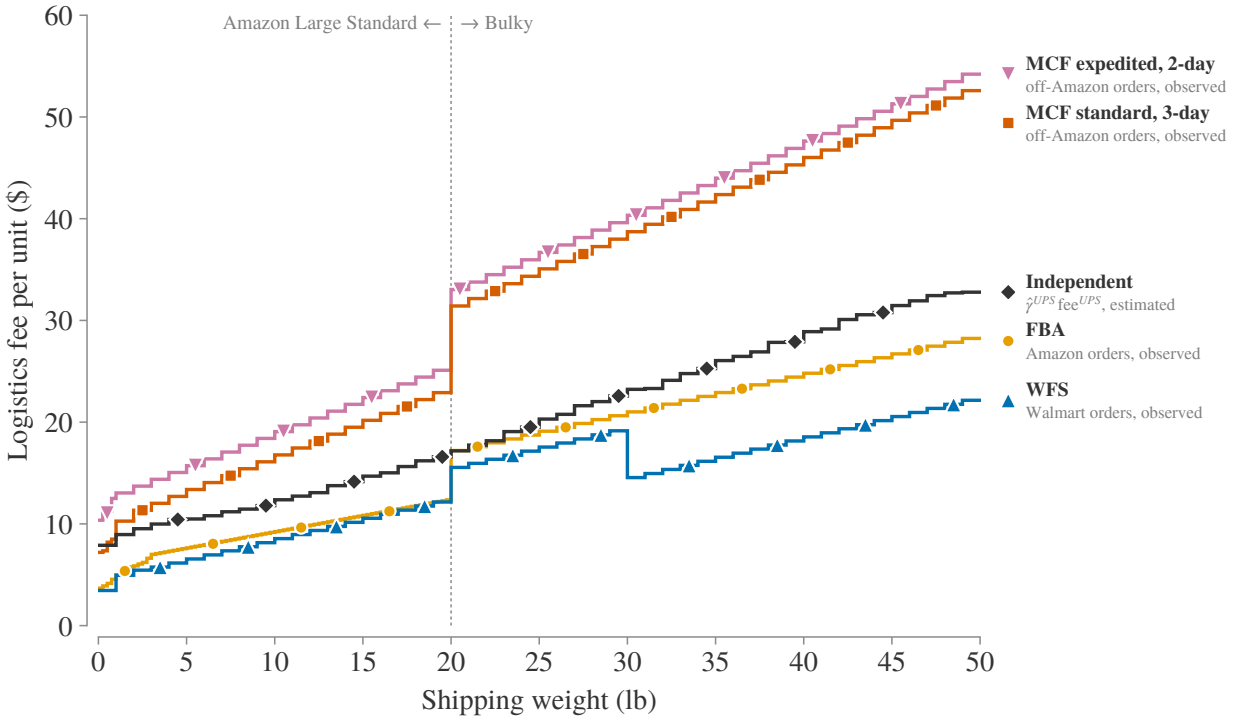
## F.2 Additional Marginal Cost Estimation Results

Table A8: Marginal Cost Function Estimates

| Variables | (1) OLS | (2) OLS | (3) OLS | (4) OLS | (5) IV | (6) IV | (7) IV | (8) IV |
|---|---|---|---|---|---|---|---|---|
| $\text{fee}^{\ell} \times \ell_{sf}$ | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 |
| $\text{fee}^{\text{UPS}} \times (1 - \ell_{sf})$ | 0.601*** | 0.604*** | 0.603*** | 0.607*** | 0.601*** | 0.609*** | 0.598*** | 0.607*** |
| | (0.093) | (0.093) | (0.090) | (0.090) | (0.090) | (0.090) | (0.090) | (0.090) |
| Scale Economy ($\log \bar{Q}$) | −0.597* | | −1.044*** | | −1.811** | | −2.764*** | |
| | (0.320) | | (0.325) | | (0.775) | | (0.615) | |
| Scale Economy ($\bar{Q}$) | | −0.063*** | | −0.118*** | | −0.317*** | | −0.144*** |
| | | (0.022) | | (0.026) | | (0.093) | | (0.047) |
| shwgt | 3.951*** | 3.963*** | 4.242*** | 4.224*** | 4.246*** | 4.205*** | 4.252*** | 4.222*** |
| | (0.227) | (0.226) | (0.260) | (0.260) | (0.259) | (0.259) | (0.259) | (0.259) |
| $\text{shwgt}^2$ | −0.047*** | −0.048*** | −0.046*** | −0.046*** | −0.046*** | −0.046*** | −0.047*** | −0.046*** |
| | (0.009) | (0.009) | (0.009) | (0.009) | (0.009) | (0.009) | (0.009) | (0.009) |
| $\text{shwgt}^3$ | 0.000*** | 0.000*** | 0.000*** | 0.000*** | 0.000*** | 0.000*** | 0.000*** | 0.000*** |
| | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) |
| Stainless Steel | 6.970*** | 7.051*** | 5.129*** | 5.218*** | 5.119*** | 5.347*** | 5.108*** | 5.235*** |
| | (1.157) | (1.152) | (1.360) | (1.365) | (1.353) | (1.368) | (1.354) | (1.361) |
| Aluminum | −3.491** | −3.269** | −0.328 | −0.288 | −0.394 | −0.373 | −0.477 | −0.299 |
| | (1.388) | (1.349) | (1.567) | (1.565) | (1.555) | (1.564) | (1.568) | (1.561) |
| Plastic | 3.826*** | 3.796*** | 1.244 | 1.228 | 1.236 | 1.182 | 1.225 | 1.222 |
| | (1.230) | (1.234) | (1.169) | (1.172) | (1.164) | (1.171) | (1.165) | (1.168) |
| Major Brand | 9.286*** | 8.932*** | 8.730*** | 8.266*** | 9.388*** | 8.996*** | 10.205*** | 8.362*** |
| | (1.946) | (1.884) | (1.913) | (1.898) | (2.176) | (2.009) | (1.976) | (1.894) |
| Category FE | No | No | Yes | Yes | Yes | Yes | Yes | Yes |
| Month FE | No | No | Yes | Yes | Yes | Yes | Yes | Yes |
| IV | No | No | No | No | Own | Own | Rival | Rival |
| Observations | 4,092 | 4,092 | 4,092 | 4,092 | 4,092 | 4,092 | 4,092 | 4,092 |
| $R^2$ | 0.569 | 0.569 | 0.590 | 0.590 | 0.590 | 0.588 | 0.587 | 0.590 |
| Weak IV $F$-stats | | | | | 49.4 | 22.6 | 94.5 | 50.6 |

*Notes.* This table shows coefficient estimates on every regressor of multiple specifications. Column (8) is the main specification, reported in Table 4. Columns (1)–(4) are OLS; columns (5)–(8) are IV with the (log or level) quantity treated as endogenous. Columns (5) and (6) use own-product demand shifters (ratings, reviews, and search rank) together with the category-level Google Trends index. Columns (7) and (8) use instruments that exclude own product characteristics: the leave-one-out sums of within-firm and competing-firm ratings and reviews, the number of first-party retail products, and the Google Trends index. The coefficient on $\text{fee}^{\ell} \times \ell_{sf}$ is constrained to one. The independent logistics fee ($\text{fee}^{\text{UPS}}$) is the posted 2025 UPS Ground rate (zone average) at the product's shipping weight. The unit of scale economy is 1,000 units. Heteroskedasticity-robust standard errors in parentheses. The reported weak-IV $F$ is the rank-based Wald $F$-statistic of Kleibergen and Paap (2006). ***, **, and * indicate the coefficient estimate is statistically significant at 1, 5, and 10% level, respectively.

Figure A12: Logistics Fee Comparison up to 50 lb with Amazon MCF




*Notes.* This figure shows the posted per-unit fees of the platforms' integrated logistics (FBA and WFS), the estimated per-unit cost of independent logistics, and Amazon's multi-channel fulfillment (MCF) service across shipping weights. MCF fees apply to orders placed outside Amazon, such as Walmart orders, and are shown for standard (three-business-day) and expedited (two-business-day) delivery. FBA fees apply to Amazon orders and WFS fees to Walmart orders. All fees are for a single-unit order. Amazon's fees are shown for the large-standard size tier up to 20 lb and the bulky tier above 20 lb. The independent logistics fee is $\hat{\gamma}^{\text{UPS}}\,\text{fee}^{\text{UPS}}$, where $\hat{\gamma}^{\text{UPS}} = 0.607$ is the main specification estimate from Table 4 and $\text{fee}^{\text{UPS}}$ is the posted 2025 UPS Ground rate.

# G Simulation Details

## G.1 Pricing Simulations with Endogenous Marginal Costs

Rewrite the seller's first-order conditions in (12):

$$\mathbf{p} \;=\; \widetilde{\mathbf{mc}}(\bar{\mathbf{Q}}(p)) \;-\; \boldsymbol{\Omega}(\mathbf{p})^{-1}\mathbf{sh}(\mathbf{p}), \tag{G.1}$$

where $\boldsymbol{\Omega}$ is $J \times J$ matrix whose $(j,k)$ element is $1\{j \in \mathcal{J}_s\} \times 1\{k \in \mathcal{J}_s\} \times (-\partial \mathrm{sh}_j/\partial p_k)$. $\mathbf{sh}_s$ stacks the corresponding market shares. $\widetilde{\mathbf{mc}}(\bar{\mathbf{Q}}(p))$ is an equilibrium object.

I use the equation (G.1) to conduct the fixed-point iteration to find counterfactual equilibrium prices. To be specific, suppose $\mathbf{p}^{(n)}$ be the prices at iteration $n$. Then, I update the prices to attain $\mathbf{p}^{(n+1)}$ according to the following rule: $\lambda \in (0,1]$:

$$\mathbf{p}^{(n+1)} \;=\; (1-\lambda)\,\mathbf{p}^{(n)} \;+\; \lambda\Big[\widetilde{\mathbf{mc}}\big(\mathbf{p}^{(n)}\big) - \boldsymbol{\Omega}\big(\mathbf{p}^{(n)}\big)^{-1}\mathbf{sh}\big(\mathbf{p}^{(n)}\big)\Big].$$

Algorithm A1 makes the order of operations explicit; I run the algorithm jointly across all sellers' products in each market. I initialize at observed prices, $\mathbf{p}^{(0)} = \mathbf{p}^{\mathrm{obs}}$, and use $\lambda = 0.5$.

**Algorithm A1** Fixed-point iteration for equilibrium prices with economies of scale

**Require:** market size vector $\mathbf{M}$; functions $\mathbf{sh}(\cdot)$, $\tilde{\mathbf{mc}}(\cdot)$ and $\Omega(\cdot)$; initial $\boldsymbol{p}^{(0)}$; step $\lambda \in (0,1]$; tol $> 0$; max_iter.

$n \leftarrow 0,\ \Delta \leftarrow \infty$
**while** $\Delta >$ tol **and** $n <$ max_iter **do**
  $\mathbf{sh}^{(n)} \leftarrow \mathbf{sh}\big(\mathbf{p}^{(n)}\big)$
  $\mathbf{Q}^{(n)} \leftarrow \mathbf{M}\mathbf{sh}^{(n)}$
  $\tilde{\mathbf{mc}}^{(n)} \leftarrow \tilde{\mathbf{mc}}(\bar{\mathbf{Q}}^{(n)})$
  $\boldsymbol{\Omega}^{(n)} \leftarrow \boldsymbol{\Omega}\big(\boldsymbol{p}^{(n)}\big)$
  $\widehat{\mathbf{p}} \leftarrow \tilde{\mathbf{mc}}^{(n)} + \big(\boldsymbol{\Omega}^{(n)}\big)^{-1}\mathbf{sh}^{(n)}$
  $\mathbf{p}^{(n+1)} \leftarrow (1-\lambda)\,\mathbf{p}^{(n)} + \lambda\,\widehat{\mathbf{p}}$
  $\Delta \leftarrow \big\|\mathbf{p}^{(n+1)} - \mathbf{p}^{(n)}\big\|_{\infty};\quad n \leftarrow n+1$
**end while**
**return** $\mathbf{p}^{(n+1)}$

# H More on Entry and Logistics Choices with Fixed Cost

Table A9: Entry and Logistics Choice Set of Seller $s$

| | Logistics service used | | Decision vector $\mathbf{d}_s$ |
|---|---|---|---|
| Entry decision | Amazon side | Walmart side | $(e_{sAmz}, e_{sWmt}, \ell_{sAmz}, \ell_{sWmt})$ |
| Amazon only | Ind | – | (1, 0, 0, 0) |
| | FBA | – | (1, 0, 1, 0) |
| Walmart only | – | Ind | (0, 1, 0, 0) |
| | – | WFS | (0, 1, 0, 1) |
| Multihoming | Ind | Ind | (1, 1, 0, 0) |
| | FBA | Ind | (1, 1, 1, 0) |
| | Ind | WFS | (1, 1, 0, 1) |
| | FBA | WFS | (1, 1, 1, 1) |

*Notes.* This table shows the eight feasible entry and logistics decisions available to an independent seller $s$ in the first stage (Section 4.2.2). $e_{sf} = 1$ if the seller enters platform $f$, and $\ell_{sf} = 1$ if it uses the platform's integrated logistics on that platform (FBA on Amazon, WFS on Walmart). Ind denotes independent logistics and – a platform the seller does not enter. Combinations with $\ell_{sf} = 1$ and $e_{sf} = 0$ are infeasible and excluded. Rows are ordered as in Figure 7 and Table A10.

## H.1 Identification Detail

Suppose a seller $s$ chooses $\mathbf{d}_s^* \in \mathcal{D}$ as an optimal decision as in (15). Holding others' decision fixed as $\mathbf{d}_{-s}$, the revealed preference rationale with the decision rule in (15) implies that for any deviation $\mathbf{d}_s' \neq \mathbf{d}_s^*$,

$$\begin{aligned} &\mathbb{E}_{(\xi,\omega)}\left[\Pi_s(\mathbf{d}_s^*, \mathbf{d}_{-s})\right] \geq \mathbb{E}_{(\xi,\omega)}\left[\Pi_s(\mathbf{d}_s', \mathbf{d}_{-s})\right] \\ \Leftrightarrow\ &\mathbb{E}_{(\xi,\omega)}\left[\pi_s(\mathbf{d}_s^*, \mathbf{d}_{-s}))\right] - \bar{FC}_s(\mathbf{d}_s^*) - \nu_s(\mathbf{d}_s^*) \geq \mathbb{E}_{(\xi,\omega)}\left[\pi_s(\mathbf{d}_s', \mathbf{d}_{-s}))\right] - \bar{FC}_s(\mathbf{d}_s') - \nu_s(\mathbf{d}_s') \end{aligned} \tag{H.1}$$

where $\bar{FC}_s(\mathbf{d}) = \sum_{j\in\mathcal{J}_s} \bar{FC}(\mathbf{d}) \equiv |\mathcal{J}_s|\bar{FC}(\mathbf{d})$ and $\nu_s(\mathbf{d}) = \sum_{j\in\mathcal{J}_s} \nu_j(\mathbf{d})$ for any $\mathbf{d} \in \mathcal{D}$. $|\mathcal{J}_s|$ means the number of products in the set $\mathcal{J}_s$.

For any pair $(\mathbf{d}, \mathbf{d}') \in \mathcal{D} \times \mathcal{D}$, denote $\Delta\bar{\pi}_s(\mathbf{d}, \mathbf{d}') \equiv \mathbb{E}_{(\xi,\omega)}\left[\pi_s(\mathbf{d}, \mathbf{d}_{-s}))\right] - \mathbb{E}_{(\xi,\omega)}\left[\pi_s(\mathbf{d}', \mathbf{d}_{-s}))\right]$ and $FC_s(\mathbf{d}) - FC_s(\mathbf{d}') \equiv \Delta FC_s(\mathbf{d}, \mathbf{d}') = \Delta\bar{FC}_s(\mathbf{d}, \mathbf{d}') + \Delta\nu_s(\mathbf{d}, \mathbf{d}')$. We can attain the first base inequality condition for identifying fixed costs by rearranging the terms in (H.1) as:

$$\Delta\bar{\pi}_s(\mathbf{d}, \mathbf{d}') \geq \Delta\bar{FC}_s(\mathbf{d}, \mathbf{d}') + \Delta\nu_s(\mathbf{d}, \mathbf{d}') \qquad \text{for } \mathbf{d} = \mathbf{d}_s^* \text{ and } \mathbf{d}' \neq \mathbf{d}_s^* \tag{H.2}$$

This inequality means that the differences between the expected variable profits from the observed optimal choice $d_s^*$ and any deviation $d_s'$ is larger than incurred fixed cost differences of them. We can similarly formulate the second base inequality by using the flip side:

$$\Delta\bar{\pi}_s(\mathbf{d}, \mathbf{d}') \leq \Delta\bar{FC}_s(\mathbf{d}, \mathbf{d}') + \Delta\nu_s(\mathbf{d}, \mathbf{d}') \qquad \text{for } \mathbf{d} \neq \mathbf{d}_s^* \text{ and } \mathbf{d}' = \mathbf{d}_s^* \tag{H.3}$$

which comes from the decision rule that every non-optimal decision is weakly dominated by the optimal decision.

Although these inequalities (H.2) and (H.3) are highly informative as an upper and lower bound of $\Delta\bar{FC}_s(\mathbf{d}, \mathbf{d}')$, the *selection bias* hinders us from directly using the above inequality conditions. Specifically, there can be bias from the fact that *conditional* expectation of differences of structural error term is different from *unconditional* expectation of them as $\mathbb{E}\left[\Delta\nu_s(\mathbf{d}, \mathbf{d}')|d = d^*\right] \neq \mathbb{E}\left[\Delta\nu_s(\mathbf{d}, \mathbf{d}')\right] = 0$ and $\mathbb{E}\left[\Delta\nu_s(\mathbf{d}, \mathbf{d}')|d \neq d^*\right] \neq \mathbb{E}\left[\Delta\nu_s(\mathbf{d}, \mathbf{d}')\right] = 0$. Notably, given that we have identified demand and marginal cost functions, which in turn allows us to treat expected variable profits as observed quantities, this selection bias is the only remaining identification challenge. To circumvent this, we need an additional assumption to take unconditional expectations on the structural error terms.

I follow the approach by Eizenberg (2014) and assume that the support of differences in fixed costs falls within the range of the supremum and infimum of expected variable profit differences:

$$\Delta FC_s(\mathbf{d}, \mathbf{d}') \subset \left[\inf_s \Delta\bar{\pi}_s(\mathbf{d}, \mathbf{d}'), \sup_s \Delta\bar{\pi}_s(\mathbf{d}, \mathbf{d}')\right] \text{ for any } s \in \mathcal{S} \tag{H.4}$$

With this assumption, I can define the following lower and upper bounds: for any choice $\mathbf{d} \in \mathcal{D}$

and its deviation $\mathbf{d}' \in \mathcal{D} \setminus \{\mathbf{d}\}$,

$$\Delta FC_s^L = \begin{cases} \Delta\bar{\pi}_s(\mathbf{d}, \mathbf{d}') & \text{if } \mathbf{d} \neq \mathbf{d}_s^* \\ \inf_s \Delta\bar{\pi}_s(\mathbf{d}, \mathbf{d}') & \text{if } \mathbf{d} = \mathbf{d}_s^* \end{cases} \qquad \Delta FC_s^U = \begin{cases} \Delta\bar{\pi}_s(\mathbf{d}, \mathbf{d}') & \text{if } \mathbf{d} = \mathbf{d}_s^* \\ \sup_s \Delta\bar{\pi}_s(\mathbf{d}, \mathbf{d}') & \text{if } \mathbf{d} \neq \mathbf{d}_s^* \end{cases} \tag{H.5}$$

This support restriction allows me to construct following inequality for any $d$ and $d'$ as

$$\Delta FC_s^L(\mathbf{d}, \mathbf{d}') \leq \Delta FC_s(\mathbf{d}, \mathbf{d}') \leq \Delta FC_s^U(\mathbf{d}, \mathbf{d}') \tag{H.6}$$

As this inequality holds for every seller $s$, we can take *unconditional* expectations:

$$\mathbb{E}\left[\Delta FC_s^L(\mathbf{d}, \mathbf{d}')\right] \leq \Delta\overline{FC}_s(\mathbf{d}, \mathbf{d}') \leq \mathbb{E}\left[\Delta FC_s^U(\mathbf{d}, \mathbf{d}')\right] \tag{H.7}$$

where $\Delta\overline{FC}_s(\mathbf{d}, \mathbf{d}') = \mathbb{E}\left[\Delta FC_s(\mathbf{d}, \mathbf{d}')\right] = \mathbb{E}\left[\Delta\overline{FC}_s(\mathbf{d}, \mathbf{d}') + \Delta\nu_s(\mathbf{d}, \mathbf{d}')\right]$. I can use this inequality (H.7) with unconditional expectation to construct a moment inequality estimator for the fixed cost parameters.

### H.1.1 Estimation of the Fixed Costs

**Sample Analogue** I form the sample analogue of (H.7) at the product-month level. I first compute the profit difference $\Delta\bar{\pi}_s(\mathbf{d}, \mathbf{d}')$ for every pair of $(\mathbf{d}, \mathbf{d}') \in \mathcal{D}$, holding others' decision vectors fixed at the observed choice. According to (H.5), I compute the $\Delta FC_s^L$ and $\Delta FC_s^U$ by taking minimum and maximum among computed $\Delta\bar{\pi}_s(\mathbf{d}, \mathbf{d}')$. This allows me to compute the sample analogue version of (H.7) for all $s \in \mathcal{S}^{\text{Ind}}$ as:

$$\begin{aligned}
&\Delta\overline{FC}_s^L(\mathbf{d}, \mathbf{d}') \;\leq\; \overline{FC}_s(\mathbf{d}, \boldsymbol{\theta}) - \overline{FC}_s(\mathbf{d}', \boldsymbol{\theta}) \;\leq\; \Delta\overline{FC}_s^U(\mathbf{d}, \mathbf{d}') && \Leftrightarrow \\
&\Delta\overline{FC}_s^L(\mathbf{d}, \mathbf{d}') \;\leq\; |\mathcal{J}_s| \left(\overline{FC}(\mathbf{d}, \boldsymbol{\theta}) - \overline{FC}(\mathbf{d}', \boldsymbol{\theta})\right) \;\leq\; \Delta\overline{FC}_s^U(\mathbf{d}, \mathbf{d}') && \Leftrightarrow \\
&\frac{\Delta\overline{FC}_s^L(\mathbf{d}, \mathbf{d}')}{|\mathcal{J}_s|} \;\leq\; \overline{FC}(\mathbf{d}, \boldsymbol{\theta}) - \overline{FC}(\mathbf{d}', \boldsymbol{\theta}) \;\leq\; \frac{\Delta\overline{FC}_s^U(\mathbf{d}, \mathbf{d}')}{|\mathcal{J}_s|} && \text{(H.8)}
\end{aligned}$$

Therefore, for each $(\mathbf{d}, \mathbf{d}')$, I can take average of lower and upper bounds of (H.8) and define

$$\begin{aligned}
\Delta\overline{FC}^L_{(\mathbf{d},\mathbf{d}')} &\equiv \frac{1}{N_{(\mathbf{d},\mathbf{d}')}} \sum_{s \in \{s | \mathbf{d}_s^* \in \{\mathbf{d},\mathbf{d}'\}\}} \frac{\Delta\overline{FC}_s^L(\mathbf{d}, \mathbf{d}')}{|\mathcal{J}_s|} \\
\Delta\overline{FC}^U_{(\mathbf{d},\mathbf{d}')} &\equiv \frac{1}{N_{(\mathbf{d},\mathbf{d}')}} \sum_{s \in \{s | \mathbf{d}_s^* \in \{\mathbf{d},\mathbf{d}'\}\}} \frac{\Delta\overline{FC}_s^U(\mathbf{d}, \mathbf{d}')}{|\mathcal{J}_s|}
\end{aligned}$$

where $N_{(\mathbf{d},\mathbf{d}')}$ denotes the effective sample size that excludes sellers whose observed choices are neither $\mathbf{d}$ nor $\mathbf{d}'$. These give us both lower and upper bounds on the difference in the fixed costs as:

$$\Delta\overline{FC}^L_{(\mathbf{d},\mathbf{d}')} \;\leq\; \Delta\overline{FC}_{(\mathbf{d},\mathbf{d}')} \;\leq\; \Delta\overline{FC}^U_{(\mathbf{d},\mathbf{d}')} \qquad \text{(H.9)}$$

which we can construct linear programming estimator for the moment inequality estimation.

## H.2 Additional Fixed Cost Estimation Results

Table A10: Fixed-Cost Estimates by Entry and Logistics Decision: Estimated Sets and Confidence Intervals

| | Logistics service used | | Estimated set of $\overline{FC}(\mathbf{d})$ | 95% confidence interval |
|---|---|---|---|---|
| Entry decision | Amazon side | Walmart side | ($1,000) | ($1,000) |
| Amazon only | Ind | – | [ 0.0, 0.0] | [ 0.0, 0.0] |
| | FBA | – | [−17.5, 47.9] | [−43.9, 78.7] |
| Walmart only | – | Ind | [−31.7, 0.0] | [−59.0, 10.4] |
| | – | WFS | [−31.7, 19.0] | [−59.0, 51.6] |
| Multihoming | Ind | Ind | [ −7.3, 56.0] | [−36.3, 100.3] |
| | FBA | Ind | [ 33.7, 76.8] | [ 4.8, 115.6] |
| | Ind | WFS | [ 3.4, 35.1] | [−25.1, 68.0] |
| | FBA | WFS | [ 15.6, 47.6] | [−12.9, 90.4] |

*Notes.* This table shows the estimated sets of the mean fixed costs $\overline{FC}(\mathbf{d})$ for each entry and logistics decision $\mathbf{d}$, and their 95% confidence intervals, in $1,000 per product-month. The estimates are normalized and relative to $\overline{FC} \equiv 0$ for the Amazon-only decision with independent logistics. The logistics columns report the service used on each platform: Ind, FBA, WFS and – denote independent logistics, Amazon logistics (Fulfillment by Amazon), Walmart logistics (Walmart Fulfillment Service), and platform not entered. Confidence intervals follow Imbens and Manski (2004) with 5,000 bootstrap resamples of sellers. The sets are those plotted in Figure 7; see its notes and Appendix Section H.1.1 for the estimation details.

# I Analysis on Fee Schedules of Amazon, UPS and FedEx

## I.1 Correlations of Fee Changes

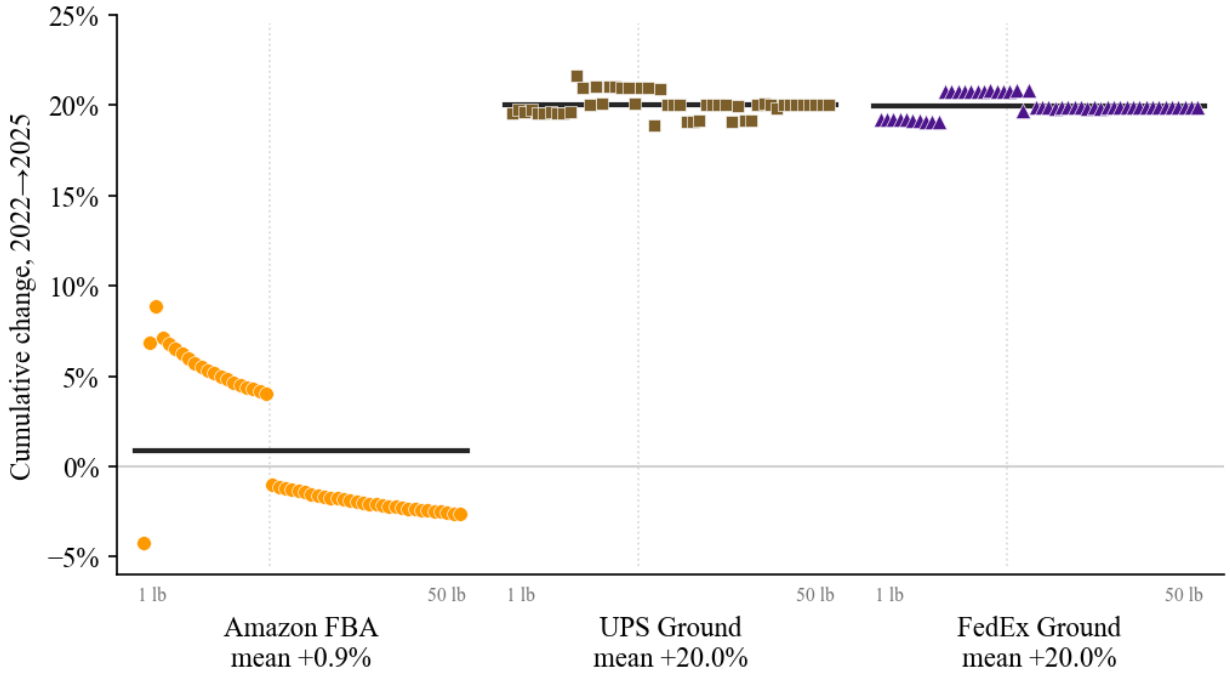
In this section, I examine correlations between the change of FBA fees and that of shipping rates of other major carriers: UPS and FedEx. I use the posted fee schedules of FBA, UPS and FedEx for the years $t \in \{2022, 2023, 2024, 2025\}$. I evaluate them on a common grid, so the unit of observation is a parcel weight class $j \in \{1, \ldots, 50\}$ lb in year $t$. Each cell records Amazon's (non-peak) logistics fee $\text{fee}^{\ell}_{jt\text{Amz}}$ and the UPS and FedEx Ground rates $\text{fee}^{\text{UPS}}_{jt}$ and $\text{fee}^{\text{FedEx}}_{jt}$, evaluated at zone 5 for a parcel.

For parcel $j$ and logistics provider $k \in \{\text{Amz}, \text{UPS}, \text{FedEx}\}$, let the cumulative rate change be

$$g^k_j \;=\; 100\left(\frac{\text{fee}^k_{j,2025}}{\text{fee}^k_{j,2022}} - 1\right), \qquad \text{fee}^{\text{Amz}}_{jt} \equiv \text{fee}^{\ell}_{jt\text{Amz}}. \tag{I.1}$$

Figure A13 shows the cumulative changes across parcel weights. The carriers co-move with one another, not with Amazon. Amazon's per-parcel changes are correlated little with either carrier's, while the two carriers are strongly correlated with each other.

Figure A13: Cumulative Fee Changes by Parcel Weight, 2022–2025




*Notes.* This figure shows the cumulative changes in logistics fees from 2022 to 2025, $g_j^k$, across shipping weights for Amazon's logistics service (FBA), UPS Ground, and FedEx Ground. Each column corresponds to one logistics service, and within each column, weight classes $j \in \{1, \ldots, 50\}$ lb run from 1 lb (left) to 50 lb (right). Orange circles, brown squares, and purple triangles denote FBA, UPS, and FedEx, respectively. The black horizontal lines mark the mean change of each service across weight classes. The dotted vertical lines mark the 20 lb shipping-weight cutoff from the large standard tier to the bulky tier of FBA (the small oversize tier before 2024). FBA fees are non-peak fees, and the UPS and FedEx Ground rates are evaluated at zone 5.

## I.2 Panel Regressions

I further conduct the panel regression analysis to control for weight class and time trend. Let $\text{fee}_{jt}^{k}$ denote carrier rate. I estimate the two-way fixed effects specification for $k \in \{\text{UPS}, \text{FedEx}\}$ as:

$$\log(\text{fee}_{jt}^{k}) = \alpha_j + \gamma_t + \eta \log(\text{fee}_{jt\text{Amz}}^{\ell}) + u_{jt}, \tag{I.2}$$

where $\alpha_j$ and $\gamma_t$ are fixed effects. Table A11 reports the results. The estimated elasticities are $\hat{\eta} = 0.017$ (s.e. $= 0.024$) for UPS and $\hat{\eta} = -0.001$ (s.e. $= 0.021$) for FedEx, which are statistically insignificant.

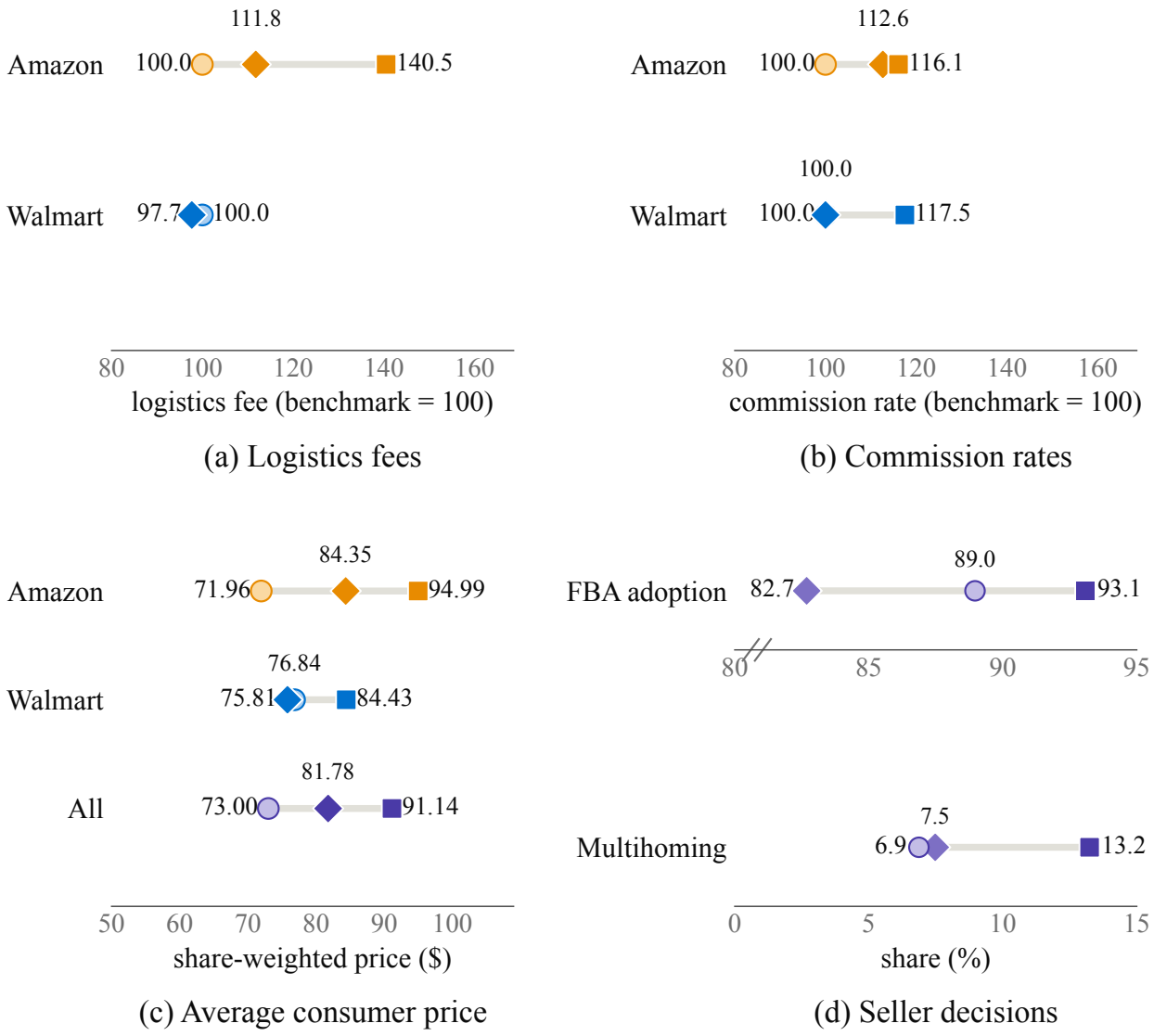
Table A11: Panel Regressions of Carrier Rates on Amazon's Logistics Fee

| Variables | (1) $\log(\text{fee}_{jt}^{\text{UPS}})$ | (2) $\log(\text{fee}_{jt}^{\text{FedEx}})$ |
|---|---|---|
| $\log(\text{fee}_{jt\text{Amz}}^{\ell})$ | 0.017 | −0.001 |
| | (0.024) | (0.021) |
| Weight class FE | Yes | Yes |
| Year FE | Yes | Yes |
| Within $R^2$ | 0.009 | 0.000 |
| Observations | 200 | 200 |

*Notes.* This table reports the two-way fixed effects estimates of the elasticity $\eta$ in (I.2). The unit of observation is a parcel weight class $j \in \{1, \ldots, 50\}$ lb in year $t \in \{2022, \ldots, 2025\}$. Amazon's logistics fee is the non-peak FBA fee, and the UPS and FedEx Ground rates are evaluated at zone 5. Standard errors clustered by weight class in parentheses. $^{***}$, $^{**}$, and $^{*}$ indicate the coefficient estimate is statistically significant at 1, 5, and 10% level, respectively.

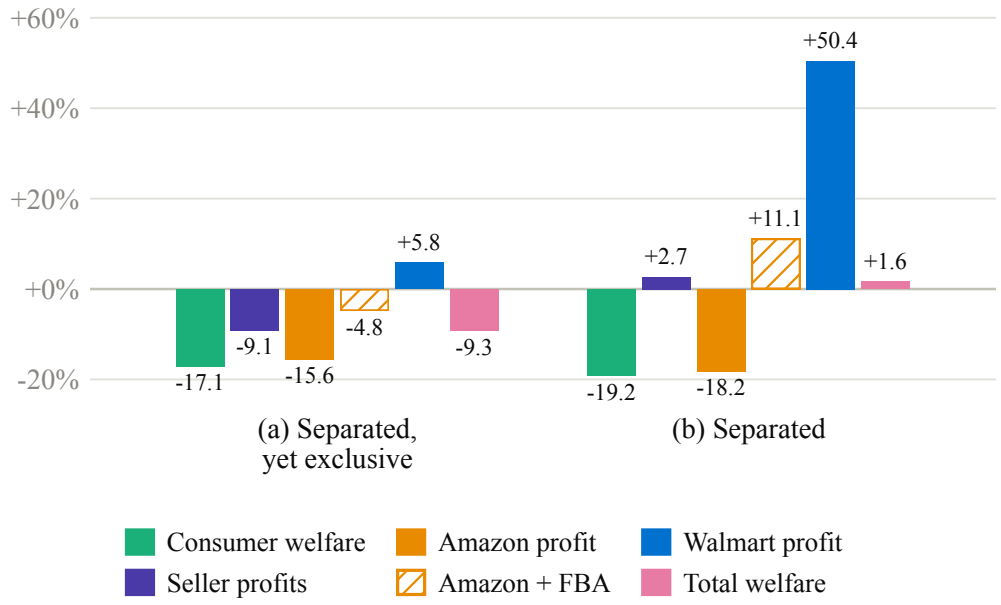
# J More on Counterfactuals

Figure A14: Decomposing FBA Separation Effects: Pricing, FBA Adoption, and Multihoming

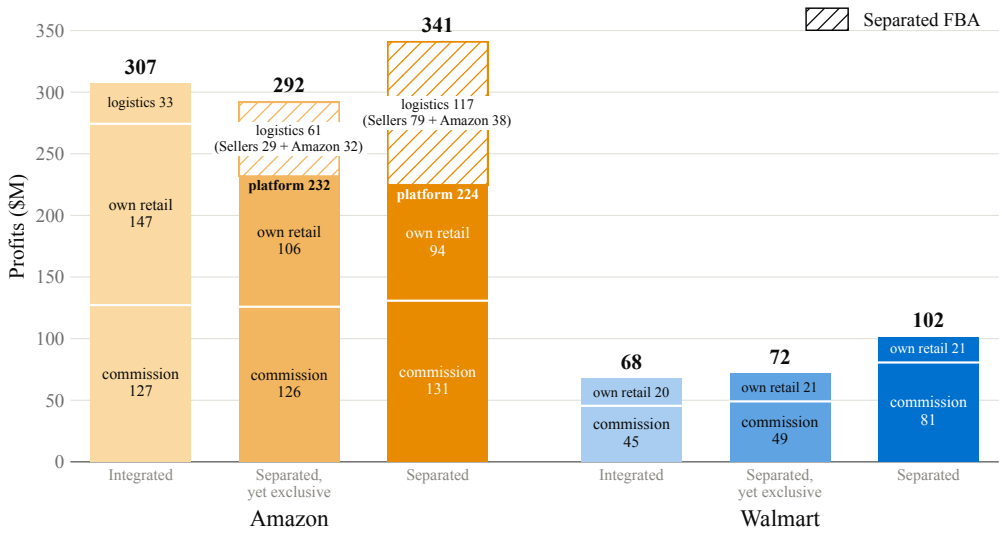



*Notes.* This figure compares the integrated benchmark with two counterfactuals that separate Amazon's logistics service (FBA) from its platform. In the benchmark, Amazon and Walmart set their commission rates and logistics fees in equilibrium. Under "Separated, yet exclusive," an independent operator runs FBA, but FBA serves only Amazon's marketplace. Under "Separated," the independent FBA serves both marketplaces. Panel (a) shows the logistics fees as indices, with the benchmark level set to 100. The fee of Walmart's logistics service (WFS) under "Separated" is omitted because no seller adopts WFS in that equilibrium. Panel (b) shows the commission rates in percent. Panel (c) shows the average consumer price on each marketplace and across both marketplaces, weighted by market shares. Panel (d) shows the share of third-party listings fulfilled by FBA and the share of seller-months listed on both marketplaces (multihoming), in percent; the horizontal axis for FBA adoption starts at 80%.

Figure A15: Decomposing FBA Separation Effects: Welfare and Platform Profits



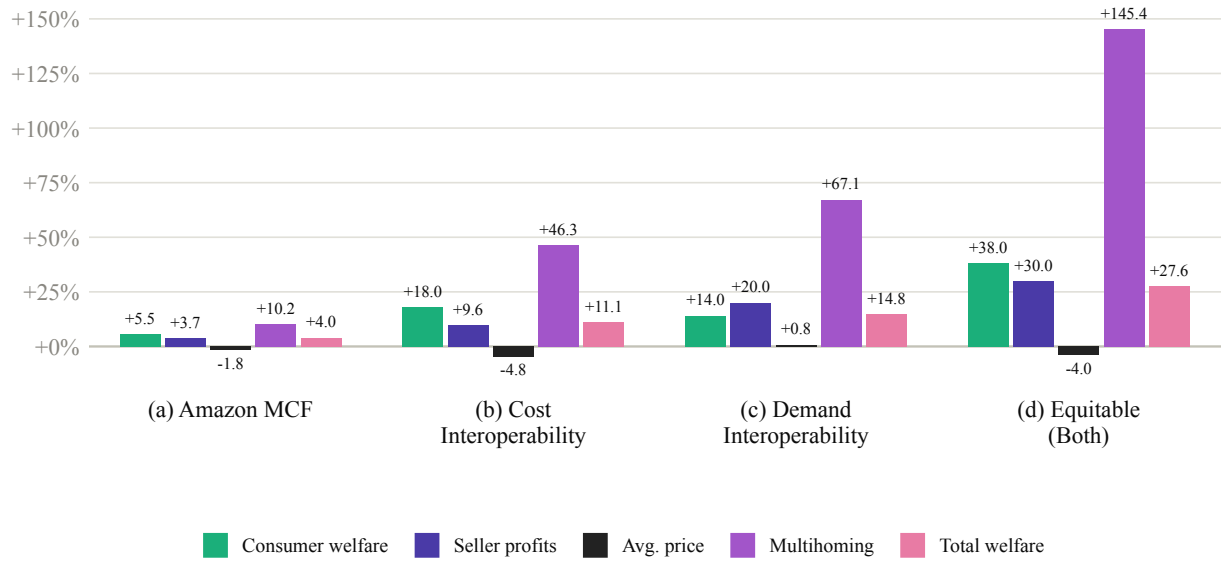

(a) Welfare




(b) Platform Profits

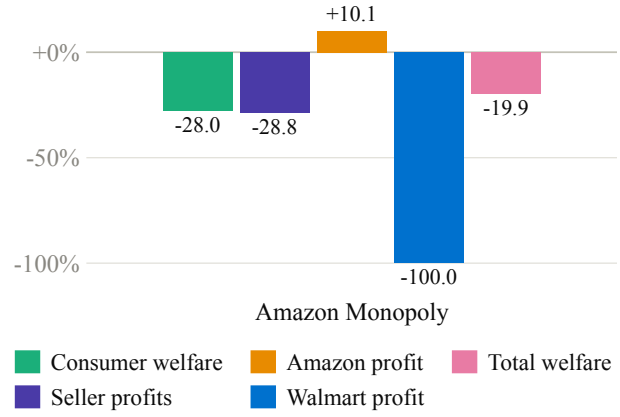
*Notes.* This figure compares the integrated benchmark with two counterfactuals that separate Amazon's logistics service (FBA) from its platform. Under "Separated, yet exclusive," an independent operator runs FBA, but FBA serves only Amazon's marketplace. Under "Separated," the independent FBA serves both marketplaces. Panel (a) shows percentage changes relative to the integrated benchmark. Amazon profit excludes the profit of the separated FBA; "Amazon + FBA" (hatched) adds it. Panel (b) decomposes platform profits ($M) into commissions, own retail, and logistics profits. The hatched block is the profit of the separated FBA; within it, "Sellers" and "Amazon" denote the profits earned from independent sellers' adoption and from Amazon's own retail, respectively. Under separation, "platform" is Amazon's profit excluding the separated FBA. Walmart's logistics profit is below $2M in every state and is not labeled.

Figure A16: Interoperable Logistics Counterfactuals

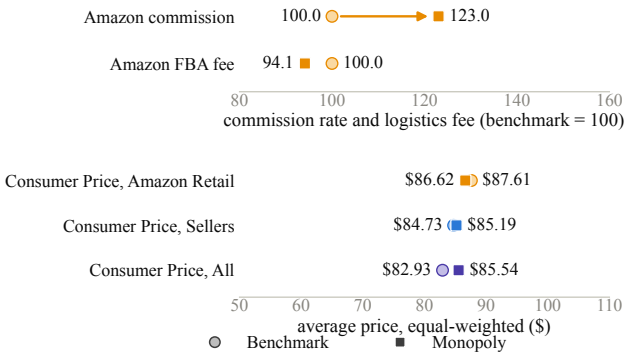



*Notes.* This figure shows the welfare and market outcomes of the interoperability analyses. Consumer welfare, seller profits, average prices, the share of multihoming sellers, and total welfare are shown as percent changes relative to the benchmark. All platform logistics fees and commission rates are held at their integrated benchmark levels in every panel. Panel (a) introduces Amazon's multi-channel fulfillment (MCF) service: multihoming sellers can fulfill Walmart orders through Amazon's logistics, pooling inventory, while paying the Amazon's posted multi-channel fee; the service offers two-business-day shipping, similar to Walmart's logistics. Panel (b) removes the cost-side exclusivity: the Walmart-side orders are charged the same fee that Amazon-side orders pay to Amazon logistics, instead of the higher multi-channel fee. Panel (c) removes the demand-side exclusivity: Prime subscribers attain the same benefits from Amazon logistics on the Walmart side as on the Amazon side, while the Walmart-side orders still pay the multi-channel fee. Panel (d) imposes equitable interoperability, combining the interventions of panels (b) and (c).

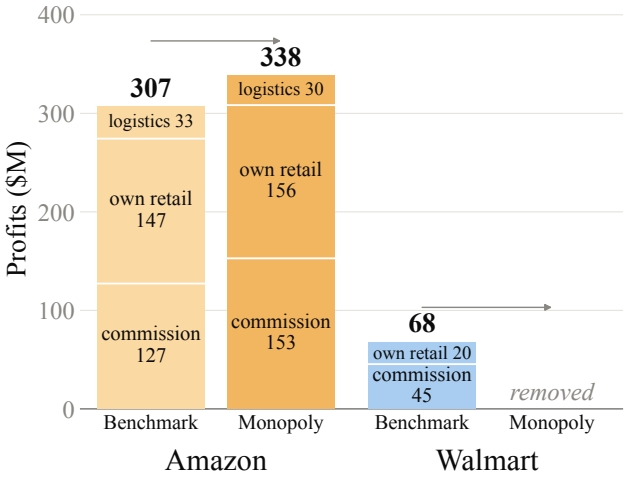
Figure A17: The Effects of an Amazon Monopoly



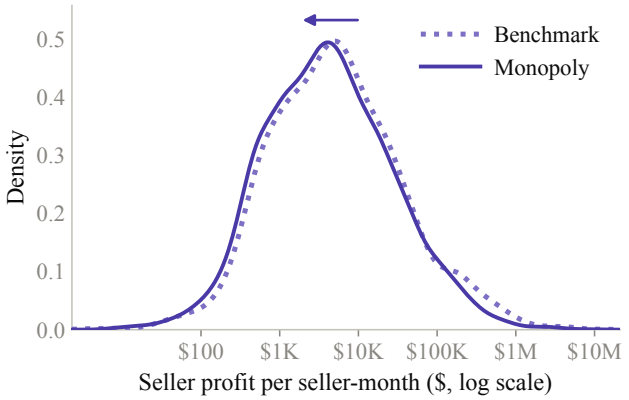

(a) Welfare




(b) Pricing




(c) Platform Profits by Parts




(d) Distribution of Seller Profits

*Notes.* This figure compares the integrated benchmark with the counterfactual that removes the Walmart marketplace and Walmart's logistics service (WFS), in which Amazon sets its commission rate and logistics fee as a monopolist. Panel (a) shows percentage changes relative to the integrated benchmark. Panel (b) shows Amazon's commission rate and logistics fee as indices, with the benchmark level set to 100, and the average consumer prices of Amazon's own retail products, of independent sellers' products, and of all products; each price is a simple average over listings, and the benchmark averages include Walmart's listings. Walmart's commission rate and logistics fee are omitted because Walmart is removed. Panel (c) decomposes platform profits ($M) into commissions, own retail, and logistics profits. Panel (d) plots kernel density estimates of seller profit per seller-month, computed on the same 3,142 seller-months in both states; the horizontal axis is on a logarithmic scale.